

SEVEN OPEN UNIT EQUILATERAL TRIANGLES DO NOT COVER A REGULAR UNIT HEXAGON

JINEON BAEK, JEEWON KIM

ABSTRACT. We prove that the regular hexagon of side length one cannot be covered by seven open equilateral triangles of side length one. Equivalently, no regular hexagon of side length greater than one is covered by seven closed unit equilateral triangles. Seven distinguished points force one center role and six vertex roles. An exhaustive classification of those roles reduces the proof to four mechanisms: perimeter and conditional-skeleton trace budgets; exact propagation of boundary and radial demands; normalized area loss; and a center-independent direct nine-point obstruction. The last mechanism forces six radial and three asymmetric witnesses into the center role. A reader-facing guide first gives the entire proof architecture; simplified body sections then present the geometric mechanisms and terminal implications, while only the irreducible exact calculations and certificates remain in technical appendices. The CE1 and CE2 center roles are treated by one signed normal form, so their traces, radial exits, propagation chains, and length budgets are derived only once. The terminal mixed-overlap signs are certified by exact rational reductions and computer-verified global Bernstein identities, with no interval subdivision or branch-and-bound.

CONTENTS

1. A Guide to the Proof	3
1.1. The theorem	3
1.2. A geometric dictionary	3
1.3. The exhaustive routing table	5
1.4. The four strategies	6
2. Structural Reduction to Finite Cases	7
2.1. Open, closed, and scaled formulations	7
2.2. The center classification	8
2.3. Vertex types and the T3-like normalization	9
2.4. Reach coordinates, supercritical rows, and gaps	11
2.5. Midpoint forcing	13
3. Signed CE1/CE2 Reductions	15
3.1. One signed center normal form	15
3.2. Boundary contribution and the diameter transfer	18
3.3. The local propagation interface	18
3.4. The common one-gap interface	19
4. Trace Budgets and Length Contradictions	22
4.1. Conditional skeleton trace caps	22

2020 *Mathematics Subject Classification.* 52C15, 52A40, 51M04.

Key words and phrases. equilateral triangles, geometric covering, regular hexagon, boundary propagation, area loss, caliper certificate.

4.2.	Vd1/Vd2 corner estimates	25
4.3.	Boundary trace caps	27
4.4.	Terminal length contradictions	28
4.5.	Common perimeter and skeleton budgets	28
5.	Strategy 2: Exact Demand Propagation	32
5.1.	The universal selected- T_+ curve	32
5.2.	Threshold routing	33
5.3.	One signed five-row chain	34
5.4.	The remaining exact-demand branches	35
6.	Area-Loss Obstructions	35
6.1.	The local square-loss inequalities	36
6.2.	The direct T3-like loss	38
6.3.	The cyclic area certificates	40
7.	Strategy 4: Direct Nine-Point Forcing	41
7.1.	Direct forcing of the disk and three frontier points	41
7.2.	Newton inner points	42
7.3.	The cap-chain lemma	42
7.4.	The four verified overlaps	43
7.5.	Enclosure and the center-independent contradiction	43
8.	Exhaustive Assembly	44
	Appendix A. Short Exact Vd1/Vd2 Placements	44
A.1.	A common adjacent-rescuer theorem	48
	Appendix B. Exact Local Demand Calculus	49
B.1.	The exact local feasible set	50
B.2.	The exact capped map	52
	Appendix C. Cyclic Propagation in the All-Vd0 Cases	53
C.1.	The exactly-one-supercritical all-Vd0 branch	53
	Appendix D. Nonsupercritical, T3-like, and Vd1/Vd2 Demand Branches	59
D.1.	The exactly-one-T3 branch	59
D.2.	The nonsupercritical CE1/CE2 boundary packages	60
D.3.	Two reusable capped-loss inequalities	67
D.4.	Half-edge radial envelope	74
D.5.	The exactly-one-Vd1/Vd2 placements	76
D.6.	Exact-demand terminal routing	77
	Appendix E. The Direct Nine-Point Obstruction	78
E.1.	The finite caliper criterion	78
E.2.	Corner-clipped AB -unions	79
E.3.	Exact-one handoffs and common demands	81
E.4.	Six directly forced radial points	81
E.5.	Tangent reduction and exact mixed overlaps	86
E.6.	The exact global certificate for the mixed overlaps	87
E.7.	Ray order and four overlaps	90
	Appendix F. Nonstandard Symbols	92

1. A GUIDE TO THE PROOF

This section gives the reader the entire proof architecture before any long calculation. It introduces the geometric objects and case labels, states the exhaustive routing table, and previews the four mechanisms that close its rows. Section 2 proves that the labels and reductions are exhaustive, and Section 3 gives the common signed center model. The reader-facing proofs then appear in Sections 4–7, before the final audit in Section 8; the irreducible exact calculations are retained in the appendices.

1.1. The theorem. Let H be the regular hexagon of side length one.

Theorem 1.1 (Main theorem). *The hexagon H cannot be covered by seven open equilateral triangles of side length one.*

The compactness and scaling argument in Proposition 2.1 gives the equivalent closed form.

Corollary 1.2 (Expanded closed formulation). *For every $L > 1$, the regular hexagon H_L of side length L cannot be covered by seven closed unit equilateral triangles.*

1.2. A geometric dictionary. Indices are taken in $\mathbb{Z}/6\mathbb{Z}$. Put

$$O = (0, 0), \quad V_i = \left(\cos \frac{i\pi}{3}, \sin \frac{i\pi}{3} \right), \quad H = \text{conv}\{V_0, \dots, V_5\}.$$

Write $e_{i,i+1} = [V_i, V_{i+1}]$ for a boundary edge, $r_i = [O, V_i]$ for a radial arm, and $M_i = V_i/2$. The radial target and the full skeleton are

$$D = \bigcup_{i=0}^5 r_i, \quad S = \partial H \cup D,$$

with lengths $\mathcal{H}^1(\partial H) = \mathcal{H}^1(D) = 6$ and $\mathcal{H}^1(S) = 12$.

Seven distinguished points force seven distinct roles: U_C contains O and U_i contains V_i . Their closures are denoted by T_C and T_i . The distinction is essential because closing an open trace may add an endpoint or a tangency.

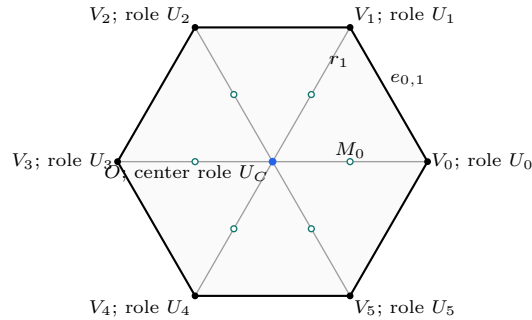


FIGURE 1. Exact target geometry in the normalization above. The labels show the seven forced role anchors; they do not prescribe the orientations of the covering triangles.

The center type records the number of boundary edges on which T_C has a positive-length trace. These numbers are 0, 1, 2, giving respectively CE0, CE1, CE2.

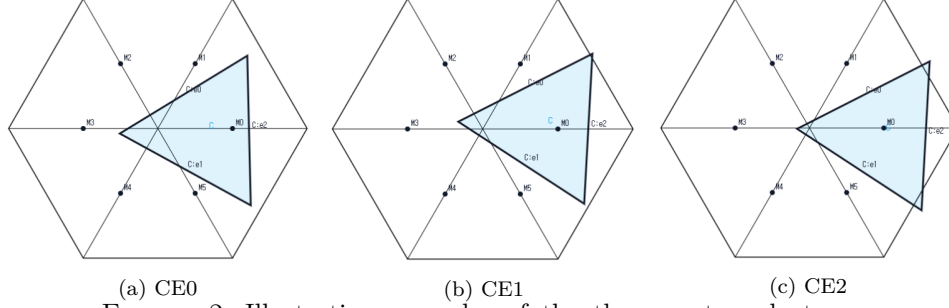


FIGURE 2. Illustrative examples of the three center-role types. The screenshots are not to scale, and no proof depends on visual inspection.

For a vertex role, let o be the number of its triangle vertices outside H and n the number of adjacent arms met in positive length. The exhaustive names are

type	(o, n)
Vd0	$(1, 0), (2, 0), (3, 0)$
Vd1	$(1, 1)$
Vd2	$(1, 2)$
T3-like	$(2, 1)$.

These classifications are mathematical rather than pictorial; the proofs of exhaustiveness appear in Propositions 2.3 and 2.5.

For an original open vertex role, let A_i, B_i, C_i be its actual maximal incoming boundary, outgoing boundary, and inward radial reaches. Thus row i is *supercritical* when $A_i + B_i > 1$, and

$$N_+ = |\{i : A_i + B_i > 1\}|.$$

Lowercase (a_i, b_i, c_i) will always denote smaller selected demands, never the actual maxima used to define N_+ .

For the CE1/CE2 case tree, call a vertex role *short* if it is supercritical or has positive-length trace on an adjacent radial arm. Let m be the number of roles of type Vd1, Vd2, or T3-like, and let q be the number of short roles. Lemma 4.16 proves the identity

$$q = N_+ + m.$$

Thus the first CE1/CE2 routing row below removes every branch with three short roles before any finer split is made. In particular, the midpoint rescuer of Lemma 4.5 puts every $N_+ \geq 2$ branch in that row.

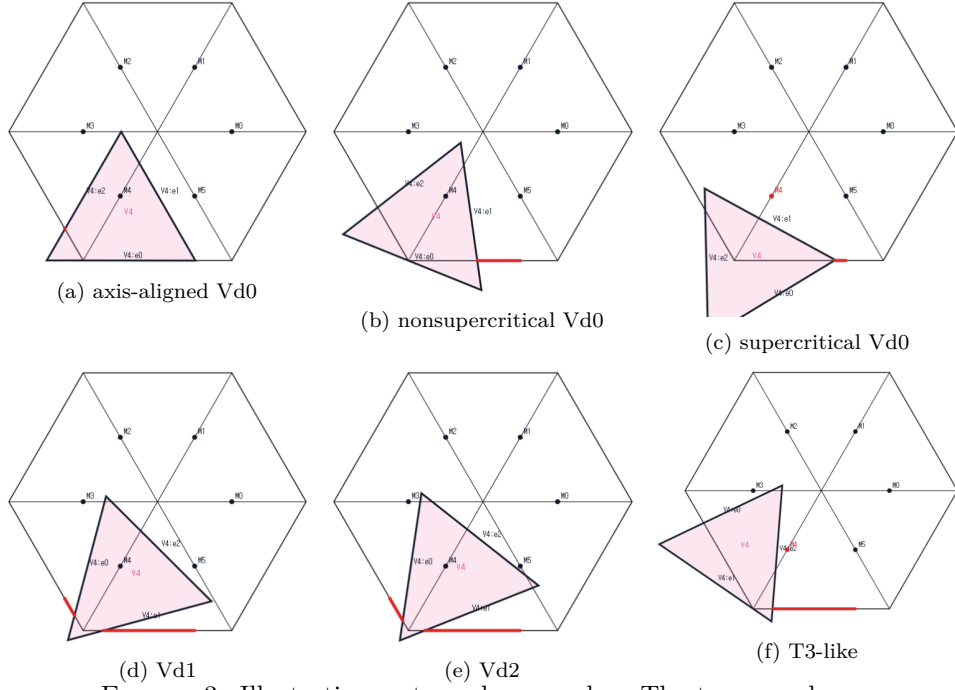


FIGURE 3. Illustrative vertex-role examples. The top row shows three Vd0 configurations; the bottom row shows the remaining three role types. The screenshots are not to scale, and no proof depends on visual inspection.

On $e_{i,i+1}$, parametrized from V_i to V_{i+1} , the two incident open traces are $[0, B_i)$ and $(1 - A_{i+1}, 1]$. Their nonempty complement

$$[B_i, 1 - A_{i+1}]$$

is a boundary gap. Equality gives a singleton gap, which remains uncovered because both incident traces are open at that point.

1.3. The exhaustive routing table. Table 1 is the logical spine of the proof. The CE0 rows are disjoint as written. The CE1/CE2 rows are applied from top to bottom: after the first row one has $q \leq 2$, so each later refinement is mutually exclusive. The one row marked 1+2 is explicitly assigned jointly to Strategies 1 and 2. Exhaustiveness is proved in Section 2, and the final audit appears in Section 8.

center and row count	exhaustive refinement	route
CE0, $N_+ = 0$	all vertex types	1
CE0, $N_+ = 1$	all six roles Vd0	4
CE0, $N_+ = 1$	a Vd1 or Vd2 role	1
CE0, $N_+ = 1$	no Vd1/Vd2 role and a T3-like role	3
CE0, $N_+ \geq 2$	all vertex types	3
CE1/CE2	$q \geq 3$	1
CE1/CE2, $N_+ = 0$	all six roles Vd0	2
CE1/CE2, $N_+ = 0$	a Vd1 or Vd2 role	1
CE1/CE2, $N_+ = 0$	no Vd1/Vd2 role and a T3-like role	2
CE1/CE2, $N_+ = 1$	all Vd0, zero gaps	4
CE1/CE2, $N_+ = 1$	all Vd0, a positive gap	2
CE1/CE2, $N_+ = 1$	exactly one T3-like, no Vd1/Vd2	2
CE1/CE2, $N_+ = 1$	exactly one Vd1/Vd2, no T3-like	1 / 1+2

TABLE 1. The complete terminal routing. The CE1/CE2 rows are read in order. Route j means Strategy j . In the last row the first entry is for CE1 and the second for CE2; $1 + 2$ is the single hybrid package.

1.4. The four strategies.

Strategy 1: one-dimensional trace budgets. Choose either ∂H or the conditional skeleton S . Bound the trace of each role on that target according to its center class, vertex type, and supercritical status. In every routed branch the seven upper bounds add to strictly less than the target length. The target, rather than the case label, is the organizing idea. Section 4 gives the trace bounds and terminal contradictions.

Strategy 2: exact demand propagation. A center trace supplies boundary inputs and complementary radial demands. For a nonsupercritical vertex row, a monotone capped map converts an incoming boundary demand into a lower bound on the next row. Iterating around the hexagon returns to row 0 with a demand exceeding its exact remaining capacity. Section 5 gives the common actual-row propagation mechanism; Appendices B–D retain the exact piecewise map and the branch-specific scalar certificates.

Strategy 3: area loss. Normalize a unit triangle to area one, so H has area six. Each selected boundary handoff forces a square amount of a vertex triangle outside the local hexagonal wedge. Two ascents, or one ascent together with a T3-like row, force total loss greater than one; after adding the center triangle, the available inside area is strictly less than six. Section 6 proves the local inequalities and both cyclic sums.

Strategy 4: a direct nine-point obstruction. In the zero-gap, all-Vd0, exact-one state, the five nonsupercritical handoffs form one strict descent chain. It produces common lower demands that force six radial points and three asymmetric points into U_C . Their convex hull cannot fit in an open unit equilateral triangle: an exact caliper argument gives enclosing side at least one, whereas compact containment in U_C would give side strictly below one. The reader-level construction and enclosure argument are in Section 7; Appendix E contains the exact frontier and overlap certificates.

The proof uses no global noncoverage claim for S , no interval subdivision, and no branch-and-bound. The last mixed-overlap step is an exact computer-verified

Bernstein certificate; its precise mathematical input, output, and rational reduction are proved in Subsection E.6.

2. STRUCTURAL REDUCTION TO FINITE CASES

Throughout, indices are taken in $\mathbb{Z}/6\mathbb{Z}$. Put

$$O = (0, 0), \quad V_i = \left(\cos \frac{i\pi}{3}, \sin \frac{i\pi}{3} \right), \quad H = \text{conv}\{V_0, \dots, V_5\}.$$

Thus H is the regular hexagon of side length one. We use the notation

$$e_{i,i+1} = [V_i, V_{i+1}], \quad r_i = [O, V_i], \quad M_i = \frac{1}{2}V_i,$$

and, when needed,

$$D = \bigcup_{i=0}^5 r_i, \quad S = \partial H \cup D, \quad S_{1/2} = \partial H \cup \{O, M_0, \dots, M_5\}.$$

The arms r_i meet only at O , up to sets of one-dimensional measure zero, and consequently

$$(1) \quad \mathcal{H}^1(\partial H) = 6, \quad \mathcal{H}^1(D) = 6, \quad \mathcal{H}^1(S) = 12.$$

For $L > 0$, $H_L := LH$ denotes the concentric regular hexagon of side length L . All symmetries used below belong to the dihedral group D_6 ; applying a symmetry always transports the indices, orientations, and incoming and outgoing labels together.

2.1. Open, closed, and scaled formulations.

Proposition 2.1 (Compactness and scaling equivalence). *The following assertions are equivalent.*

- (1) *The set H is covered by seven open equilateral triangles of side length 1.*
- (2) *For some $\varepsilon > 0$, the set H is covered by seven closed equilateral triangles of side length $1 - \varepsilon$.*
- (3) *For some $L > 1$, the set H_L is covered by seven closed equilateral triangles of side length 1.*

Proof. Suppose that open unit triangles $U_1, \dots, U_7$ cover H . Define

$$m_j(p) = \text{dist}(p, \mathbb{R}^2 \setminus U_j), \quad m(p) = \max_{1 \leq j \leq 7} m_j(p).$$

The continuous function m is positive on the compact set H , so $\eta := \min_{p \in H} m(p) > 0$. Choose

$$0 < \rho < \min \left\{ \eta, \frac{\sqrt{3}}{6} \right\}.$$

Moving each side of U_j inward through distance ρ gives a closed equilateral triangle K_j of side $1 - 2\sqrt{3}\rho$. Every $p \in H$ has $m_j(p) \geq \eta > \rho$ for some j , and hence belongs to K_j . This proves (1) $\Rightarrow$ (2), with $\varepsilon = 2\sqrt{3}\rho$.

Conversely, enlarging a closed triangle of side $1 - \varepsilon$ about its incenter to an open triangle of side 1 preserves containment, proving (2) $\Rightarrow$ (1). Finally, dilation by $(1 - \varepsilon)^{-1}$ identifies (2) with (3), with

$$L = (1 - \varepsilon)^{-1} > 1.$$

□

We conduct the geometric argument in the open side-one model. Denote the original open triangles by U_C and U_i for $0 \leq i \leq 5$, and put

$$T_C = \overline{U_C}, \quad T_i = \overline{U_i}.$$

The distinction between U_i and T_i will be maintained: closure may add endpoints or tangencies, although it does not change area or one-dimensional trace measure.

Lemma 2.2 (Distinct roles). *The seven triangles can be labelled so that*

$$O \in U_C, \quad V_i \in U_i \quad (0 \leq i \leq 5),$$

and these seven roles are distinct. Consequently

$$O \in \text{int}(T_C), \quad V_i \in \text{int}(T_i).$$

Proof. The seven distinguished points $O, V_0, \dots, V_5$ are pairwise at distance at least 1. Any two points of an open unit equilateral triangle are at distance strictly less than its diameter 1. Thus one open triangle contains at most one distinguished point. Selecting a covering triangle for each of the seven points gives an injection from seven points to seven triangles, hence a bijection and the asserted labelling. □

2.2. The center classification. For a closed center role define

$$N_E(T_C) = |\{i : T_C \cap e_{i,i+1} \text{ contains a positive-length interval}\}|.$$

Point contacts, endpoint contacts, and tangencies do not contribute. We say that T_C has type CE0, CE1, or CE2 according as $N_E(T_C)$ is 0, 1, or 2.

Proposition 2.3 (Exhaustive center types). *Every center role has exactly one of the types CE0, CE1, and CE2. Equivalently,*

$$N_E(T_C) \leq 2.$$

Proof. Among any three edges of the six-cycle ∂H , two have no common endpoint. It is therefore enough to show that relative-interior points of two nonadjacent edges are more than unit distance apart. Up to symmetry there are two cases. For $0 < t, s < 1$, put

$$x = (1 - t)V_0 + tV_1.$$

If $y = (1 - s)V_2 + sV_3$, direct calculation gives

$$\|x - y\|^2 - 1 = (1 - t)(2 - t) + s(s + t) > 0.$$

If instead $y = (1 - s)V_3 + sV_4$ and $u = t + s$, then

$$\|x - y\|^2 - 1 = (u - 1)^2 + 2 > 0.$$

A unit equilateral triangle has diameter 1. It therefore cannot contain relative-interior points on two nonadjacent hexagon edges. Three positive-length edge traces would supply such a pair, a contradiction. □

2.3. Vertex types and the T3-like normalization. For an original vertex role T_i , let

$$o(T_i) = |\{\text{vertices of } T_i \text{ outside } H\}|$$

and let $n(T_i)$ be the number of adjacent arms r_{i-1}, r_{i+1} having positive-length intersection with T_i . We use the names

type	(o, n)
Vd0	$(1, 0), (2, 0), (3, 0)$
Vd1	$(1, 1)$
Vd2	$(1, 2)$
T3-like	$(2, 1)$.

Lemma 2.4 (Local wedge). *Every original vertex role has $1 \leq o \leq 3$. If it has positive-length intersection with one adjacent arm, at least one of its vertices belongs to H ; if it has positive-length intersection with both adjacent arms, at least two of its vertices belong to H .*

Proof. Normalize to $i = 0$ and write

$$X = V_0 + x(V_1 - V_0) + y(V_5 - V_0).$$

Then $V_0 = (0, 0)$,

$$\|(x, y)\|^2 = x^2 + y^2 - xy,$$

and, within the closed unit ball about V_0 , membership in H is equivalent to membership in the wedge

$$W = \{(x, y) : x \geq 0, y \geq 0\}.$$

Indeed,

$$x^2 + y^2 - xy = \left(y - \frac{x}{2}\right)^2 + \frac{3x^2}{4} = (x - y)^2 + xy$$

shows that $x, y \geq 0$ and norm at most one imply the remaining hexagon inequalities. The adjacent arms are

$$r_1 = \{(1, y) : 0 \leq y \leq 1\}, \quad r_5 = \{(x, 1) : 0 \leq x \leq 1\}.$$

Suppose $T_0 \cap r_1$ has positive length, and let m be the largest x -coordinate of a vertex of T_0 . If $m > 1$, a maximizing vertex (m, y) lies in the unit ball and hence satisfies $0 < y < 1$; it belongs to H . If $m = 1$, positive-length contact with the supporting line $x = 1$ forces a side of T_0 to lie there, and both endpoints satisfy $1 + y^2 - y \leq 1$, hence belong to H . The same argument applies to r_5 . If both arms are met but only one triangle vertex belonged to H , that vertex would have to have both $x > 1$ and $y > 1$. This is impossible because $(x - y)^2 + xy > 1$.

Finally, if all three triangle vertices belonged to the convex set H , then $T_0 \subset H$. This is incompatible with the extreme point V_0 lying in the interior of T_0 . Hence $o \geq 1$. $\square$

Proposition 2.5 (Exhaustive vertex types). *Every original vertex role has exactly one of the four types Vd0, Vd1, Vd2, and T3-like.*

Proof. We have $n \in \{0, 1, 2\}$. Lemma 2.4 gives $o \leq 2$ when $n \geq 1$, and $o \leq 1$ when $n = 2$. Together with $o \geq 1$, the possibilities are

n	possible o
0	1, 2, 3
1	1, 2
2	1,

which are precisely the four named classes. $\square$

The next normalization is useful for trace estimates, but its final warning is essential.

Proposition 2.6 (T3-like closed-trace domination). *Let T be the closure of an original T3-like V_i -role. There is a translate $\hat{T}$ of T , with the same orientation and side length, such that*

$$V_i \in \text{relint}(a \text{ side of } \hat{T}), \quad T \cap H \subsetneq \hat{T} \cap H,$$

and $\hat{T}$ remains of type $(o, n) = (2, 1)$. Every old open interior point in $H \setminus \{V_i\}$ remains an interior point of $\hat{T}$. The point V_i itself becomes a boundary point, so $\hat{T}$ is a closed-trace majorant and is not a replacement open role.

Proof. Use the wedge coordinates of Lemma 2.4. Set

$$D_t = 1 - t + t^2, \quad z = \sqrt{D_t}, \quad 0 \leq t \leq 1.$$

After relabelling triangle vertices, every orientation belongs to one of the following two families:

	first side vector	second side vector
I	$z^{-1}(1, t)$	$z^{-1}(1 - t, 1)$
II	$z^{-1}(t, -(1 - t))$	$z^{-1}(1, t)$.

For Type I write

$$U = \alpha + y - tx, \quad V = \beta + x - (1 - t)y, \quad T = \{U \geq 0, V \geq 0, U + V \leq z\}.$$

Here $\alpha, \beta > 0$ and $\alpha + \beta < z$. Direct inversion shows that if exactly two triangle vertices are outside W , the sole wedge vertex has both coordinates strictly below 1: in the two possible cases its coordinates are bounded respectively by $(1 - t)/z$ and t/z , with the endpoint strictness supplied by $\alpha > 0$ or $\beta > 0$. The support argument in Lemma 2.4 then excludes positive-length contact with either $x = 1$ or $y = 1$. Thus Type I cannot be T3-like.

For Type II write

$$U = \alpha + (1 - t)x + ty, \quad V = \beta + tx - y, \quad T = \{U \geq 0, V \geq 0, U + V \leq z\}.$$

Reflecting in $x = y$ if necessary, assume the supported adjacent arm is r_1 . The two outside vertices have respectively a negative x - and a negative y -coordinate, while the unique wedge vertex is

$$Q = \left(\frac{z - \alpha - t\beta}{D_t}, \frac{t(z - \alpha) + (1 - t)\beta}{D_t} \right).$$

The endpoint cases $t = 0, 1$ give no positive adjacent interval, so $0 < t < 1$. The exact axis reaches are

$$A = \beta, \quad B = z - \alpha - \beta,$$

and T3-like support is equivalent to

$$Q_x > 1, \quad Q_y \leq 1.$$

Define $\widehat{T}$ by replacing α with 0 and keeping t, β fixed. Since the linear map $(x, y) \mapsto (U, V)$ is nonsingular, this changes only the translation. At the origin, $\widehat{U} = 0$ and $0 < \widehat{V} = \beta < z$, so V_0 is in the relative interior of a side. For $(x, y) \in W$,

$$\widehat{U} = (1-t)x + ty \geq 0, \quad \widehat{V} = V, \quad \widehat{U} + \widehat{V} = U + V - \alpha.$$

Thus $T \cap W \subset \widehat{T} \cap W$. The old x -axis reach is B and the new one is $B + \alpha$; because $B < 1$ and $\alpha > 0$, the inclusion is strict inside H .

The two outside vertices remain outside, whereas V_0 and the new wedge vertex lie in H , so $o = 2$. Moreover $\widehat{Q}_x > Q_x > 1$. From $Q_x > 1$ one gets $t\beta < z - \alpha - D_t$, and hence

$$tD_t(1 - \widehat{Q}_y) > D_t(1 - z) + (1 - t)\alpha > 0.$$

Thus $\widehat{Q}_y < 1$ and $n = 1$. Finally, for every nonzero point of W , $\widehat{U} > 0$; the displayed slack relations show that every old interior point other than the origin remains interior. $\square$

2.4. Reach coordinates, supercritical rows, and gaps. For the original vertex role U_i , define its actual maximal incoming, outgoing, and radial reaches by

$$\begin{aligned} A_i &= \sup\{t \in [0, 1] : V_i + t(V_{i-1} - V_i) \in U_i\}, \\ B_i &= \sup\{t \in [0, 1] : V_i + t(V_{i+1} - V_i) \in U_i\}, \\ C_i &= \sup\{t \in [0, 1] : V_i + t(O - V_i) \in U_i\}. \end{aligned}$$

Thus the closure T_i contains the corresponding endpoints. We reserve lowercase (a_i, b_i, c_i) for selected or prescribed lower-bound demands whenever actual and selected values occur together. A realizing role then satisfies

$$A_i \geq a_i, \quad B_i \geq b_i, \quad C_i \geq c_i.$$

An actual row is supercritical if $A_i + B_i > 1$, and

$$(2) \quad N_+ = |\{i : A_i + B_i > 1\}|.$$

This definition never counts a smaller demand pair in place of the actual row.

Parametrize $e_{i,i+1}$ by

$$X_i(x) = V_i + x(V_{i+1} - V_i), \quad 0 \leq x \leq 1.$$

The two incident open traces are

$$U_i \cap e_{i,i+1} = [0, B_i), \quad U_{i+1} \cap e_{i,i+1} = (1 - A_{i+1}, 1]$$

in this coordinate. A *boundary gap* on this edge is the nonempty closed complement between these two traces,

$$[B_i, 1 - A_{i+1}] \quad \text{when } B_i \leq 1 - A_{i+1}.$$

In particular, equality gives a singleton gap. This convention records the actual open traces; a point missing from both open roles is not erased merely because both closures contain it.

Lemma 2.7 (Exhaustiveness of the gap split). *On each edge the complement of the two incident vertex-role traces is either empty or one closed interval; a singleton interval counts as a gap. Every edge having a gap has positive-length intersection with the open center role. Consequently a CE0 center permits no gaps, a CE1 center permits at most one, and a CE2 center permits at most two.*

Proof. The two incident traces are intervals issuing from the two endpoints, so their complement is exactly the interval displayed above when nonempty. A nonincident vertex role cannot contain a relative-interior point of the edge: that point is more than unit distance from the role's distinguished vertex. Thus every point of a gap lies in U_C . Because U_C is open, its intersection with that edge contains a relative neighborhood and hence has positive length. Different gap edges therefore give different edges counted by $N_E(T_C)$. Proposition 2.3 gives the three stated bounds, including singleton gaps. $\square$

Lemma 2.8 (T3-like rows are nonsupercritical). *Every original T3-like row satisfies*

$$(3) \quad A_i + B_i \leq 1.$$

In particular, it is not counted by N_+ .

Proof. The proof of Proposition 2.6 showed that a T3-like role has Type II orientation with

$$A_i = \beta, \quad B_i = z - \alpha - \beta, \quad z = \sqrt{1 - t + t^2} \leq 1.$$

Thus $A_i + B_i = z - \alpha \leq 1$. The same conclusion holds after reflection. $\square$

We record the precise transfer from actual rows to strict handoff demands. Its hypothesis that the six vertex roles themselves cover the boundary is part of the statement.

Proposition 2.9 (Strict boundary handoffs). *Assume that $U_0, \dots, U_5$ cover ∂H . On each edge choose*

$$x_i \in (1 - A_{i+1}, B_i)$$

and define

$$a_i = 1 - x_{i-1}, \quad b_i = x_i.$$

Then $0 < a_i < A_i$, $0 < b_i < B_i$, the two selected anchors lie in U_i , and

$$a_i^2 + a_i b_i + b_i^2 < 1.$$

Moreover:

- (1) *if exactly one actual row, indexed by p , is supercritical, every strict selection has exactly one selected supercritical row, also indexed by p ;*
- (2) *if at least two actual rows are supercritical, some simultaneous strict selection has at least two selected supercritical rows.*

Proof. Because V_i is interior to a diameter-one closed triangle,

$$0 < A_i < 1, \quad 0 < B_i < 1.$$

No nonincident vertex role contains a relative-interior point of $e_{i,i+1}$, since such a point is more than unit distance from its distinguished vertex. The two displayed incident traces therefore cover the edge and, being relatively open and nonempty, overlap. Hence each interval $(1 - A_{i+1}, B_i)$ is nonempty. The anchor and distance

assertions follow from the choice and from the fact that two points of U_i are at distance strictly less than one.

Put $\ell_i = 1 - A_{i+1}$ and $u_i = B_i$. Then

$$(4) \quad A_i + B_i > 1 \iff \ell_{i-1} < u_i, \quad a_i + b_i > 1 \iff x_{i-1} < x_i.$$

If actual row i is nonsupercritical, then

$$x_i < u_i \leq \ell_{i-1} < x_{i-1},$$

so every selected version of it is strictly subcritical. Also

$$(5) \quad \sum_{i=0}^5 (a_i + b_i) = \sum_{i=0}^5 (1 - x_{i-1} + x_i) = 6.$$

If p is the sole actual supercritical index, the other five selected sums are strictly below one; equation (5) forces the p th sum above one. This proves (1).

For (2), first take two nonadjacent supercritical indices p, q . For each $r \in \{p, q\}$ choose

$$\ell_{r-1} < z_r < u_r,$$

then choose

$$x_{r-1} \in (\ell_{r-1}, \min\{u_{r-1}, z_r\}), \quad x_r \in (\max\{\ell_r, z_r\}, u_r).$$

The two pairs of variables are disjoint and give ascents at p, q . Any set of at least three indices on a six-cycle contains two nonadjacent indices. It remains to consider exactly two adjacent supercritical indices $p, p+1$. For the other four nonsupercritical rows, successive inequalities $\ell_i < u_i \leq \ell_{i-1}$ give

$$\ell_{p-1} < \ell_{p+1}.$$

Together with supercriticality at $p, p+1$, this implies

$$\max\{\ell_{p-1}, \ell_p\} < \min\{u_p, u_{p+1}\}.$$

Choose x_p between these bounds and then choose $x_{p-1} < x_p < x_{p+1}$ in their respective overlap intervals. By (4), both adjacent rows are selected supercritical. $\square$

2.5. Midpoint forcing. We first need a local fact for vertex roles. Take unit vectors

$$u = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right), \quad v = \left(\frac{1}{2}, -\frac{\sqrt{3}}{2}\right).$$

Lemma 2.10 (Self-midpoint obstruction). *Let a closed unit triangle contain*

$$0, \quad au, \quad bv, \quad \frac{u+v}{2}.$$

Then $a + b \leq 1$. Consequently, for every vertex role,

$$M_i \in T_i \implies A_i + B_i \leq 1.$$

If the two demanded boundary points and the midpoint are contained with a positive margin, the conclusion is strict.

Proof. Suppose $a + b > 1$ and set

$$K = \text{conv} \left\{ 0, au, \frac{u+v}{2}, bv \right\}.$$

For outward unit normals n_0, n_1, n_2 separated by 120° , the least equilateral triangle with those normals and containing K has side length

$$\frac{2}{\sqrt{3}} \sum_{j=0}^2 h_K(n_j).$$

The finite-caliper theorem, Theorem E.1, reduces its minimum to an outward normal of one of the four edges of K .

For the edge from 0 to au , direct projection gives

$$S_{0A} = \max \left\{ \frac{\sqrt{3}}{4}, \frac{\sqrt{3}a}{2} \right\} + \frac{\sqrt{3}b}{2} > \frac{\sqrt{3}}{2};$$

if $a \geq 1/2$ this is $(\sqrt{3}/2)(a+b)$, and if $a < 1/2$ then $b > 1/2$. The edge through bv is symmetric. For the edge from au to $(u+v)/2$, put $D_a = \sqrt{4a^2 - 2a + 1}$. Projection gives

$$S_{AM} = \begin{cases} \frac{\sqrt{3}(a+b)}{2D_a}, & a \leq \frac{1}{2}, \\ \frac{\sqrt{3}(2a^2+b)}{2D_a}, & a > \frac{1}{2}. \end{cases}$$

The first value is strictly above $\sqrt{3}/2$ because $D_a \leq 1$. In the second case, $b > 1-a$ and

$$(2a^2 - a + 1)^2 - D_a^2 = a^2(2a - 1)^2 \geq 0,$$

so again $S_{AM} > \sqrt{3}/2$. The fourth edge is symmetric. Every enclosing equilateral triangle therefore has side greater than one, a contradiction.

If equality $a + b = 1$ occurred while all relevant points had positive margin, both boundary demands could be increased slightly; the closed conclusion would then be violated. $\square$

The corresponding center fact is exact and uses the interior condition at O .

Proposition 2.11 (Unique center midpoint). *Let T_C be a closed unit equilateral triangle with $O \in \text{int}(T_C)$. If T_C has positive-length intersection with a boundary edge of H , then it contains exactly one of $M_0, \dots, M_5$.*

Proof. Normalize the positive trace by a dihedral symmetry. Proposition 3.1 gives a single signed side-slack model for both CE1 and CE2 and proves directly that its midpoint set is $\{M_0\}$. Undoing the normalization proves the assertion. $\square$

Remark 2.12. The hypothesis $O \in \text{int}(T_C)$ cannot be weakened to $O \in T_C$: the triangle $\text{conv}\{O, V_0, V_1\}$ has a boundary-edge overlap and contains both M_0 and M_1 .

Proposition 2.13 (Exhaustive structural reduction). *Every hypothetical cover by seven open unit equilateral triangles admits the distinct center and vertex roles used in Table 1, and its actual reaches determine exactly one row of that table.*

Proof. Lemma 2.2 gives the seven roles. Propositions 2.3 and 2.5 give the exhaustive and mutually exclusive center and vertex types. The actual reaches determine exactly one of $N_+ = 0$, $N_+ = 1$, and $N_+ \geq 2$. For CE0 these data and the vertex classification give the five displayed rows.

Now suppose the center is CE1 or CE2. It contains exactly one radial midpoint by Proposition 2.11. Lemma 4.16 gives $q = N_+ + m$, and Lemma 4.5, together with the disjoint count, shows that $N_+ \geq 2$ forces a third short role. Hence every such branch either enters the first CE1/CE2 row or has $q \leq 2$ and $N_+ \leq 1$. If $N_+ = 0$, the vertex classification gives, in order, all Vd0, at least one Vd1/Vd2, or no such role and at least one T3-like. If $N_+ = 1$, then $q \leq 2$ implies $m \leq 1$: the roles are all Vd0, exactly one T3-like, or exactly one Vd1/Vd2. In the all-Vd0 case, Lemma 2.7 supplies the exhaustive zero-gap/positive-gap split. These alternatives are disjoint, proving the claim. $\square$

3. SIGNED CE1/CE2 REDUCTIONS

This section gives the common center model and the shared one-gap proof interface. The exact local admissible set and the four-label capped map are proved in Appendix B; only their monotonicity, duality, and threshold consequences are used here.

3.1. One signed center normal form.

Proposition 3.1 (Signed CE1/CE2 center normal form). *Let T_C be a closed unit equilateral triangle with $O \in \text{int}(T_C)$ and a positive-length boundary trace. Normalize such a trace to $e_{0,1}$ and use affine coordinates*

$$X = V_0 + b(V_1 - V_0) + a(V_5 - V_0).$$

There are parameters

$$0 < R < 1, \quad W = 1 - R, \\ E = \sqrt{1 - RW}, \quad \eta = 1 - E, \quad P = E(1 - E),$$

and nonnegative center slacks α, δ such that, with

$$k = \eta + \alpha + \delta,$$

the center triangle is

$$(6) \quad \boxed{\begin{array}{l} F_0 = R + \alpha - a + Wb, \\ F_1 = Rb + Wa - k, \\ F_2 = W + \delta - b + Ra, \end{array}} \quad T_C = \{F_0, F_1, F_2 \geq 0\}.$$

Define

$$(7) \quad \Delta_R = P - \alpha - W\delta, \quad \Delta_L = P - R\alpha - \delta.$$

Then

$$(8) \quad T_C \cap e_{0,1} = \left[\frac{k}{R}, W + \delta \right], \quad \mathcal{H}^1(T_C \cap e_{0,1}) = \frac{\Delta_R}{R},$$

and the possible companion trace is

$$(9) \quad T_C \cap e_{5,0} = \left[\frac{k}{W}, R + \alpha \right] \quad \text{when } \Delta_L > 0, \quad \mathcal{H}^1(T_C \cap e_{5,0}) = \frac{[\Delta_L]_+}{W}.$$

The normalization requires $\Delta_R > 0$, and

$$(10) \quad T_C \text{ is CE1} \iff \Delta_L \leq 0, \quad T_C \text{ is CE2} \iff \Delta_L > 0.$$

The six center exits, measured from O , are

$$(11) \quad \boxed{\begin{aligned} d_0^C &= E - \alpha - \delta, & d_1^C &= \frac{\delta}{R}, & d_2^C &= \delta, \\ d_3^C &= \min \left\{ \frac{\alpha}{R}, \frac{\delta}{W} \right\}, & d_4^C &= \alpha, & d_5^C &= \frac{\alpha}{W}. \end{aligned}}$$

For closures of original open center roles one has $\alpha, \delta > 0$.

Proof. Write

$$T_C \cap e_{0,1} = [s, t], \quad 0 \leq s < t \leq 1.$$

The two sides through these endpoints have a positive slope ratio. Reflect across the perpendicular bisector of $e_{0,1}$ if necessary and call the ratio $R \leq 1$; put $W = 1 - R$. Direct equations for the three side lines give the unit-equilateral edge-cut slacks

$$F_1 = Rb + Wa - Rs,$$

$$F_2 = -b + Ra + t,$$

$$F_0 = Wb - a + E + Rs - t.$$

Here $E = \sqrt{1 - RW}$ is forced by the unit side-length relation. The case $R = 1$ would give $F_0(O) = s - t < 0$, contrary to $O \in \text{int}(T_C)$. Thus $0 < R < 1$.

Set

$$\alpha = F_0(O), \quad \delta = F_2(O).$$

Then

$$t = W + \delta, \quad Rs = \eta + \alpha + \delta = k,$$

and substitution gives (6). The gradients are the three unit-equilateral side-normal directions and

$$F_0 + F_1 + F_2 = E,$$

so this is the same unit equilateral triangle.

On $e_{0,1}$ one has $a = 0$. The active inequalities are

$$Rb \geq k, \quad b \leq W + \delta,$$

which gives (8); its length is

$$W + \delta - \frac{k}{R} = \frac{P - \alpha - W\delta}{R}.$$

On $e_{5,0}$ one has $b = 0$, and the active inequalities are

$$Wa \geq k, \quad a \leq R + \alpha.$$

Their signed interval length is

$$R + \alpha - \frac{k}{W} = \frac{P - R\alpha - \delta}{W},$$

which proves (9).

The positive right trace gives

$$\delta < \frac{P}{W} = \frac{ER}{1 + E} < \frac{R}{2}.$$

On $e_{1,2}$, parameterized by $b = 1 + q$, $a = q$, the slack F_2 equals

$$\delta - R - Wq < 0.$$

Thus a second positive trace cannot occur on $e_{1,2}$. The CE classification says that a center triangle has at most two positive boundary traces and that two such traces are adjacent. Hence $e_{5,0}$ is the only possible companion edge, and its signed length proves (10). Equality $\Delta_L = 0$ is only a point contact and is therefore CE1.

For the radial exits, substitute the six ray parametrizations in (6). On r_1 the decreasing slack is $\delta - Rq$; on r_2 it is $\delta - q$; on r_3 the two decreasing slacks are $\alpha - Rq$ and $\delta - Wq$. Reflection gives r_4, r_5 , and on r_0 the decreasing slack is

$$E - \alpha - \delta - q.$$

This proves (11).

Finally, the third center slack is $E - \alpha - \delta$. Since

$$W(\alpha + \delta) \leq \alpha + W\delta < P,$$

we have

$$E - \alpha - \delta > E - \frac{P}{W} = \frac{E(E - R)}{W} > \frac{1}{2};$$

the last inequality follows from

$$2E(E - R) - W = (E - R)^2 > 0.$$

Together with

$$\alpha < P < \min \left\{ \frac{R}{2}, \frac{W}{2} \right\}, \quad \delta < \frac{R}{2},$$

the midpoint tests give the exact midpoint set $\{M_0\}$. The closures of original open center roles have all three center slacks positive, so $\alpha, \delta > 0$ there. $\square$

Remark 3.2 (Legacy variables). For CE1, take

$$\lambda = R, \quad s = \frac{k}{R}, \quad t = W + \delta, \quad C_0 = \alpha, \quad C_2 = \delta.$$

For CE2, take

$$x = \frac{k}{W}, \quad u = R + \alpha, \quad y = \frac{k}{R}, \quad v = W + \delta.$$

If $S = x + y$ and $D = \sqrt{x^2 + xy + y^2}$, then

$$S = \frac{k}{RW}, \quad D = \frac{Ek}{RW},$$

and the old coupling equation is automatic:

$$(u + v)S - xy = D.$$

Thus the CE1 and CE2 formula lists are two coordinate readings of Proposition 3.1.

3.2. Boundary contribution and the diameter transfer.

Corollary 3.3 (Common center-boundary formula). *The full center-boundary contribution is*

$$(12) \quad L_{\partial H}(T_C) = \frac{\Delta_R}{R} + \frac{[\Delta_L]_+}{W}.$$

In CE1,

$$L_{\partial H}(T_C) \leq P \leq \frac{\sqrt{3}}{2} - \frac{3}{4}.$$

In CE2,

$$L_{\partial H}(T_C) = \frac{E}{1+E} - \frac{E^2(\alpha + \delta)}{RW} < \frac{1}{2}.$$

Proof. Equation (12) is (8) plus (9). If $\Delta_L \leq 0$, then $R\alpha + \delta \geq P$. Multiplying by W and using $\alpha \geq RW\alpha$ gives

$$\alpha + W\delta \geq WP.$$

Thus $\Delta_R \leq RP$, so the boundary contribution is at most P . Since $E \geq \sqrt{3}/2$ and $E(1-E)$ decreases on this interval, the CE1 cap follows.

If $\Delta_L > 0$, direct addition gives

$$\begin{aligned} L_{\partial H}(T_C) &= \frac{P - \alpha - W\delta}{R} + \frac{P - R\alpha - \delta}{W} \\ &= \frac{P - E^2(\alpha + \delta)}{RW}. \end{aligned}$$

Here $W + R^2 = W^2 + R = E^2$, and $P/(RW) = E/(1+E)$. The last display is therefore strictly less than $1/2$. $\square$

Lemma 3.4 (Adjacent-edge diameter transfer). *For $0 \leq q \leq 1$, put*

$$\beta(q) = \frac{-q + \sqrt{4 - 3q^2}}{2}, \quad \lambda_{\circ}(q) = 1 - \beta(q).$$

If a unit equilateral triangle reaches parameters q, b on two adjacent boundary edges, then

$$b \leq \beta(q).$$

For $q > 0$,

$$\beta(q) < 1 - \frac{q}{2}, \quad \lambda_{\circ}(q) > \frac{q}{2}.$$

Proof. The squared distance between the two boundary points is $q^2 + qb + b^2$, so it is at most the squared diameter 1. Solving the quadratic in b gives $b \leq \beta(q)$. For $q > 0$, $\sqrt{4 - 3q^2} < 2$, which gives both strict linear comparisons. $\square$

3.3. The local propagation interface. Let

$$u = \left(\frac{1}{2}, \frac{\sqrt{3}}{2} \right), \quad v = \left(\frac{1}{2}, -\frac{\sqrt{3}}{2} \right),$$

and for $0 \leq a, b, c \leq 1$ put

$$K(a, b, c) = \text{conv}\{0, au, bv, c(u+v)\}.$$

The local admissible set is

$$\mathcal{A} = \{(a, b, c) : K(a, b, c) \text{ lies in a closed unit equilateral triangle}\}.$$

For $a, c \in [0, 1]$ define

$$\begin{aligned} B_c(a) &= \max\{b : (a, b, c) \in \mathcal{A}\}, \quad F_c(a) = \min\{B_c(a), 1 - a\}, \\ G_c(a) &= 1 - F_c(a). \end{aligned}$$

Propositions B.2 and B.3 prove that $\mathcal{A}$ is coordinatewise down-closed, F_c is nonincreasing, and G_c is nondecreasing and extensive:

$$(13) \quad G_c(z) \geq z.$$

They also prove the exact reflection duality

$$(14) \quad G_c(a) \leq z \iff G_c(1 - z) \leq 1 - a.$$

Suppose an actual nonsupercritical row has maximal reaches (A_i, B_i, C_i) and

$$A_i \geq z, \quad C_i \geq c_i.$$

The same role realizes (z, B_i, c_i) , so

$$B_i \leq F_{c_i}(z).$$

If the next boundary handoff gives $A_{i+1} \geq 1 - B_i$, then

$$(15) \quad A_{i+1} \geq 1 - B_i \geq 1 - F_{c_i}(z) = G_{c_i}(z).$$

Every composition below is shorthand for repeated applications of this inequality to specified actual rows.

For the low-root thresholds used below, put

$$(16) \quad e(d) = \frac{1-d}{2} \left(1 - \sqrt{4(1-d)^2 - 3} \right) \quad \left(0 < d < 1 - \frac{\sqrt{3}}{2} \right).$$

For the low-radial order transition, put

$$(17) \quad h(c) = \frac{2c}{1 + \sqrt{16c^2 - 3}} \quad \left(\frac{1}{2} < c \leq \frac{\sqrt{3}}{2} \right).$$

The exact map calculation in (79)–(80) supplies the quoted bounds and threshold implication for e ; Proposition B.3 identifies h as the low-radial order transition.

3.4. The common one-gap interface.

Proposition 3.5 (Common five-row one-gap reduction). *Assume every vertex role is $Vd0$, $N_+ = 1$, and the normalized right center trace contains a V -gap, possibly a singleton. In the CE2 case assume the companion trace contains no V -gap. Put*

$$X = R - \delta, \quad H = \frac{k}{2R}, \quad m = \min \left\{ \frac{\alpha}{R}, \frac{\delta}{W} \right\},$$

and

$$\begin{aligned} c_1 &= 1 - \frac{\delta}{R}, & c_2 &= 1 - \delta, & c_3 &= 1 - m, \\ c_4 &= 1 - \alpha, & c_5 &= 1 - \frac{\alpha}{W}, & c_0 &= k. \end{aligned}$$

Let

$$z_0 = X, \quad z_i = G_{c_i}(z_{i-1}) \quad (1 \leq i \leq 5),$$

and $Z = z_5$. Then the actual reaches satisfy

$$(18) \quad a_0 \geq Z, \quad a_0 \leq B_k \left(\frac{k}{R} \right) < 1 - H.$$

Consequently it is enough to prove

$$(19) \quad (G_{1-\delta} \circ G_{1-m} \circ G_{1-\alpha})(H) > W + \delta.$$

Proof. The exact-midpoint argument makes rows $1, \dots, 5$ nonsupercritical and row 0 uniquely supercritical. Gap containment gives

$$b_0 \geq \frac{k}{R}, \quad a_1 \geq X.$$

Suppose inductively that $a_i \geq z_{i-1}$. Radial coverage gives actual radial reach at least c_i , and the proof-safe capped map gives

$$b_i \leq F_{c_i}(z_{i-1}).$$

The next boundary handoff therefore gives

$$a_{i+1} \geq 1 - b_i \geq G_{c_i}(z_{i-1}) = z_i.$$

For the final handoff, the CE1 companion trace is empty and the CE2 companion trace is gap-free. Hence

$$a_0 \geq z_5 = Z.$$

The same row has boundary reach at least k/R and radial reach at least k . Reflection and down-closedness of $\mathcal{A}$ give

$$\left(\frac{k}{R}, a_0, k\right) \in \mathcal{A},$$

so $a_0 \leq B_k(k/R)$. Lemma 3.4 gives

$$B_k(k/R) \leq \beta(k/R) < 1 - \frac{k}{2R} = 1 - H,$$

which proves (18).

The first and fifth maps are extensive and the middle maps are nondecreasing, so it is enough to prove

$$(G_{c_4} \circ G_{c_3} \circ G_{c_2})(X) > 1 - H.$$

Three applications of the exact duality

$$G_c(a) \leq z \iff G_c(1 - z) \leq 1 - a$$

show that its negation is equivalent to the negation of (19). This proves the reduction. $\square$

The CE1 sign $\Delta_L \leq 0$ is completed by the selected- T_+ chord proof in Proposition 5.3. We now give the shorter CE2 scalar clause.

Lemma 3.6 (CE2 total slack and two thresholds). *Assume $\Delta_R > 0$ and $\Delta_L > 0$, and put*

$$T = \alpha + \delta, \quad Q = \frac{\eta + T}{2R}.$$

Then

$$(20) \quad T < \frac{\eta}{E},$$

and

$$(21) \quad \min\{e(\alpha), e(\delta)\} < Q.$$

Proof. The CE2 inequalities are

$$\alpha + W\delta < P, \quad R\alpha + \delta < P.$$

Multiply the first by W , the second by R , and add. Since $W + R^2 = W^2 + R = E^2$, this gives

$$E^2T < P = E\eta,$$

which is (20).

Let $d = \min\{\alpha, \delta\}$. Since $d \leq T/2$ and the rational low-root bound is increasing,

$$\min\{e(\alpha), e(\delta)\} < \frac{2d}{1-2d} \leq \frac{T}{1-T}.$$

It remains to prove

$$\frac{T}{1-T} < \frac{\eta + T}{2R}.$$

Define

$$\Phi(t) = (\eta + t)(1 - t) - 2Rt.$$

This function is concave, so its minimum on $[0, \eta/E]$ is at an endpoint. Clearly $\Phi(0) = \eta > 0$. Direct simplification using $E^2 = 1 - R + R^2$ gives

$$\Phi\left(\frac{\eta}{E}\right) = \frac{R(E - R)^2}{E^2} > 0.$$

Thus $\Phi(T) > 0$, proving the desired comparison and (21). $\square$

Proposition 3.7 (Short CE2 one-gap completion). *Under the CE2 hypotheses of Proposition 3.5, the one-gap state is impossible in either orientation.*

Proof. The right-gap initial input is $z_0 = R - \delta$. The strict CE2 domain gives

$$Wz_0 = RW - W\delta = \eta + P - W\delta > \eta + \alpha.$$

For

$$\pi(q) = (\eta + q)(1 - 2q) - 2qW,$$

one has

$$\pi(0) = \eta > 0, \quad \pi(P) = \eta(\eta + 2ER^2) > 0.$$

Concavity therefore gives $\pi(\alpha) > 0$, and the low-root bound yields

$$(22) \quad z_0 > \frac{\eta + \alpha}{W} > \frac{2\alpha}{1 - 2\alpha} > e(\alpha).$$

If $e(\alpha) < Q$, extensivity carries (22) to the row-4 map $G_{1-\alpha}$, which gives an output at least $1 - e(\alpha) > 1 - Q$. All later maps preserve it. If $e(\alpha) \geq Q$, Lemma 3.6 gives $e(\delta) < Q$, and

$$z_0 > e(\alpha) \geq Q > e(\delta).$$

The row-2 map $G_{1-\delta}$ then gives an output above $1 - Q$, again preserved by all later maps. Thus $Z > 1 - Q = 1 - H$, contradicting (18).

Reflection exchanges

$$R \leftrightarrow W, \quad \alpha \leftrightarrow \delta$$

and reverses the row order. The signed CE2 domain and the proof above are invariant under this exchange, so the left-gap orientation is impossible as well. $\square$

Remark 3.8 (Why the CE1 scalar clause remains separate). The geometric one-gap interface is common, but the CE2 estimate (21) is not valid on the full CE1 sign domain. Thus the merger stops at Proposition 3.5; the CE1 selected- T_+ clause and the CE2 threshold clause are genuinely different scalar proofs of the same target (19).

4. TRACE BUDGETS AND LENGTH CONTRADICTIONS

This section develops the boundary and conditional-skeleton trace bounds used by the length mechanism.

For a closed triangle T and a rectifiable target E , write

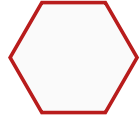
$$L_E(T) = \mathcal{H}^1(T \cap E).$$

The passage from an original open role U to its closure $T = \overline{U}$ can only enlarge the trace. Hence a cover of E by the open roles implies

$$(23) \quad \mathcal{H}^1(E) \leq L_E(T_C) + \sum_{i=0}^5 L_E(T_i).$$

All estimates in this section concern full traces. They therefore remain valid for any assigned measurable subsets of those traces.

The strategy uses only the perimeter ∂H and the full skeleton S , whose measures were recorded in (1). Every skeleton estimate below is conditional on its stated center and row hypotheses; no unconditional noncoverage assertion for S is made.



∂H : length 6



S : length 12

the full-skeleton estimate is used only under its stated branch hypotheses

FIGURE 4. The two one-dimensional targets. The full skeleton is used only under the hypotheses of the corresponding trace budget.

4.1. Conditional skeleton trace caps. We now develop the skeleton estimate used only when the center is CE1 or CE2. The first lemma is the center contribution.

Lemma 4.1 (Center skeleton cap). *If T_C is a CE1 or CE2 center role and contains exactly one radial midpoint, then*

$$L_S(T_C) < \frac{3}{2}.$$

Proof. Normalize the unique midpoint as M_0 and a positive boundary overlap to $e_{0,1}$. Use the signed side slacks from Proposition 3.1, and put

$$\lambda = R, \quad \rho = E, \quad X = \alpha, \quad Y = \delta.$$

Thus $0 < \lambda < 1$ and $\rho = \sqrt{1 - \lambda + \lambda^2}$. Put

$$P = \rho(1 - \rho),$$

so the positive-length condition $s < t$ is equivalent to

$$(24) \quad X + (1 - \lambda)Y < P.$$

The boundary trace on $e_{0,1}$ is

$$\ell_{01} = 1 - \lambda + Y - \frac{X + Y + 1 - \rho}{\lambda},$$

and the possible second trace on $e_{5,0}$ has length

$$\ell_{50} = \left[\frac{P - (\lambda X + Y)}{1 - \lambda} \right]_+.$$

Parametrizing the six radial arms by distance from O , direct substitution in the three side slacks gives the upper bounds

$$\begin{array}{c|cccccc} i & 0 & 1 & 2 & 3 & 4 & 5 \\ \hline \mathcal{H}^1(T_C \cap r_i) & \rho - X - Y & Y/\lambda & Y & \min\{X/\lambda, Y/(1-\lambda)\} & X & X/(1-\lambda). \end{array}$$

Adding boundary and radial contributions yields

$$L_S(T_C) \leq K_\lambda + \mathcal{E}(X, Y),$$

where

$$K_\lambda = 1 - \lambda + \rho - \frac{1 - \rho}{\lambda}$$

and

$$\mathcal{E}(X, Y) = -\frac{X}{\lambda} + \frac{X}{1 - \lambda} + Y + \min\left\{\frac{X}{\lambda}, \frac{Y}{1 - \lambda}\right\} + \left[\frac{P - (\lambda X + Y)}{1 - \lambda} \right]_+.$$

Splitting according to which term realizes the minimum and whether the positive part vanishes gives, in all four cases,

$$(25) \quad \mathcal{E}(X, Y) \leq \frac{P}{1 - \lambda}.$$

For example, if $X/\lambda \leq Y/(1 - \lambda)$, the expression without the positive part is $X/(1 - \lambda) + Y$; if the positive part is active, the excess over $P/(1 - \lambda)$ is $X - \lambda Y/(1 - \lambda) \leq 0$. In the opposite order, the corresponding excess is $Y - (1 - \lambda)X/\lambda \leq 0$. If the positive part vanishes, (24) gives the same bound strictly.

It remains only to estimate a one-variable expression:

$$\frac{3}{2} - K_\lambda - \frac{P}{1 - \lambda} = \frac{1 + A - 2\rho A}{2\lambda(1 - \lambda)}, \quad A = 1 + \lambda - \lambda^2.$$

Both sides in $1 + A > 2\rho A$ are positive, and

$$(1 + A)^2 - 4A^2\rho^2 = \lambda^2(1 - \lambda)^2(5 + 4\lambda - 4\lambda^2) > 0.$$

Thus $K_\lambda + P/(1 - \lambda) < 3/2$, proving the lemma. $\square$

Lemma 4.2 (Positive-support skeleton cap). *Let T_i be the closure of an original open vertex role, so $V_i \in \text{int}(T_i)$. If T_i has positive-length intersection with an adjacent radial arm, then*

$$L_S(T_i) < \frac{3}{2}.$$

Proof. Normalize to $i = 0$ and translate the hexagon by V_0 :

$$\tilde{H} = V_0 + H, \quad \tilde{O} = V_0, \quad W_j = V_0 + V_j.$$

Then $W_3 = O$, $W_2 = V_1$, and $W_4 = V_5$. The five original skeleton components on which a unit triangle containing V_0 can have positive-length trace are

$$[V_0, V_1], [V_0, V_5], [O, V_1], [O, V_0], [O, V_5].$$

All five belong to the skeleton $\tilde{S}$ of $\tilde{H}$. Since $V_0 = \tilde{O}$ is interior to T_0 , the triangle is a legitimate center role for $\tilde{H}$. Its positive adjacent-arm trace is a positive boundary-edge trace of $\tilde{H}$, so it is CE1 or CE2 by Proposition 2.3. Proposition 2.11 and Lemma 4.1 therefore give

$$L_{\tilde{S}}(T_0) < \frac{3}{2}.$$

A nonlocal component of S is more than unit distance from the interior point V_0 , apart from zero-measure endpoints. Hence $L_S(T_0) \leq L_{\tilde{S}}(T_0)$. $\square$

Lemma 4.3 (No-support skeleton cap). *If a vertex role has $A_i + B_i \leq 1$ and no positive-length trace on either adjacent radial arm, then*

$$L_S(T_i) \leq 2.$$

Proof. Its boundary contribution is $A_i + B_i \leq 1$ by (28). For $P = sV_j \in r_j \setminus \{O\}$,

$$\|V_i - P\|^2 = 1 + s^2 - 2s \cos \frac{(j-i)\pi}{3} > 1$$

when $j \notin \{i-1, i, i+1\}$. The two adjacent arms have zero-length trace by hypothesis, so only the own unit arm can contribute positive length. This adds at most one. $\square$

For the remaining cap we use one specialized part of the exact local equilateral enclosure criterion. In local directions u, v meeting at 120° , suppose a closed unit triangle contains

$$0, \quad au, \quad bv, \quad c(u+v),$$

with $0 \leq a \leq b \leq 1$ and $a + b > 1$. The support-line calculation gives the selected supercritical cell

$$(26) \quad \begin{aligned} a^2 + ab + b^2 &\leq 1, & c &\leq \frac{1}{2}, \\ (a^2 - 1)c^2 + (2ab^2 + b)c + (b^4 - b^2) &\leq 0, \end{aligned}$$

where c lies on the connected component containing $c = 0$. To see the selector directly, the quadratic is negative at $c = 0$, positive at $c = 1/2$, and opens downward. For the middle assertion, four times its value at $1/2$ is increasing in a and, at $a = 1 - b$, equals $b^2(2b - 1)^2$; it is therefore positive when $a + b > 1$. Hence $c \leq 1/2$ selects its smaller-root component. Formula (26) follows by fixing in the support sum the three active contacts $\{B\}$, $\{B, c(u+v)\}$, and $\{A\}$, and squaring the positive unit-side equation.

Lemma 4.4 (Supercritical skeleton cap). *If $A_i + B_i > 1$, then*

$$L_S(T_i) < \frac{3}{2}.$$

Proof. By Proposition 2.5, a vertex role is Vd0, Vd1, Vd2, or T3-like. Proposition 4.6 makes every Vd1/Vd2 row nonsupercritical, and Lemma 2.8 does the same for T3-like rows. Hence a supercritical role is Vd0 and, by definition, has no positive-length trace on either adjacent arm. Thus, after normalizing and writing $a = A_i$, $b = B_i$, the skeleton trace has length $a + b + c$, where c is the own-arm reach. Reflect so that $a \leq b$, and put

$$s = a + b, \quad \delta = b - a, \quad c_0 = \frac{3}{2} - s.$$

The diameter constraint gives

$$1 < s \leq \frac{2}{\sqrt{3}}, \quad 0 \leq \delta \leq \sqrt{4 - 3s^2}.$$

Let $P(c)$ denote the quadratic in (26). Substitution and multiplication by 16 give

$$\begin{aligned} 16P(c_0) = Q(s, \delta) &= \delta^4 + 8\delta^3s - 6\delta^3 + 14\delta^2s^2 - 18\delta^2s + 5\delta^2 \\ &\quad - 8\delta s^3 + 30\delta s^2 - 34\delta s + 12\delta \\ &\quad + s^4 - 6s^3 - 19s^2 + 60s - 36. \end{aligned}$$

Its derivative factors as

$$\frac{\partial Q}{\partial \delta} = 2(2\delta + 4s - 3)(\delta^2 + \delta(4s - 3) + (s - 1)(2 - s)) \geq 0.$$

Therefore

$$Q(s, \delta) \geq Q(s, 0) = (s - 1)(s^3 - 5s^2 - 24s + 36) > 0.$$

Indeed, the cubic is decreasing on $[1, 2/\sqrt{3}]$ and at its right endpoint equals $88/3 - 136\sqrt{3}/9 > 0$. Hence $P(c_0) > 0$. Since (26) selects the smaller-root component, $P(c) \leq 0$ forces $c < c_0$. Thus $a + b + c < 3/2$. $\square$

Lemma 4.5 (A positive-support rescuer is forced). *Suppose T_C is CE1 or CE2 and, after symmetry, contains exactly M_0 . If at least two vertex rows are supercritical, then a vertex role distinct from a chosen supercritical role has positive-length adjacent-arm support.*

Proof. Choose a supercritical index $s \neq 0$. Lemma 2.10 gives $M_s \notin T_s$, and $M_s \notin T_C$ by the unique-midpoint property. In the original open cover, M_s is therefore contained by some other open vertex role. Diameter one restricts this role to U_{s-1} or U_{s+1} . Because it contains an interior point of the adjacent arm, its intersection with that arm has positive length. $\square$

4.2. Vd1/Vd2 corner estimates. We next isolate the much stronger trace loss caused by Vd1 and Vd2 geometry. For the following two results use the temporary coordinates

$$X = V_0 + x(V_5 - V_0) + y(V_1 - V_0).$$

Thus $V_0 = (0, 0)$, $V_5 = (1, 0)$, $V_1 = (0, 1)$, $O = (1, 1)$, and $\|(x, y)\|^2 = x^2 + y^2 - xy$.

Proposition 4.6 (Vd1/Vd2 corner normal form). *Let T be the closure of an original Vd1 or Vd2 role at V_0 , and let a, b be its exact reaches toward V_5, V_1 . There is a unique $t > 0$ such that, with $d = \sqrt{t^2 + t + 1}$,*

$$T = \left\{ (x, y) : \begin{aligned} x - (t+1)y &\leq a, \\ ty - (t+1)x &\leq tb, \\ tx + y &\leq d - a - tb \end{aligned} \right\}.$$

The raw traces, in coordinates measured from the indicated outer vertex, are

$$\begin{array}{l|l} r_0 & 0 \leq s \leq \frac{d - a - tb}{t + 1} \\ r_1 & \frac{t(1 - b)}{t + 1} \leq s \leq \frac{d - a - tb - 1}{t} \\ r_5 & \frac{1 - a}{t + 1} \leq s \leq d - a - tb - t. \end{array}$$

Intersecting with $[0, 1]$ gives the actual traces. Moreover,

$$(27) \quad a + b < \frac{1}{2}.$$

Proof. A Vd1/Vd2 triangle has one vertex W outside H and two vertices U, Z in H . Since V_0 is an interior convex combination of them, both coordinates of W are negative. The two axis endpoints $(a, 0), (0, b)$ lie on distinct sides incident with W . Put $p = U - W$, $q = Z - W$. Both vectors have positive coordinates and satisfy

$$\|p\| = \|q\| = \|p - q\| = 1.$$

The order in which the two sides cross the positive axes selects the 60° rotation

$$q = (p_x - p_y, p_x), \quad p_x > p_y > 0.$$

Writing $t = (p_x - p_y)/p_y$ gives uniquely

$$p = \frac{(t+1, 1)}{d}, \quad q = \frac{(t, t+1)}{d}.$$

The two side lines through the axis endpoints and the parallel third side are exactly those in (4.6). Substitution of $(s, s), (s, 1), (1, s)$ gives (4.6).

If the r_1 trace has positive length, its lower endpoint is smaller than its upper endpoint. Clearing denominators gives

$$(t+1)a + tb < (t+1)(d-1) - t^2.$$

Since the left side is at least $t(a+b)$,

$$a + b < \frac{(t+1)(d-1) - t^2}{t} < \frac{1}{2}.$$

For the last inequality, both sides are positive and

$$\left(t^2 + \frac{3t}{2} + 1\right)^2 - (t+1)^2(t^2 + t + 1) = \frac{t^2}{4} > 0.$$

Reflection proves the same conclusion when the supported arm is r_5 . $\square$

Lemma 4.7 (Vd2 neighboring-midpoint cap). *If an original Vd2 role contains either neighboring midpoint, then its exact boundary reaches satisfy*

$$a + b < \frac{1}{3}.$$

Proof. In Proposition 4.6, reflection sends (t, a, b, M_1, M_5) to $(1/t, b, a, M_5, M_1)$, so assume $t \geq 1$ and put $U = a + tb$.

If $M_5 = (1, 1/2)$ is contained, the third half-plane in (4.6) gives

$$a + b \leq U \leq d - t - \frac{1}{2} \leq \sqrt{3} - \frac{3}{2} < \frac{1}{3};$$

the middle expression decreases for $t \geq 1$.

If $M_1 = (1/2, 1)$ is contained, the second and third half-planes give

$$b \geq \frac{t-1}{2t}, \quad a + b \leq S(t) := d - t - \frac{1}{2t}.$$

Positive-length contact with the other adjacent arm r_5 gives, by (4.6),

$$a + b < R(t) := \frac{2(t+1)d - 3t^2 - t - 2}{2t}.$$

Direct differentiation shows that S is increasing and R decreasing on $[1, \infty)$. For example, the sign calculation for S' reduces to

$$(2t+1)^2 t^4 - (t^2+t+1)(2t^2-1)^2 = (t+1)(t^3+3t^2-1) > 0,$$

while $R' < 0$ follows from $d > t + 1/2$. Splitting at $t = 4/3$ yields

$$\begin{aligned} 1 \leq t \leq \frac{4}{3} : \quad a+b &\leq S(4/3) = \frac{8\sqrt{37}-41}{24} < \frac{1}{3}, \\ t \geq \frac{4}{3} : \quad a+b &< R(4/3) = \frac{7\sqrt{37}-39}{12} < \frac{1}{3}. \end{aligned}$$

The final inequalities follow respectively from $8\sqrt{37} < 49$ and $7\sqrt{37} < 43$. $\square$

4.3. Boundary trace caps. The signed center normal form gives the CE1 and CE2 center contributions from one formula; no separate maximal-interval calculation is needed here.

Theorem 4.8 (Boundary trace table). *For closures of original open roles, the following bounds hold:*

<i>role</i>	<i>hypothesis</i>	$L_{\partial H}$
T_C	CE0	0
T_C	CE1	$\leq \frac{\sqrt{3}}{2} - \frac{3}{4}$
T_C	CE2	$< \frac{1}{2}$
T_i	Vd0, $A_i + B_i \leq 1$	$\leq \frac{1}{2}$
T_i	Vd0, $A_i + B_i > 1$	$\leq \frac{2}{\sqrt{3}}$
T_i	Vd1/Vd2	$< \frac{1}{2}$
T_i	T3-like	< 1 .

Proof. For every vertex role,

$$(28) \quad L_{\partial H}(T_i) = A_i + B_i.$$

Indeed, the two incident traces are initial intervals of these lengths. A relative-interior point of any nonincident edge is more than unit distance from V_i , whereas V_i is interior to the diameter-one triangle T_i . This excludes any other positive-length boundary trace.

The CE0 row is the definition. Both remaining center rows are the two class specializations of the common signed formula in Corollary 3.3.

For a nonsupercritical Vd0 row, (28) is enough. For a supercritical row, its incident endpoints are at squared distance $A_i^2 + A_i B_i + B_i^2 \leq 1$, and hence

$$\frac{3}{4}(A_i + B_i)^2 \leq 1.$$

This proves the $2/\sqrt{3}$ cap. The Vd1/Vd2 row is (27). Finally, the Type II calculation in Proposition 2.6 gives for an original T3-like role

$$A_i + B_i = z - \alpha < z < 1$$

because $0 < t < 1$ and $\alpha > 0$. $\square$

4.4. Terminal length contradictions.

Proposition 4.9 (Boundary-length branches). *The following branches are impossible:*

- (1) *CE0 with $N_+ = 0$;*
- (2) *CE0 with $N_+ = 1$ and at least one Vd1/Vd2 role;*
- (3) *CE1 or CE2 with $N_+ = 0$ and at least one Vd1/Vd2 role;*
- (4) *CE1 with $N_+ = 1$ and at least one Vd1/Vd2 role;*
- (5) *CE2 with $N_+ = 1$ and at least two Vd1/Vd2 roles.*

Proof. In CE0, the center open role contains no relative-interior boundary point. No nonincident vertex role can contain such a point either. Hence on every edge the two incident open vertex traces form two nonempty relatively open sets covering a connected interval; they overlap in positive length. Thus

$$1 < B_i + A_{i+1}$$

on every edge. Summing gives $6 < \sum_i (A_i + B_i)$, contradicting $N_+ = 0$. This proves (1).

Items (2)–(5) are respectively items (1)–(4) of the common budget Corollary 4.17. That corollary applies the one master perimeter deficit to the center cap and the indicated strict Vd1/Vd2 caps. $\square$

Proposition 4.10 (Vd2 neighboring-midpoint branch). *Assume a CE2, $N_+ = 1$ configuration has exactly one Vd1/Vd2 role, that role is Vd2 and contains a neighboring midpoint, and all other vertex roles are Vd0. Then the boundary cannot be covered.*

Proof. This is item (5) of Corollary 4.17, using Lemma 4.7. $\square$

Proposition 4.11 (Branches closed by Strategy 1). *Direct length-sum obstructions close the following routing rows and hybrid subcase:*

- (1) *CE0, $N_+ = 0$;*
- (2) *CE0, $N_+ = 1$, with a Vd1/Vd2 role;*
- (3) *CE1/CE2 with $q = N_+ + m \geq 3$;*
- (4) *CE1/CE2, $N_+ = 0$, with a Vd1/Vd2 role;*
- (5) *CE1, $N_+ = 1$, with exactly one Vd1/Vd2 role and no T3-like role;*
- (6) *the Vd2 neighboring-midpoint subcase of the CE2 hybrid row.*

Proof. Items (1), (2), (4), and (5) are Proposition 4.9. Item (3) is the short-role count and the three-short-role theorem in Corollary 4.17. Item (6) is Proposition 4.10. This is only the indicated subcase of the exceptional CE2 row; its remaining placements are handled by the exact-demand strategy. $\square$

4.5. Common perimeter and skeleton budgets. The signed center formula also removes repeated length arithmetic. We first state a generic perimeter deficit, then count all short roles before entering the CE1/CE2 case tree.

Lemma 4.12 (Master perimeter deficit). *Put*

$$\theta = \frac{2}{\sqrt{3}} - 1.$$

Suppose a branch has a center contribution L_C , n supercritical Vd0 rows, $m \geq 1$ distinguished nonsupercritical rows with strict boundary caps $q_1, \dots, q_m < 1$, and $6 - n - m$ remaining rows whose boundary contributions are at most 1. If

$$(29) \quad L_C + n\theta \leq \sum_{j=1}^m (1 - q_j),$$

then the seven roles do not cover ∂H .

Proof. The supercritical Vd0 cap is $2/\sqrt{3}$. Hence the total boundary contribution is strictly less than

$$L_C + n\frac{2}{\sqrt{3}} + \sum_{j=1}^m q_j + (6 - n - m) = 6 + L_C + n\theta - \sum_{j=1}^m (1 - q_j) \leq 6.$$

The strictness follows from $m \geq 1$ and the strict distinguished-row caps. A cover of the perimeter would give the opposite weak inequality by subadditivity. $\square$

Lemma 4.13 (Small center slacks in the one-Vd1/Vd2 branch). *Assume there is exactly one supercritical Vd0 row, exactly one Vd1/Vd2 row, and four nonsupercritical Vd0 rows. Then every candidate surviving the perimeter budget is CE2 and satisfies*

$$(30) \quad \boxed{\alpha + \delta < \frac{1}{24}, \quad \alpha + \delta < \frac{\min\{R, W\}}{6}.}$$

Proof. The Vd1/Vd2 boundary cap is strict and equals $1/2$. If the perimeter were covered, Lemma 4.12 would force

$$L_C + \theta > \frac{1}{2}.$$

Put

$$\kappa = \frac{1}{2} - \theta = \frac{3}{2} - \frac{2}{\sqrt{3}}.$$

Thus $L_C > \kappa$. The CE1 cap in Corollary 3.3 gives

$$L_C + \theta \leq \left(\frac{\sqrt{3}}{2} - \frac{3}{4} \right) + \left(\frac{2}{\sqrt{3}} - 1 \right) < \frac{1}{2},$$

so every survivor is CE2.

Set $T = \alpha + \delta$. The CE2 formula gives

$$T < M(E) := \frac{RW}{E^2} \left(\frac{E}{1+E} - \kappa \right).$$

Since $RW = 1 - E^2$,

$$M(E) = \frac{1-E}{E^2} (E - \kappa(1+E)),$$

and

$$M'(E) = -\frac{E-2\kappa}{E^3} < 0 \quad \left(\frac{\sqrt{3}}{2} \leq E < 1 \right).$$

Here $2\kappa < \sqrt{3}/2$, because it is equivalent to $6\sqrt{3} < 11$, whose square is $108 < 121$. Therefore

$$T < M \left(\frac{\sqrt{3}}{2} \right) = \frac{8\sqrt{3}}{9} - \frac{3}{2} < \frac{1}{24}.$$

The last inequality is equivalent to $64\sqrt{3} < 111$, and follows after squaring from $12288 < 12321$.

For the second estimate, use $E/(1+E) < 1/2$:

$$T < \frac{RW}{E^2} \left(\frac{1}{2} - \kappa \right) = \frac{RW}{E^2} \theta.$$

Since

$$E^2 = W + R^2 = R + W^2,$$

one has $RW/E^2 \leq R$ and $RW/E^2 \leq W$. Also $\theta < 1/6$, equivalently $12 < 7\sqrt{3}$, whose square is $144 < 147$. This proves (30). $\square$

Lemma 4.14 (CE2 initial endpoints dominate the total slack). *Assume $\Delta_R > 0$ and $\Delta_L > 0$. Put*

$$T = \alpha + \delta, \quad x = \frac{\eta + T}{W}, \quad y = \frac{\eta + T}{R}.$$

Then

$$(31) \quad \boxed{T < \frac{1}{2} \min\{x, y\}.$$

Proof. Equation (20) gives $T < \eta/E$. Moreover

$$E^2 - (2R - 1)^2 = 3RW > 0,$$

so $|2R - 1| < E$ and therefore $|2R - 1|T < \eta$. Hence

$$x - 2T = \frac{\eta - (1 - 2R)T}{W} > 0,$$

and

$$y - 2T = \frac{\eta - (2R - 1)T}{R} > 0.$$

This is (31). $\square$

Theorem 4.15 (Three short vertex roles). *Call a vertex role short if it is supercritical or has positive-length intersection with an adjacent radial arm. If an exact- M_0 CE1/CE2 branch contains at least three short vertex roles, then the seven roles do not cover the full skeleton S .*

Proof. The center skeleton cap is

$$L_S(T_C) < \frac{3}{2}.$$

Every short role also has skeleton contribution strictly less than $3/2$: this is the supercritical cap for a supercritical role and the translated center-role cap for an original positive-support role. Every remaining role is nonsupercritical and has no adjacent-ray support, so its skeleton contribution is at most 2.

If there are $q \geq 3$ short roles, then

$$\begin{aligned} L_S(T_C) + \sum_{i=0}^5 L_S(T_i) &< \frac{3}{2} + q\frac{3}{2} + (6-q)2 \\ &= \frac{27-q}{2} \\ &\leq 12. \end{aligned}$$

The full skeleton has length 12, so subadditivity for a cover gives a contradiction. $\square$

Lemma 4.16 (Short-role count). *Let m be the number of vertex roles of type Vd1, Vd2, or T3-like, and let q be the number of short vertex roles. Then*

$$(32) \quad \boxed{q = N_+ + m.}$$

Consequently, every exact- M_0 CE1/CE2 branch with $N_+ + m \geq 3$ is impossible.

Proof. The exhaustive vertex classification says that Vd0 is exactly the class with no positive-length adjacent-arm support, while Vd1, Vd2, and T3-like are exactly the three classes with such support. Proposition 4.6 gives $A_i + B_i < 1/2$ for Vd1/Vd2, and Lemma 2.8 gives $A_i + B_i \leq 1$ for T3-like. Thus none of these m roles is supercritical. Conversely, every supercritical role is Vd0, so it has no positive adjacent-arm support. The two disjoint sources of short roles are therefore the N_+ supercritical roles and the m non-Vd0 roles. This proves (32); Theorem 4.15 gives the last assertion. $\square$

Corollary 4.17 (Length branches closed by the common budgets). *The following branches are impossible.*

- (1) CE0, $N_+ = 1$, and at least one Vd1/Vd2 row;
- (2) $N_+ = 0$ and at least one Vd1/Vd2 row, for either CE1 or CE2;
- (3) CE1, $N_+ = 1$, and at least one Vd1/Vd2 row;
- (4) CE2, $N_+ = 1$, and at least two Vd1/Vd2 rows;
- (5) CE2, $N_+ = 1$, with a Vd2 row containing a neighboring midpoint;
- (6) CE1 or CE2 with $N_+ + m \geq 3$, where m is the number of Vd1, Vd2, or T3-like roles;
- (7) CE1 or CE2 with $N_+ \geq 2$.

Proof. For item 1, take $L_C = 0$, one supercritical cap $2/\sqrt{3}$, and one strict Vd1/Vd2 cap $1/2$ in Lemma 4.12; the needed inequality is

$$\frac{2}{\sqrt{3}} - 1 < \frac{1}{2}.$$

For item 2, the center contribution is strictly less than $1/2$ and the chosen Vd1/Vd2 row has strict cap $1/2$; the other five rows contribute at most 1 each. This is Lemma 4.12 with $n = 0$ and $q_1 = 1/2$.

For item 3, the unique supercritical row is Vd0, while the CE1 center cap, the Vd1/Vd2 cap, and the four remaining unit caps give

$$\left(\frac{\sqrt{3}}{2} - \frac{3}{4}\right) + \frac{1}{2} + \frac{2}{\sqrt{3}} + 4 < 6.$$

For item 4, the corresponding CE2 sum is strictly less than

$$\frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \frac{2}{\sqrt{3}} + 3 < 6.$$

For item 5, the neighboring-midpoint Vd2 cap is $1/3$, and the total is strictly less than

$$\frac{1}{2} + \frac{1}{3} + \frac{2}{\sqrt{3}} + 4 < 6.$$

Item 6 is Lemma 4.16 and Theorem 4.15.

Finally suppose $N_+ \geq 2$. Choose a supercritical index $s \neq 0$. Its own midpoint M_s is not covered by that row, and the exact- M_0 center does not cover it. Diameter locality leaves only rows $s-1, s, s+1$ as possible vertex roles containing M_s . Thus one of the adjacent rows contains M_s and, because it is the closure of an original open role, has positive-length support on the adjacent radial arm. By Lemma 4.16, this rescuer is nonsupercritical and distinct from the two supercritical rows. Thus there are three short roles, and the theorem applies. $\square$

5. STRATEGY 2: EXACT DEMAND PROPAGATION

This section presents the reusable mechanism without repeating separate CE1 and CE2 coordinate systems. Section 3 develops the signed center model and the local propagation interface; the exact admissible-set catalogue and the remaining scalar certificates are in Appendix B.

We now build on the maps $\mathcal{A}$, B_c , F_c , and G_c defined in that section. In particular, every composition below is interpreted through the actual-row implication (15).

5.1. The universal selected- T_+ curve. The exact capped map has four genuine labels: Full, L , T_- , and selected T_+ . Its only nonlinear transition is independent of the row deficit.

Proposition 5.1 (Universal selected- T_+ curve). *Fix $0 < d < 1/2$, put $c = 1 - d$, and suppose an input p belongs to a selected T_+ arc of G_c . Write*

$$q = G_c(p), \quad \nu = q - p, \quad x = \frac{q - d}{1 - d}.$$

Then

$$(33) \quad \nu = \psi(x) := 1 - \sqrt{1 - x + x^2},$$

and

$$(34) \quad q = d + (1 - d)x, \quad p = d + (1 - d)x - \psi(x).$$

Consequently $p \mapsto G_{1-d}(p)$ is increasing and strictly concave on every selected T_+ arc.

Proof. The selected quadratic is

$$(1 - q)(q - d) = (1 - d)^2 \nu(2 - \nu).$$

Substituting $q = d + (1 - d)x$ gives

$$x(1 - x) = \nu(2 - \nu).$$

The selected component uses the smaller root, which is (33). If $r(x) = 1 - x + x^2$, then

$$\psi'(x) = \frac{1-2x}{2\sqrt{r(x)}}, \quad \psi''(x) = -\frac{3}{4r(x)^{3/2}} < 0.$$

Thus

$$p_d(x) = d + (1-d)x - \psi(x)$$

is increasing and strictly convex. Its inverse is increasing and strictly concave, while $q = d + (1-d)x$ is affine in that inverse. $\square$

With $e(d)$ defined by (16), the selected high-radial arc has endpoints (d, d) and $(e(d), d + e(d))$, so concavity gives

$$(35) \quad G_{1-d}(p) \geq p + \frac{d}{e(d) - d}(p - d).$$

In the low-radial range, if $t = h(1-d)$ is the order-transition value, the endpoints are (d, d) and $(t, 1-t)$, and

$$(36) \quad G_{1-d}(p) \geq p + \frac{1-2t}{t-d}(p - d).$$

For algebraic calculations put $z = \psi(x)/x$. Then

$$(37) \quad x = \frac{1-2z}{1-z^2}, \quad \psi(x) = \frac{z(1-2z)}{1-z^2}.$$

In the historical notation $x = 1 - \beta$ and $m_\beta = \sqrt{\beta^2 - \beta + 1}$,

$$(38) \quad \beta = \frac{z(2-z)}{1-z^2}, \quad m_\beta = \frac{1-z+z^2}{1-z^2}.$$

5.2. Threshold routing. For $0 < d < 1 - \sqrt{3}/2$, the exact high-demand threshold is

$$(39) \quad z > e(d) \implies G_{1-d}(z) \geq 1 - e(d).$$

Lemma 5.2 (Threshold routing). *Let $z_j = G_j(z_{j-1})$, where every G_j is nondecreasing and extensive. If $G_k = G_{1-d}$ and $z_{k-1} > e(d)$, then*

$$z_j \geq 1 - e(d) \quad (j \geq k).$$

If a chain contains later occurrences of G_{1-d_1} and G_{1-d_2} and

$$z_0 > e(d_1), \quad \min\{e(d_1), e(d_2)\} < Q,$$

then one of those two rows forces every subsequent output above $1 - Q$.

Proof. Equation (39) gives the first triggered output, and extensivity preserves it. For the two-threshold statement, if $e(d_1) < Q$, trigger at the first row. Otherwise

$$z_0 > e(d_1) \geq Q > e(d_2),$$

so trigger at the second row. $\square$

The lemma is used only after Equation (15) has identified each formal input as a lower bound for an actual row.

5.3. One signed five-row chain. Use the signed center data of Proposition 3.1. Put

$$X = R - \delta, \quad H = \frac{\eta + \alpha + \delta}{2R}, \quad m = \min \left\{ \frac{\alpha}{R}, \frac{\delta}{W} \right\}.$$

The five nonsupercritical demands are

$$1 - \frac{\delta}{R}, \quad 1 - \delta, \quad 1 - m, \quad 1 - \alpha, \quad 1 - \frac{\alpha}{W}.$$

A gap in the normalized right trace starts the same actual chain $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 0$ for CE1 and CE2. The common interface in Proposition 3.5 reduces both classes to

$$(40) \quad (G_{1-\delta} \circ G_{1-m} \circ G_{1-\alpha})(H) > W + \delta.$$

The only remaining distinction is the sign of $\Delta_L = P - R\alpha - \delta$.

Proposition 5.3 (Simplified CE1 one-gap obstruction). *The CE1, $N_+ = 1$, all-Vd0, one-gap state is impossible.*

Proof. Here $\Delta_L \leq 0$. The low-root estimates give $e(\alpha) < X$ and $H > \alpha$. At the row with deficit α , the Full labels are impossible. The L and T_- labels immediately force an output above $1 - X$.

On the selected T_+ label, the center inequalities force $X > 1/2$. Applying (35) first at deficit α and then the appropriate high- or low-radial chord at deficit m gives lower bounds

$$\begin{aligned} p_1 &> (2 - 4\alpha)H - (1 - 4\alpha)\alpha =: L_1, \\ p_2 &> (2 - 5m)L_1 - (1 - 5m)m =: L_2. \end{aligned}$$

The exact terminal calculation in Appendix C proves

$$(41) \quad \delta < \frac{1}{10}, \quad p_2 > L_2 > 2\delta + 3\delta^2 > e(\delta).$$

Threshold routing at the row with deficit δ therefore gives an output above $1 - X$, proving (40). The common actual-row interface then yields

$$a_0 \geq Z > B_k(k/R) \geq a_0,$$

a contradiction. All inequalities used in the selected branch, including the proof that $X > 1/2$ and the terminal estimate (41), are proved in the cited appendix. $\square$

Proposition 5.4 (Simplified CE2 one-gap obstruction). *The CE2, $N_+ = 1$, all-Vd0, exactly-one-gap state is impossible in either orientation.*

Proof. Here $\Delta_L > 0$. Proposition 3.7 proves

$$R - \delta > e(\alpha), \quad \min\{e(\alpha), e(\delta)\} < H.$$

If $e(\alpha) < H$, threshold routing triggers at the row with deficit α . Otherwise

$$R - \delta > e(\alpha) \geq H > e(\delta),$$

and it triggers at the row with deficit δ . In either case the return output exceeds $1 - H$, contradicting the common supercritical-row bound. Reflection exchanges R with W , α with δ , and reverses the row order, so the second orientation is identical. $\square$

5.4. The remaining exact-demand branches. The CE2 two-gap state uses the same endpoint deficits α/W and δ/R , but the two gaps share row T_0 ; the paired capped-loss theorem is therefore retained. The $N_+ = 0$ all-Vd0 proof uses the same one-side loss and the common three-row boundary budget. The T3-like high sheet is the universal curve (33), with the rational parameter (38); only its branch-specific endpoint estimates remain in Appendix D.

For the one-Vd1/Vd2 branch, Lemma 4.13 first gives

$$\alpha + \delta < \frac{1}{24}, \quad \alpha + \delta < \frac{\min\{R, W\}}{6}.$$

These bounds prove the adjacent and nonadjacent placements in Lemmas A.1 and A.2. The common hiding argument for a T3-like or Vd1 adjacent rescuer is Lemma A.3.

Proposition 5.5 (Branches closed by exact demand). *Strategy 2 closes all routing-table rows assigned to exact demand propagation:*

- (1) *the $N_+ = 0$ all-Vd0 and T3-like-without-Vd1/Vd2 branches for CE1 and CE2;*
- (2) *every positive-gap $N_+ = 1$ all-Vd0 state;*
- (3) *the exactly-one-T3-like $N_+ = 1$ state with no Vd1/Vd2 role;*
- (4) *the complementary exact placements in the CE2, $N_+ = 1$, exactly-one-Vd1/Vd2 branch.*

Singleton gaps are included.

Proof. The first item is Propositions D.2 and D.5. For the second, the one-gap states are Propositions 5.3 and 5.4; the two-gap state and the no-gap transfer are assembled in Proposition C.2. The third item is Proposition D.1. The fourth is Proposition A.5. In every positive-gap chain, Equation (15) supplies the passage from scalar maps to actual role reaches. $\square$

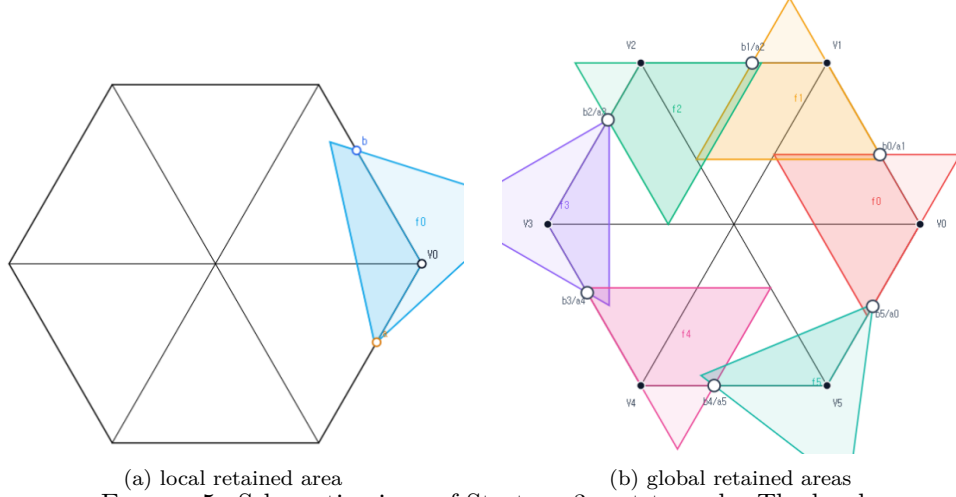
6. AREA-LOSS OBSTRUCTIONS

This section proves the local inequalities and cyclic estimates used in the area-loss argument.

Let

$$A_\Delta = \frac{\sqrt{3}}{4}$$

be the area of a unit equilateral triangle. Throughout this section areas are divided by A_Δ . Thus a unit triangle has normalized area one and H has normalized area six.



(a) local retained area (b) global retained areas
 FIGURE 5. Schematic views of Strategy 3, not to scale. The local panel shows one vertex role with boundary reaches a, b , and the global panel shows the six cyclic roles. Only the darker portions represent area retained inside H ; the proof uses the inequalities below, not the picture.

6.1. The local square-loss inequalities. Fix a vertex V_i . Let P_a and P_b be the points at distances a and b from V_i on its incoming and outgoing boundary edges. For a closed unit equilateral triangle T containing V_i, P_a, P_b , put

$$G(T) = \frac{\text{area}(T \setminus H)}{A_\Delta}.$$

Equivalently, if

$$f(a, b) = \max_T \frac{\text{area}(T \cap H)}{A_\Delta}, \quad G(a, b) = 1 - f(a, b),$$

then $G(a, b)$ is the least possible normalized loss. Reflection in the radial axis interchanges a and b , and increasing either demand can only decrease f .

Lemma 6.1 (Local wedge reduction). *After rotating indices to $i = 0$, write*

$$X = V_0 + x(V_1 - V_0) + y(V_5 - V_0), \quad W = \{(x, y) : x \geq 0, y \geq 0\}.$$

If a closed unit equilateral triangle T contains V_0 , then

$$T \cap H = T \cap W.$$

Normalized Euclidean area is twice the corresponding (x, y) -coordinate area.

Proof. The two basis vectors have length one and angle 120° , so

$$\|(x, y)\|^2 = x^2 + y^2 - xy, \quad |\det(V_1 - V_0, V_5 - V_0)| = \frac{\sqrt{3}}{2}.$$

The hexagon is

$$H = \{0 \leq x, y \leq 2, |x - y| \leq 1\} \subset W.$$

Conversely, if $(x, y) \in T \cap W$, then the diameter-one condition gives $x^2 + y^2 - xy \leq 1$. Hence

$$|x - y|^2 \leq x^2 + y^2 - xy \leq 1, \quad \frac{3}{4}x^2 \leq x^2 + y^2 - xy,$$

and the same estimate holds for y . Thus $|x - y| \leq 1$ and $0 \leq x, y \leq 2/\sqrt{3} < 2$, proving $(x, y) \in H$. The determinant identity proves the area assertion. $\square$

Theorem 6.2 (Unconditional local square loss). *For every feasible pair $0 \leq a, b \leq 1$ and every realizing closed unit equilateral triangle T ,*

$$(42) \quad G(T) \geq \min(a, b)^2.$$

If $a + b > 1$, then

$$(43) \quad G(T) \geq \max(a, b)^2.$$

Consequently the same inequalities hold for $G(a, b) = 1 - f(a, b)$.

Proof. In the coordinates of Lemma 6.1, physical rotation through 60° is $R(x, y) = (x - y, x)$. Put

$$0 \leq t \leq 1, \quad D = 1 - t + t^2, \quad z = \sqrt{D}.$$

After choosing one of the six directed sides, every orientation has one of the following two forms:

$$(44) \quad \begin{array}{c|cc} & d & Rd \\ \hline \text{I} & z^{-1}(1, t) & z^{-1}(1 - t, 1) \\ \text{II} & z^{-1}(t, -(1 - t)) & z^{-1}(1, t). \end{array}$$

Indeed these forms cover, respectively, the physical direction sectors $[120^\circ, 180^\circ]$ and $[60^\circ, 120^\circ]$; their endpoints cover the common boundary orientations.

For Type I introduce

$$U = \alpha + y - tx, \quad V = \beta + x - (1 - t)y.$$

Then T is the simplex $U, V \geq 0, U + V \leq z$. Containment of $(0, 0), (0, a), (b, 0)$ gives

$$(45) \quad \alpha \geq tb, \quad \beta \geq (1 - t)a, \quad \alpha + \beta + (1 - t)b \leq z, \quad \alpha + \beta + ta \leq z.$$

The two inequalities defining W become

$$(1 - t)U + V \geq (1 - t)\alpha + \beta, \quad U + tV \geq \alpha + t\beta.$$

For fixed t , lowering α or β therefore enlarges $T \cap W$. The outside loss is minimized at $\alpha = tb, \beta = (1 - t)a$. At that placement the portion outside W has vertices, in the (U, V) -plane,

$$(0, 0), \quad (a + tb, 0), \quad (tb, (1 - t)a), \quad (0, b + (1 - t)a).$$

Since the Jacobian of $(x, y) \mapsto (U, V)$ has absolute value D ,

$$(46) \quad G(T) \geq G_I(a, b; t) = \frac{(1 - t)a^2 + tb^2 + 2t(1 - t)ab}{D}.$$

If $a \leq b$, then

$$D(G_I - a^2) = t\{b^2 - a^2 + (1 - t)(a^2 + 2ab)\} \geq 0;$$

the reflected factorization proves $G_I \geq b^2$ when $b \leq a$. This proves (42) in Type I.

Suppose now that $a + b > 1$ and $a \leq b$. Set

$$r = \frac{a}{b}, \quad t_0 = \frac{1 - r^2}{1 + 2r}.$$

Since $b > 1/(1 + r)$, feasibility in (46) and (45) excludes $0 \leq t < t_0$: for $0 < t < t_0$,

$$\left(\frac{1 + r(1 - t)}{1 + r} \right)^2 - D = \frac{t(1 + 2r)(t_0 - t)}{(1 + r)^2} > 0,$$

so $b + (1 - t)a > z$, while $t = 0$ would give $a + b \leq 1$. Hence $t \geq t_0$, and

$$D(G_I - b^2) = (1 - t)b^2\{r^2 - 1 + t(2r + 1)\} \geq 0.$$

If $b \leq a$, put $r = b/a$ and $t_1 = (r^2 + 2r)/(1 + 2r)$. The identical reflected calculation excludes $t > t_1$ and gives

$$D(G_I - a^2) = ta^2\{r^2 + 2r - t(1 + 2r)\} \geq 0.$$

Thus (43) holds in Type I.

For Type II put

$$U = \alpha + (1 - t)x + ty, \quad V = \beta + tx - y.$$

Again T is the simplex $U, V \geq 0, U + V \leq z$. Its exact reaches on the positive y - and x -axes are

$$(47) \quad A = \beta, \quad B = z - \alpha - \beta, \quad A + B \leq z \leq 1.$$

In particular Type II is impossible when $a + b > 1$. Its wedge intersection is the quadrilateral with vertices

$$(0, 0), \quad (B, 0), \quad \left(\frac{B + (1 - t)A}{D}, \frac{A + tB}{D} \right), \quad (0, A).$$

Hence

$$(48) \quad G(T) = G_{II}(A, B; t) = 1 - \frac{(1 - t)A^2 + tB^2 + 2AB}{D}.$$

Let $S = A + B$. If $A \leq B$, write $A = rS$, $B = (1 - r)S$ with $0 \leq r \leq 1/2$. Since $S^2 \leq D$, and with

$$C = (1 - t)r^2 + t(1 - r)^2 + 2r(1 - r),$$

we have $G_{II} \geq 1 - C$. The exact factorization

$$1 - C - r^2D = (1 - t)(1 - 2r + tr^2) \geq 0$$

gives $G_{II} \geq r^2D \geq A^2$. Reflection gives $G_{II} \geq B^2$ when $B \leq A$. Since $A \geq a$ and $B \geq b$, this proves (42). The two orientation types are exhaustive, so the theorem follows. $\square$

6.2. The direct T3-like loss. The next theorem uses the classification data (o, n) from Section 2: o counts triangle vertices outside H , and n counts adjacent radial arms met in positive length. Thus a T3-like role has $(o, n) = (2, 1)$.

Theorem 6.3 (Direct T3-like loss). *Let T be a closed unit V_i -triangle of T3-like type, and let a, b be lower bounds for its adjacent boundary reaches. Then*

$$(49) \quad a + b \leq 1.$$

Moreover, for $0 \leq m \leq 1/2$,

$$(50) \quad a, b \geq m \implies G(T) \geq 2m - 4m^2.$$

Proof. The first assertion is Lemma 2.8. For the loss bound apply the closed-trace domination in Proposition 2.6. It gives a translated Type-II triangle $\hat{T}$ of the same area with

$$T \cap H \subset \hat{T} \cap H, \quad G(T) \geq G(\hat{T}).$$

In the notation of that proposition, $0 < t < 1$, $D = 1 - t + t^2$, $z = \sqrt{D}$, and after reflection the translated majorant has exact reaches

$$A = \beta, \quad B = z - \beta$$

and unique wedge vertex

$$Q = \left(\frac{B + (1-t)A}{D}, \frac{A + tB}{D} \right), \quad Q_x > 1, \quad Q_y \leq 1.$$

Thus $z - tA > D$, so

$$(51) \quad A < L := \frac{z - D}{t} = \frac{z(1-z)}{t}.$$

The exact normalized loss is

$$G(\hat{T}) = G(z, A) = \frac{tA^2 + (1-t)(z-A)^2}{D}.$$

This decreases for $A < L < (1-t)z$, and therefore

$$(52) \quad G(T) \geq G(\hat{T}) > G(z, L).$$

Set $r = (1-z)/t$. Then $0 < r < 1/2$, and eliminating t, z gives

$$t = \frac{1-2r}{1-r^2}, \quad z = \frac{r^2-r+1}{1-r^2},$$

$$L = A_0(r) := \frac{r(r^2-r+1)}{1-r^2}, \quad z-L = B_0(r) := \frac{r^2-r+1}{1+r}.$$

The limiting normalized inside area and loss are

$$F_0(r) = \frac{(r^2-r+1)^2}{1-r^2}, \quad G_0(r) = 1 - F_0(r).$$

Since the exact reach A is at least the required demand $a \geq m$, (51) gives $A_0(r) > m$. Direct simplification yields

$$G_0(r) - \{2A_0(r) - 4A_0(r)^2\} = \frac{r^2(5r^4 - 8r^3 + 13r^2 - 8r + 2)}{(1-r^2)^2} \geq 0,$$

because the last polynomial is at least $9r^2 - 8r + 2 \geq 2/9$ on $[0, 1/2]$. Also $A_0(r) \leq r$, and

$$G_0(r) - \frac{1}{4} = \frac{(1-2r)(2r^3 - 3r^2 + 6r - 1)}{4(1-r^2)} \geq 0 \quad \text{when } A_0(r) \geq \frac{1}{4};$$

the cubic is increasing and has value $11/32$ at $r = 1/4$.

Finally $\psi(u) = 2u - 4u^2$ is increasing on $[0, 1/4]$ and never exceeds $1/4$ on $[0, 1/2]$. If $A_0(r) \leq 1/4$, then $G_0(r) \geq \psi(A_0(r)) \geq \psi(m)$; if $A_0(r) \geq 1/4$, then $G_0(r) \geq 1/4 \geq \psi(m)$. Together with (52), this proves (50). The reflected crossing is identical. $\square$

6.3. The cyclic area certificates. The selected handoffs of Proposition 2.9 have the form

$$(a_i, b_i) = (1 - x_{i-1}, x_i), \quad 0 < x_i < 1,$$

with cyclic indices. Row i is selected supercritical exactly when $x_i > x_{i-1}$.

Lemma 6.4 (Cyclic area-loss certificate). *Let the six actual vertex triangles realize selected handoffs*

$$(a_i, b_i) = (1 - x_{i-1}, x_i), \quad 0 < x_i < 1.$$

Their total normalized loss is strictly greater than one if either

- (1) *at least two selected rows are supercritical; or*
- (2) *exactly one selected row is supercritical, every role is Vd0 or T3-like, and at least one role is T3-like.*

Proof. Put $m = \min_i x_i$ and $M = \max_i x_i$. The reflection $y_i = 1 - x_{i-1}$ swaps each row's two coordinates and preserves both alternatives and the total loss. Hence assume $m \leq 1 - M$. Every selected row coordinate is then at least m , so (42) gives loss at least m^2 for every row.

In case (1), choose an ascent out of a minimum plateau, $x_{p-1} = m < x_p$. Its row contains the coordinate $1 - m$ and loses at least $(1 - m)^2$ by (43). A second ascent has one coordinate strictly larger than $1/2$ and hence loses strictly more than $1/4$. The other four rows lose at least m^2 each, so the total loss is strictly larger than

$$4m^2 + (1 - m)^2 + \frac{1}{4} = 5 \left(m - \frac{1}{5} \right)^2 + \frac{21}{20} > 1.$$

In case (2), rotate so that the unique ascent leaves the minimum: $x_5 = m < x_0 = M$. The supercritical row loses at least $(1 - m)^2$. Lemma 2.8 shows that a T3-like row is different from it, and Theorem 6.3 gives that row loss at least $2m - 4m^2$. The remaining four rows lose at least m^2 each. Their total loss is at least

$$(1 - m)^2 + (2m - 4m^2) + 4m^2 = 1 + m^2 > 1.$$

□

Proposition 6.5 (Branches closed by area). *The following branches are impossible:*

- (1) T_C is CE0, $N_+ \geq 2$, with arbitrary vertex-role types;
- (2) T_C is CE0, $N_+ = 1$, no role is Vd1 or Vd2, and at least one role is T3-like.

Proof. In the CE0 class, the center role has no positive-length boundary trace. If the six vertex roles missed a boundary point, openness of the center role would give such a trace. Hence the vertex roles cover ∂H , so Proposition 2.9 applies.

If $N_+ \geq 2$, its at-least-two clause selects handoffs with at least two supercritical rows. Lemma 6.4 makes the six vertex triangles' total normalized inside area strictly less than five. The center triangle contributes at most one, so the total is strictly less than six.

If $N_+ = 1$, the exact-one clause preserves the unique actual supercritical index for every strict selection. The exhaustive vertex-type hypothesis gives case (2) of Lemma 6.4; the six vertex triangles contribute less than five normalized area, and the center triangle contributes at most one. Again the sum is strictly below the normalized area six of H . □

7. STRATEGY 4: DIRECT NINE-POINT FORCING

This section gives the geometric proof of the direct nine-point obstruction. The exact frontier calculations, moving-circle signs, adjacent-overlap inequalities, rational radial envelopes, and global Bernstein identities are proved in Appendix E. The body of the proof uses those verified inputs only through the forcing lemmas and the four-overlap proposition below.

Assume throughout that all six vertex roles are Vd0, that they cover ∂H , and that exactly one actual boundary row is supercritical. Rotate its index to 4. Lemma E.3 gives strict handoffs

$$(53) \quad x_3 < x_2 < x_1 < x_0 < x_5 < x_4,$$

and parameters

$$(54) \quad a = 1 - x_3, \quad b = x_4, \quad p = 1 - b, \quad q = 1 - a$$

with

$$(55) \quad 0 < a, b < 1, \quad a + b > 1, \quad a^2 + ab + b^2 < 1.$$

Every selected row has incoming demand at least p and outgoing demand at least q .

7.1. Direct forcing of the disk and three frontier points. For $(x, y) \in [0, 1]^2$, write

$$c_{\max}(x, y) = \max\{c : (x, y, c) \in \mathcal{A}\}.$$

Let

$$c_* = c_{\max}(p, q), \quad h = \frac{\sqrt{3}}{2}, \quad \eta = h(1 - c_*).$$

The six radial points

$$D_i = (1 - c_*)V_i, \quad 0 \leq i \leq 5,$$

are excluded from their matching roles by maximality, from the two adjacent roles by Vd0 locality, and from the other three roles by distance from their distinguished vertices. Lemma E.4 therefore gives

$$(56) \quad \mathcal{D}_\eta := \{X : \|X\| \leq \eta\} \subset U_C.$$

At the supercritical corner V_4 , the strict AB -frontier consists of two circle pieces and two line pieces. The two line pieces meet at Q_0 ; their first intersections with the two actual adjacent-handoff unit circles are Q_- and Q_+ . The exact moving-circle signs in Lemma E.5 exclude all three points from the two adjacent roles. Direct vertex-distance inequalities exclude the remaining three nondistinguished roles, and exact frontier membership excludes the supercritical role. Lemma E.6 therefore gives

$$(57) \quad Q_-, Q_0, Q_+ \in U_C.$$

Consequently the center role contains the compact witness set

$$(58) \quad K_{\text{wit}} = \mathcal{D}_\eta \cup \{Q_-, Q_0, Q_+\}.$$

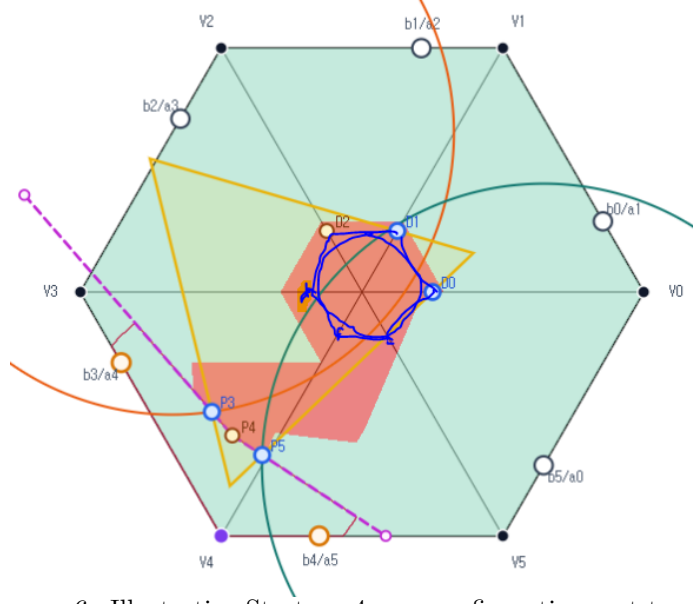


FIGURE 6. Illustrative Strategy 4 core configuration, not to scale. It is an example only; no exclusion, forcing, or enclosure inequality is inferred from the picture.

7.2. Newton inner points. The exact outer points Q_- and Q_+ contain two independent quadratic radicals. The exact construction in Lemma E.7 takes one Newton step inward on each frontier line. It gives points

$$A \in (Q_0, Q_-), \quad B = Q_0, \quad C \in (Q_0, Q_+)$$

whose coordinates are rational in a, b and the single radical

$$D = \sqrt{4(a^2 + ab + b^2) - 3}.$$

The same lemma gives

$$(59) \quad \widehat{K} := \text{conv}(\mathcal{D}_\eta \cup \{A, B, C\}) \subseteq \text{conv}(K_{\text{wit}}), \quad \Lambda(K_{\text{wit}}) \geq \Lambda(\widehat{K}).$$

7.3. The cap-chain lemma. For a compact set K , let

$$h_K(n) = \max_{X \in K} \langle X, n \rangle,$$

and let R denote rotation through $2\pi/3$. The least equilateral side containing K is

$$(60) \quad \Lambda(K) = \frac{1}{h} \min_{\|n\|=1} \sum_{j=0}^2 h_K(R^j n).$$

For a disk $\mathbb{D}(X, r)$ define its level- h cap

$$I(X, r) = \{n \in S^1 : \langle X, n \rangle + r \geq h\}.$$

Lemma 7.1 (Cyclic cap chain). *Let K be compact and put*

$$S = K + RK + R^2K.$$

Suppose S contains the full R -orbits of

$$\mathbb{D}(A, 2\eta), \quad \mathbb{D}(B, 2\eta), \quad \mathbb{D}(C, 2\eta), \quad \mathbb{D}(W, \eta), \quad W = C + RA.$$

Assume that the rays

$$A, B, C, W, RA$$

occur in this cyclic order, each corresponding cap is a connected arc of half-width less than $\pi/2$, and the four consecutive pairs

$$\begin{aligned} I(A, 2\eta) &\leftrightarrow I(B, 2\eta), & I(B, 2\eta) &\leftrightarrow I(C, 2\eta), \\ I(C, 2\eta) &\leftrightarrow I(W, \eta), & I(W, \eta) &\leftrightarrow I(RA, 2\eta) \end{aligned}$$

overlap across their intervening short sectors. Then $\Lambda(K) \geq 1$.

Proof. The four overlaps and the cyclic ray order make the five caps cover the sector from the ray of A to the ray of RA . Their R -orbits cover the unit circle. For every unit normal n , one of the displayed disks therefore has support at least h in direction n . Since all those disks lie in S ,

$$h_S(n) \geq h.$$

On the other hand,

$$h_S(n) = \sum_{j=0}^2 h_K(R^j n).$$

Equation (60) now gives $\Lambda(K) \geq 1$. □

7.4. The four verified overlaps. For $K = \widehat{K}$, the Minkowski sum in the cap-chain lemma contains the four required disk orbits: one witness point and two copies of the centered disk give the three radius- 2η orbits, while $C + RA$ and one disk copy give the radius- η orbit.

Proposition 7.2 (Four-overlap certificate). *Assume (55) and $c_* > 2/3$. The Newton points above satisfy*

$$\|A\|, \|C\| > \eta,$$

the rays A, B, C, W, RA occur in that cyclic order, and all four consecutive cap overlaps in Lemma 7.1 hold.

Proof. This is Proposition E.10. Its proof gives the ray order and the two adjacent overlaps analytically, then invokes the exact Gram reductions and twenty global Bernstein identities of Subsection E.6 for the two mixed overlaps. □

7.5. Enclosure and the center-independent contradiction.

Theorem 7.3 (Reader-facing witness enclosure). *For every pair (a, b) satisfying (55),*

$$\Lambda(K_{\text{wit}}(a, b)) \geq 1.$$

Proof. The disk alone gives

$$\Lambda(K_{\text{wit}}) \geq \frac{3\eta}{h} = 3(1 - c_*),$$

so the result is immediate when $c_* \leq 2/3$. Assume $c_* > 2/3$. The Newton reduction (59), Proposition 7.2, and Lemma 7.1 give

$$\Lambda(K_{\text{wit}}) \geq \Lambda(\widehat{K}) \geq 1.$$

□

Theorem 7.4 (Center-independent direct obstruction). *There is no cover of H by open unit equilateral triangles $U_C, U_0, \dots, U_5$ such that every U_i is a Vd0 role, the six vertex roles cover ∂H , and exactly one actual boundary row is supercritical.*

Proof. Direct forcing gives the compact containment $K_{\text{wit}} \subset U_C$, whereas Theorem 7.3 gives $\Lambda(K_{\text{wit}}) \geq 1$. A compact subset of an open unit equilateral triangle has positive clearance from all three sides and is therefore contained in a closed equilateral triangle of side strictly below one. This would give $\Lambda(K_{\text{wit}}) < 1$, a contradiction. $\square$

Proposition 7.5 (Branches closed by Strategy 4). *Theorem 7.4 closes*

- (1) *the CE0, $N_+ = 1$, all-Vd0 branch;*
- (2) *the zero-gap CE1 and CE2, $N_+ = 1$, all-Vd0 branches.*

Proof. In the CE0 branch, a boundary point missed by the six vertex roles would belong to the open center role and create a positive center trace, contrary to CE0. Thus the vertex roles cover ∂H . In the CE1 and CE2 branches, zero gaps mean exactly that the adjacent vertex traces cover every boundary edge. The exact-one hypothesis supplies the unique supercritical row in all three cases, so Theorem 7.4 applies. $\square$

8. EXHAUSTIVE ASSEMBLY

Proof of Theorem 1.1. Assume that seven open unit equilateral triangles cover H . Proposition 2.13 assigns the cover to exactly one row of Table 1, with N_+ computed from the actual reaches. The rows assigned to Strategies 1, 2, 3, and 4 contradict, respectively, Propositions 4.11, 5.5, 6.5, and 7.5. In the sole row marked 1 + 2, the first two propositions cover the complementary subcases stated in their conclusions. Thus every row is impossible. $\square$

Corollary 1.2 follows from Proposition 2.1.

APPENDIX A. SHORT EXACT VD1/VD2 PLACEMENTS

We now prove the two local placements that previously required separate CE2 outer-ratio calculations. The common small-slack lemma reduces both to one-line comparisons with the center exits.

Lemma A.1 (Adjacent Vd1/Vd2 placement). *Assume a reduced CE2 exact- M_0 candidate has $N_+ = 1$, row T_0 uniquely supercritical, row T_1 the unique Vd1/Vd2 row, and rows T_2, T_3, T_4, T_5 nonsupercritical Vd0. Then the perimeter together with r_1, r_2 is not covered. The reflected T_5 placement is also impossible.*

Proof. Use the signed variables of Proposition 3.1. Write

$$x = \frac{k}{W}, \quad u = R + \alpha, \quad y = \frac{k}{R}, \quad v = W + \delta.$$

The center exit on r_2 is

$$(61) \quad d_2^C = \delta.$$

Let (a_0, b_0) be the boundary reaches of the supercritical row. Then

$$(62) \quad a_0 + b_0 > 1, \quad a_0^2 + a_0 b_0 + b_0^2 \leq 1.$$

Coverage on $e_{0,1}$ forces the incoming reach of T_1 to be at least

$$A = \begin{cases} 1 - b_0, & b_0 < y, \\ \max\{0, 1 - \max\{b_0, v\}\}, & b_0 \geq y. \end{cases}$$

Coverage on $e_{5,0}$ forces the far endpoint row to reach at least

$$H = \begin{cases} 1 - a_0, & a_0 < x, \\ \max\{0, 1 - \max\{a_0, u\}\}, & a_0 \geq x. \end{cases}$$

A genuine candidate has $H > 0$. Propagating this lower bound through the ordinary nonsupercritical rows gives

$$b_5 \geq H \implies b_4 \geq H \implies b_3 \geq H \implies b_2 \geq H \implies b_1 \geq H.$$

Thus

$$a_1 \geq A, \quad b_1 \geq H.$$

The Vd1/Vd2 corner cap gives

$$(63) \quad \boxed{A + H < \frac{1}{2}}.$$

Lemma 4.13 applies and gives

$$(64) \quad \alpha + \delta < \frac{1}{24}, \quad \alpha + \delta < \frac{\min\{R, W\}}{6}.$$

We claim

$$(65) \quad \boxed{\delta < \frac{H}{3}}.$$

There are two possible positive sources of H .

If $H = 1 - u = W - \alpha$, then

$$3\delta + \alpha \leq 3(\alpha + \delta) < \frac{W}{2} < W,$$

so $3\delta < W - \alpha = H$.

Now assume $H = 1 - a_0$. The other demand A cannot equal $1 - b_0$; otherwise (63) would imply $a_0 + b_0 > 3/2$, whereas (62) gives $a_0 + b_0 \leq 2/\sqrt{3} < 3/2$. Since $b_0 < 1$ and $v < 1$, the maximum defining A is positive. Hence

$$A = 1 - v, \quad v > \frac{1}{2} + H.$$

This alternative requires $b_0 \geq y$. Since $a_0 = 1 - H$ and $b_0 \geq y$, (62) gives

$$(1 - H)^2 + (1 - H)y + y^2 \leq 1.$$

Lemma 3.4 therefore gives

$$(66) \quad H \geq \lambda_\circ(y) > \frac{y}{2} > \frac{\eta}{2R}.$$

Suppose $R \geq 3/8$. From $v = W + \delta > 1/2 + H$ and (66),

$$\delta > J(R) := R - \frac{1}{2} + \frac{\eta}{2R}.$$

We show $J(R) \geq 1/24$. If $3/8 \leq R \leq 13/24$, then $RW \geq 15/64$, so $E \leq 7/8$ and

$$\frac{\eta}{2R} = \frac{W}{2(1+E)} \geq \frac{4W}{15}.$$

Thus

$$J(R) \geq \frac{11R}{15} - \frac{7}{30} \geq \frac{1}{24}.$$

If $R \geq 13/24$, then already $R - 1/2 \geq 1/24$. Both alternatives contradict $\delta < 1/24$, so

$$R < \frac{3}{8}.$$

Now

$$(2 - 3R)^2 - E^2 = (3 - 8R)(1 - R) > 0,$$

so $E < 2 - 3R$ and

$$\frac{\eta}{R} = \frac{W}{1 + E} > \frac{1}{3}.$$

Therefore $y > 1/3$ and (66) gives $H > 1/6$. Together with $\delta < 1/24$, this yields

$$\delta < \frac{1}{24} < \frac{1}{18} < \frac{H}{3}.$$

This proves (65).

It remains to cover r_2 . Orient the Vd1/Vd2 corner normal form so that r_2 is its supported adjacent arm. For some $t > 0$ and $d = \sqrt{t^2 + t + 1}$, its upper endpoint is

$$u_2 = \frac{d - a_1 - tb_1 - 1}{t}.$$

If it bridged to the center interval, then $u_2 \geq 1 - \delta$, and hence

$$a_1 + tb_1 \leq d - 1 - t + t\delta < t\delta,$$

because $d < t + 1$. Since $b_1 \geq H$, this gives $H < \delta$, contrary to (65).

Without such a bridge, row T_2 would have to reach the center interval. The boundary handoff gives $a_2 \geq 1 - b_1$. Since $a_1 + b_1 < 1/2$ and $a_1 \geq A$,

$$a_2 > \frac{1}{2} + A.$$

Backward propagation gives $b_2 \geq H$. Put $p = 1/2 + A$. Then (63) gives

$$p \geq \frac{1}{2}, \quad 0 \leq H \leq p, \quad p + H < 1.$$

Coordinatewise down-closedness and the half-edge radial envelope give

$$c_2 \leq c_{\max}(p, H) \leq 1 - \frac{H}{3} < 1 - \delta.$$

Thus T_2 also fails to meet the center interval, and r_2 is uncovered. Reflection exchanges T_1 and T_5 , proving the reflected placement. $\square$

Lemma A.2 (Nonadjacent Vd1/Vd2 placements). *Assume the same reduced CE2 branch, but let the unique Vd1/Vd2 row be T_τ with $\tau \in \{2, 3, 4\}$. Then the perimeter together with r_τ is not covered.*

Proof. Put

$$T = \alpha + \delta, \quad x = \frac{\eta + T}{W}, \quad y = \frac{\eta + T}{R}.$$

Lemma 4.14 gives

$$(67) \quad T < \frac{1}{2} \min\{x, y\}.$$

Let B, U be the farthest already-covered extents from V_0 on the two active boundary edges, and put

$$A = 1 - B, \quad H = 1 - U.$$

Explicitly,

$$B = \begin{cases} b_0, & b_0 < y, \\ \max\{b_0, v\}, & b_0 \geq y, \end{cases} \quad U = \begin{cases} a_0, & a_0 < x, \\ \max\{a_0, u\}, & a_0 \geq x. \end{cases}$$

Boundary propagation gives

$$a_\tau \geq A, \quad b_\tau \geq H,$$

and the Vd1/Vd2 cap gives $A + H < 1/2$.

We prove

$$(68) \quad B \leq \beta(x), \quad U \leq \beta(y).$$

It suffices to prove the first inequality. If $B = v$, the center contains the points with parameters x and v on the two adjacent edges, so $x^2 + xv + v^2 \leq 1$ and $v \leq \beta(x)$. If $B = b_0$ and $a_0 < x$, then $U = a_0$ and $A + H < 1/2$ would imply $a_0 + b_0 > 3/2$, contrary to the disk inequality. Thus $a_0 \geq x$. The same disk inequality gives $b_0 \leq \beta(a_0) \leq \beta(x)$ because β is decreasing. Reflection proves the second inequality in (68).

Lemma 3.4 and (67) now give

$$A = 1 - B > \frac{x}{2} > T, \quad H = 1 - U > \frac{y}{2} > T.$$

Hence

$$(69) \quad \boxed{T < \min\{A, H\}}.$$

The center exits are

$$d_2^C = \delta < T, \quad d_4^C = \alpha < T,$$

and

$$d_3^C = \min\left\{\frac{\alpha}{R}, \frac{\delta}{W}\right\} \leq R\frac{\alpha}{R} + W\frac{\delta}{W} = T.$$

Thus

$$(70) \quad d_\tau^C < \min\{A, H\}, \quad \tau = 2, 3, 4.$$

For the Vd1/Vd2 row, the corner normal form gives

$$c_\tau = \frac{d - a_\tau - tb_\tau}{t + 1}, \quad d = \sqrt{t^2 + t + 1} < t + 1.$$

Therefore

$$c_\tau < 1 - \frac{a_\tau + tb_\tau}{t + 1} \leq 1 - \min\{A, H\}.$$

But the vertex-side entry point of the center interval is $1 - d_\tau^C > 1 - \min\{A, H\}$. Hence the own-radial trace of T_τ does not meet the center trace. Diameter locality and the Vd0 hypotheses exclude any other positive interval on r_τ , so that arm is uncovered. $\square$

A.1. A common adjacent-rescuer theorem. For $0 \leq c \leq 1/2$, put

$$B_{\text{sc}}(c) = \frac{c + \sqrt{c^2 - 8c + 4}}{2}, \quad A_{\text{sc}}(c) = 1 - B_{\text{sc}}(c).$$

The free strict-supercritical envelope says that a strict supercritical row with radial demand at least c has outgoing reach strictly below $B_{\text{sc}}(c)$, and B_{sc} is strictly decreasing.

Lemma A.3 (Common adjacent-rescuer obstruction). *Let a local rescuer T_0 have trace $[0, a]$ on $e_{5,0}$ and adjacent-radial trace $[c, u]$ on r_1 , measured from V_1 toward O . Put*

$$\varepsilon = 1 - u > 0, \quad \Theta = \frac{a}{a + \varepsilon}.$$

Assume

$$(71) \quad a \leq A_{\text{sc}}(c), \quad \Theta \leq A_{\text{sc}}(c),$$

and assume radial isolation forces the center to cover the O -side gap of length ε . If T_1 is the unique strict supercritical row and T_2, T_3, T_4, T_5 are nonsupercritical, then the perimeter and r_1 are not covered.

Proof. The center exit on r_1 is δ/R , so radial isolation gives

$$(72) \quad \delta \geq R\varepsilon.$$

Let h be the reach forced on T_5 along $e_{5,0}$. If the center has no positive companion trace, or if its companion trace does not hide the gap after coordinate a , then

$$h \geq 1 - a \geq B_{\text{sc}}(c).$$

It remains to consider the hiding order

$$\frac{k}{W} \leq a < R + \alpha.$$

The left endpoint inequality and (72) give

$$k = \eta + \alpha + \delta \leq Wa, \quad R\varepsilon \leq Wa.$$

Hence

$$R(a + \varepsilon) \leq a, \quad R \leq \Theta.$$

Also

$$\alpha \leq Wa - R\varepsilon - \eta,$$

so the far center endpoint satisfies

$$R + \alpha \leq R + Wa - R\varepsilon = a + R(u - a).$$

Because the hiding order has $R + \alpha > a$, this forces $u > a$. Therefore

$$R + \alpha \leq a + \Theta(u - a) = \frac{a}{a + \varepsilon} = \Theta \leq A_{\text{sc}}(c).$$

Thus in every ordering

$$(73) \quad h \geq B_{\text{sc}}(c).$$

Coverage of r_1 gives the supercritical row a radial demand at least c . The strict envelope and its monotonicity give

$$b_1 < B_{\text{sc}}(c).$$

Together with (73), $h > b_1$. The four remaining boundary obligations give

$$\begin{aligned} \sum_{i=2}^5 (a_i + b_i) &\geq (1 - b_1) + 1 + 1 + 1 + h \\ &= 4 + (h - b_1) > 4, \end{aligned}$$

contrary to the four nonsupercritical row caps. $\square$

Corollary A.4 (T3-like and Vd1 adjacent rescue). *The common adjacent-rescuer lemma applies both to the exactly-one-T3-like branch and to the normalized Vd1 row containing the neighboring midpoint.*

Proof. For the T3-like normal form, the local proof in Equation (85) gives

$$x_0 = \frac{aD}{R_{\text{loc}}} \leq A_{\text{sc}}(c),$$

while $a \leq x_0$. Its endpoint is

$$u = 1 - \frac{R_{\text{loc}}}{D} + a,$$

so

$$\Theta = \frac{a}{a + 1 - u} = \frac{aD}{R_{\text{loc}}} = x_0.$$

Thus (71) holds.

For the Vd1 normal form, the complete local calculation in Lemma D.11 gives

$$a \leq A_{\text{sc}}(c), \quad \frac{a}{a + 1 - u} \leq A_{\text{sc}}(c).$$

In both branches, the support-isolation and midpoint arguments force the center to cover the O -side gap before the rescuer begins. The common lemma therefore supplies the same final boundary contradiction. $\square$

Proposition A.5 (Short one-Vd1/Vd2 placement assembly). *In the CE2, $N_+ = 1$, exactly-one-Vd1/Vd2 branch with no additional positive-support row, every placement is impossible directly or reduces to the all-Vd0 boundary-loss theorem.*

Proof. If the supercritical row is T_0 , an adjacent Vd1/Vd2 row is eliminated by Lemma A.1, while a nonadjacent one is eliminated by Lemma A.2. If the unique Vd1/Vd2 row is T_0 , midpoint forcing makes the supercritical row adjacent. The Vd2 alternative is excluded by the neighboring-midpoint boundary cap; the Vd1 alternative is Corollary A.4. If neither special row is T_0 , midpoint forcing again makes them adjacent. A Vd2 neighbor rescue is impossible, while a Vd1-supercritical pair is replaced by two nonsupercritical Vd0 rows as in Lemma D.12; the resulting configuration is contradicted by Proposition D.2. The positive-support complement is already excluded by Theorem 4.15. $\square$

APPENDIX B. EXACT LOCAL DEMAND CALCULUS

This section develops the local calculus for configurations in which a coarse length sum does not by itself record enough information. The mechanism is always the same. A center trace fixes boundary inputs and complementary radial demands; an exact local map propagates these data through nonsupercritical rows; and the returning demand exceeds the remaining boundary or radial capacity.

Actual maximal reaches continue to be denoted by (A_i, B_i, C_i) , while lowercase letters denote prescribed demands. Every estimate in this section is proved analytically. In particular, no empirical branch catalogue, machine-assisted certificate, or formal root from a discarded algebraic component is used.

B.1. The exact local feasible set. Put

$$u = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right), \quad v = \left(\frac{1}{2}, -\frac{\sqrt{3}}{2}\right).$$

For $0 \leq a, b, c \leq 1$ define

$$K(a, b, c) = \text{conv}\{0, au, bv, c(u+v)\}.$$

Thus a, b are lower demands on the two incident boundary rays and c is a lower demand on the inward radial ray.

Definition B.1. The local admissible set is

$$\mathcal{A} = \{(a, b, c) \in [0, 1]^3 : K(a, b, c) \text{ is contained in a closed unit equilateral triangle}\}.$$

For $a, c \in [0, 1]$ put

$$\begin{aligned} B_c(a) &= \max\{b : (a, b, c) \in \mathcal{A}\}, \\ F_c(a) &= \min\{B_c(a), 1 - a\}, \\ G_c(a) &= 1 - F_c(a). \end{aligned}$$

The distinction between demands and actual reaches is important. If a role has actual reaches (A, B, C) and $A \geq a, C \geq c$, then (a, B, c) is a demand triple realized by the same role. If in addition $A + B \leq 1$, then

$$(74) \quad B \leq F_c(a).$$

No assertion about the type of a maximizing triangle is required.

Proposition B.2 (Exact admissible set and radial envelope). *Put*

$$s = a + b, \quad d = a^2 + ab + b^2, \quad m = \min\{a, b\}, \quad M = \max\{a, b\},$$

and

$$q = s^4 - s^2 + ab.$$

Define

$$\begin{aligned} F_L(a, b, c) &= c^4 - c^2 + mc - m^2, \\ F_T(a, b, c) &= (s^2 - 1)c^2 + Mc - M^2, \\ F_S(a, b, c) &= (m^2 - 1)c^2 + (2mM^2 + M)c + M^4 - M^2. \end{aligned}$$

Then $\mathcal{A} = \mathcal{A}_L \cup \mathcal{A}_T \cup \mathcal{A}_S$, where

$$\begin{aligned} \mathcal{A}_L &= \{d \leq 1, s \leq 1, q \leq 0, F_L \leq 0\}, \\ \mathcal{A}_T &= \{d \leq 1, s \leq 1, q \geq 0, F_T \leq 0, c \leq 2M\}, \\ \mathcal{A}_S &= \{d \leq 1, s > 1, F_S \leq 0, c \leq \tfrac{1}{2}\}. \end{aligned}$$

All variables in these three sets are also constrained to $[0, 1]$.

For $d \leq 1$, the maximal radial demand is

$$(75) \quad c_{\max}(a, b) = \begin{cases} 1, & a = b = 0, \\ C_L(m), & s \leq 1, q \leq 0, (a, b) \neq (0, 0), \\ C_T(m, M), & s \leq 1, q > 0, \\ C_S(m, M), & s > 1, \end{cases}$$

where $C_L(m)$ is the unique root in $[\sqrt{3}/2, 1]$ of

$$z^4 - z^2 + mz - m^2 = 0,$$

and

$$C_T(m, M) = \frac{2M}{1 + \sqrt{4s^2 - 3}}, \quad C_S(m, M) = \frac{M(2mM + 1) - M\sqrt{4d - 3}}{2(1 - m^2)}.$$

If $d > 1$, the radial envelope is undefined. At $q = 0$ the two subcritical formulas agree and equal s .

Proof. If $s > 0$ and $cs \leq ab$, then $c/a + c/b \leq 1$ (with the zero-coordinate cases obtained by continuity), so $c(u + v)$ lies in $\text{conv}\{0, au, bv\}$. Its minimum enclosing equilateral side is the diameter

$$\sqrt{a^2 + ab + b^2}.$$

Assume now that $cs > ab$. The four points occur cyclically as $0, bv, c(u + v), au$. For three outward unit normals separated by 120° , the required side equals $2/\sqrt{3}$ times the sum of the three support values. On an angular cell with fixed support points this sum has second derivative equal to its negative. Hence an angular minimum occurs when a normal is perpendicular to one of the four hull edges. The four resulting side lengths are

$$\begin{aligned} L_{0A} &= b + \max\{a, c\}, & L_{B0} &= a + \max\{b, c\}, \\ L_{AC} &= \begin{cases} (a^2 + bc)/\sqrt{a^2 - ac + c^2}, & a \geq c, \\ c(a + b)/\sqrt{a^2 - ac + c^2}, & a < c \leq a + b, \\ c^2/\sqrt{a^2 - ac + c^2}, & c > a + b, \end{cases} \end{aligned}$$

and L_{CB} is obtained by interchanging a, b . Thus the minimum side is the minimum of these four expressions.

Ordering a, b as $m \leq M$, squaring the positive support equations gives $F_L = 0, F_T = 0, F_S = 0$ in the three respective contact regimes; the inactive support inequalities give $q \leq 0, q \geq 0, s > 1$. Squaring creates one spurious component in each of the last two cells. In the T cell the two roots are

$$\frac{2M}{1 + \sqrt{4s^2 - 3}}, \quad \frac{2M}{1 - \sqrt{4s^2 - 3}},$$

and $c \leq 2M$ selects the first. In the S cell the smaller root lies below $1/2$ and the other above $1/2$; hence $c \leq 1/2$ is exactly the geometric selector. The L fiber is the interval from zero to its unique selected root. Solving the selected equations gives (75). This also proves that $\mathcal{A}$ is coordinatewise down-closed. $\square$

B.2. The exact capped map. For $1/2 < c < 1$ put

$$\ell(c) = \frac{c}{2} \left(1 - \sqrt{4c^2 - 3} \right), \quad r(c) = c - \ell(c)$$

when $c \geq \sqrt{3}/2$, and use the order-transition function $h(c)$ from (17). Also define

$$T_-(a, c) = \frac{\sqrt{a^2 - ac + c^2}}{c} - a,$$

$$T_+(a, c) = \frac{2(1 - a^2)c^2}{2ac^2 + c + \sqrt{(2ac^2 + c)^2 - 4(1 - c^2)(1 - a^2)c^2}}.$$

Proposition B.3 (Four-label capped map). *For $1/2 < c \leq \sqrt{3}/2$,*

$$(76) \quad F_c(a) = \begin{cases} 1 - a, & 0 \leq a \leq 1 - c, \\ T_+(a, c), & 1 - c < a < h(c), \\ h(c), & a = h(c), \\ T_-(a, c), & h(c) < a < c, \\ 1 - a, & c \leq a \leq 1. \end{cases}$$

For $\sqrt{3}/2 < c < 1$,

$$(77) \quad F_c(a) = \begin{cases} 1 - a, & 0 \leq a \leq 1 - c, \\ T_+(a, c), & 1 - c < a \leq \ell(c), \\ \ell(c), & \ell(c) < a < r(c), \\ T_-(a, c), & r(c) \leq a < c, \\ 1 - a, & c \leq a \leq 1. \end{cases}$$

Thus there are exactly four labels

$$\text{Full}, \quad L, \quad T_-, \quad T_+^{\text{hi}}.$$

At $a = \ell(c)$ in (77), $F_c(a) = r(c)$; immediately to its right the value is $\ell(c)$. This downward jump is genuine. The formal other root of the T_+ quadratic is not feasible because it violates $c \leq 2 \max\{a, F_c(a)\}$.

The maps F_c are nonincreasing, the maps $G_c = 1 - F_c$ are nondecreasing and extensive, and

$$(78) \quad G_c(a) \leq z \iff G_c(1 - z) \leq 1 - a.$$

Proof. The cap $a + b \leq 1$ eliminates the S cell of Proposition B.2. On the cap $b = 1 - a$, the T inequality reduces to $M(c - M) \leq 0$; hence the Full face is feasible exactly for $a \leq 1 - c$ or $a \geq c$. Off that face a maximal point lies on $F_L = 0$ or $F_T = 0$. The L equation has roots $\ell(c), r(c)$ and the selector $q \leq 0$ keeps only the smaller coordinate. The ordered T halves give respectively the displayed lower quadratic root T_+ and the root T_- . Their order boundary is $a = b = h(c)$, while their $q = 0$ boundaries are $(\ell(c), r(c))$ and $(r(c), \ell(c))$. These observations give (76)–(77), including all equality assignments.

Down-closedness gives monotonicity, while $F_c(a) \leq 1 - a$ gives $G_c(a) \geq a$. Finally,

$$G_c(a) \leq z \iff F_c(a) \geq 1 - z \iff (a, 1 - z, c) \in \mathcal{A}, \quad a + 1 - z \leq 1.$$

Reflection $a \leftrightarrow b$ gives (78). $\square$

Put

$$\mu = \frac{2\sqrt{3}-3}{4},$$

and, for $0 < X < 1 - \sqrt{3}/2$, write

$$e(X) = \ell(1 - X).$$

The root equation gives

$$(79) \quad \begin{aligned} X < e(X) & \quad \left(0 < X < 1 - \frac{\sqrt{3}}{2}\right), \\ e(X) & < \frac{2X}{1-2X} \quad (0 < X \leq \mu), \\ e(X) & \leq 2X + 5X^2 \quad (0 < X \leq 1/8). \end{aligned}$$

Moreover,

$$(80) \quad a > e(X) \implies G_{1-X}(a) \geq 1 - e(X) \quad \left(0 < X < 1 - \frac{\sqrt{3}}{2}\right).$$

Indeed, for

$$p_X(z) = z^2 - (1 - X)z + (1 - X)^2 - (1 - X)^4,$$

one has

$$p_X(X) = X(1 - 3X + 4X^2 - X^3) > 0.$$

Since $X < (1 - X)/2$, this puts X strictly below the smaller root $e(X)$. For $U = 2X/(1 - 2X)$, direct substitution gives

$$p_X(U) = -\frac{X^2}{(1 - 2X)^2}(4X^4 - 20X^3 + 37X^2 - 28X + 3).$$

The polynomial in parentheses is decreasing on $[0, \mu]$ and at the right endpoint equals $8217/64 - 74\sqrt{3} > 0$. Indeed, its derivative is at most $16/512 + 74/8 - 28 < 0$, and $(8217/64)^2 - 3 \cdot 74^2 = 230001/4096 > 0$. Hence $p_X(U) < 0$, which puts U strictly between the roots. Finally,

$$p_X(2X + 5X^2) = X^2(3X + 4)(8X - 1) \leq 0,$$

and $X < 2X + 5X^2 < (1 - X)/2$ on the stated range. Thus the trial point lies between the two roots, proving the third comparison. The implication (80) is read directly from (77). We shall also use $e(X) < 2X + 3X^2$ for $0 < X \leq 1/10$, obtained from $p_X(2X + 3X^2) = X^2(8X^2 + 19X - 2) < 0$; here too the trial point lies strictly between X and $(1 - X)/2$.

APPENDIX C. CYCLIC PROPAGATION IN THE ALL-Vd0 CASES

C.1. The exactly-one-supercritical all-Vd0 branch. For a center interval, call the nonempty closed set missed by its two open endpoint roles a *V-gap*. It may be a singleton. If the maximal closed center trace is $[s, t]$, open coverage gives the weak bounds $B_i \geq s$ and $A_{i+1} \geq 1 - t$; thus none of the arguments below discards a singleton gap.

Proposition C.1 (CE1 one-gap five-map obstruction). *Assume that every vertex role is Vd0, $N_+ = 1$, and the center role is CE1. If its boundary trace contains a V-gap, the roles do not cover H .*

Proof. Use the signed center variables of Proposition 3.1, and put

$$w = W, \quad A = \alpha, \quad D = \delta, \quad X = R - D, \quad H = \frac{\eta + A + D}{2R}.$$

The CE1 sign gives $D + RA \geq P$ and $RD > wA$, so

$$m = \min \left\{ \frac{A}{R}, \frac{D}{w} \right\} = \frac{A}{R}.$$

The exact strict domain needed below is

$$(81) \quad \begin{aligned} &A > 0, \quad D > 0, \quad A + wD < P, \quad D + RA \geq P, \\ &A < \frac{w}{2}, \quad D < \frac{R}{2}, \quad RD - wA > 0, \end{aligned}$$

Proposition 3.5 contains the five actual-row handoffs, the three duality steps, and the terminal diameter comparison. It remains only to prove its scalar target

$$(82) \quad (G_{1-D} \circ G_{1-m} \circ G_{1-A})(H) > 1 - X.$$

Here $A < P \leq \mu$, because $P = E(1 - E)$ and $E \geq \sqrt{3}/2$. We first prove the two input comparisons used at row 4. Since $D = R - X$, the inequality $A + wD < P$ and the identity $Rw = \eta + P$ give

$$A < wX - \eta.$$

Put

$$\varphi(q) = wq - \frac{q}{2(1+q)}.$$

The bound $0 < D < R/2$ gives $R/2 < X < R$, and φ is convex. At the endpoints,

$$\varphi(R/2) < \frac{Rw}{2} < \eta,$$

while $\varphi(R) < \eta$ is equivalent to

$$E(1 - 2R^2) < 1,$$

which follows from $E < 1$. Convexity therefore gives $\varphi(X) < \eta$, and hence

$$A < \frac{X}{2(1+X)}.$$

Using the rational low-root bound in its valid range,

$$(83) \quad e(A) < \frac{2A}{1-2A} < X.$$

We also have

$$2R(H - A) = \eta + D + (1 - 2R)A.$$

This is positive if $R \leq 1/2$. If $R > 1/2$, then $A < P = \eta E$ gives

$$2R(H - A) > \eta(1 - (2R - 1)E) > 0.$$

Thus $H > A$. The positive signed right trace gives

$$\frac{\eta + A + D}{R} < w + D < 1,$$

so $H < 1/2 < 1 - A$.

Now consult the exact high-radial catalog for $c_4 = 1 - A$. The lower and upper Full ranges would require $H \leq A$ and $H \geq 1 - A$, respectively, so both are impossible. On the L or T_- range,

$$F_{1-A}(H) \leq e(A),$$

and therefore

$$G_{1-A}(H) \geq 1 - e(A) > 1 - X$$

by (83). Thus only the T_+^{hi} range remains. Its selector gives

$$H \leq e(A) \leq 2A + 5A^2.$$

We claim that this range forces $X > 1/2$. Indeed, if $X \leq 1/2$, the selector and the lower center inequality give

$$2RH = \eta + A + D \geq \eta + P + wA = w(R + A).$$

Together with $H < 2A/(1 - 2A)$, this implies

$$f_R(A) < 0, \quad f_R(z) := w(R + z)(1 - 2z) - 4Rz.$$

But $f_R''(z) = -4w < 0$ and $f_R(0) = Rw > 0$. We check the other endpoint of the possible interval for A .

If $R \leq 1/2$, then $0 < A < P$. Direct simplification gives

$$f_R(P) = (1 - E)(4E^3 - 2E^2 + 1 - R(2E^3 - 2E^2 + 5E)).$$

The coefficient of R is positive, so

$$f_R(P) \geq \frac{1 - E}{2}(6E^3 - 2E^2 - 5E + 2).$$

For $\phi(E) = 6E^3 - 2E^2 - 5E + 2$ on $[\sqrt{3}/2, 1)$,

$$\phi'(E) = 18E^2 - 4E - 5 \geq \frac{17}{2} - 2\sqrt{3} > 0,$$

and

$$\phi\left(\frac{\sqrt{3}}{2}\right) = \frac{2 - \sqrt{3}}{4} > 0.$$

Hence $f_R(P) > 0$. Strict concavity puts f_R above its endpoint chord, so $f_R(A) > 0$ for $0 < A < P$, a contradiction.

If $R > 1/2$, the assumption $X \leq 1/2$ gives $D \geq R - 1/2$, and therefore

$$0 < A < A_0, \quad A_0 := P - w\left(R - \frac{1}{2}\right) = \frac{w}{2} - \eta.$$

Direct simplification gives

$$f_R(A_0) = \frac{1 - E}{2}(E + 11R - 3ER - 5).$$

Since $11 - 3E > 0$ and $R > 1/2$,

$$E + 11R - 3ER - 5 > \frac{11 - 3E}{2} + E - 5 = \frac{1 - E}{2} > 0.$$

Thus $f_R(A_0) > 0$. Concavity again gives $f_R(A) > 0$ on $(0, A_0)$, a contradiction. Both ranges are impossible, proving $X > 1/2$.

Put $p_1 = G_{1-A}(H)$. If row 3 is not T_+^{hi} , then $p_1 \leq A + e(A) < 3A + 5A^2 < 29/64 < 1/2$, and $2R(H - m) = \eta + D - A > \eta - P > 0$. Extensivity gives $p_1 \geq H > m$, while $p_1 < 1/2 < 1 - m$; thus both Full ranges are impossible. On L , T_- , or the low- c tie one has $F_{1-m}(p_1) \leq p_1$, so

$$G_{1-m}(p_1) \geq 1 - p_1 > 1 - X.$$

Extensivity of the final reverse map G_{1-D} preserves this strict bound and proves (82). It remains to treat the selected row-3 T_+ branch. Proposition 5.1 proves once for all that every selected T_+ arc is increasing and strictly concave.

When $d < 1 - \sqrt{3}/2$, the arc endpoints are (d, d) and $(e(d), d + e(d))$, so concavity gives

$$(84) \quad q \geq p + \frac{d}{e(d) - d}(p - d).$$

For $0 < d \leq \mu$, the valid rational low-root bound gives

$$\frac{d}{e(d) - d} > \frac{1 - 2d}{1 + 2d} > 1 - 4d > 1 - 5d.$$

This covers row 4, because its parameter is $d = A < P \leq \mu$. For the row-3 parameter $d = m$, which need not be at most μ , the remaining high-radial range is handled without the rational estimate. If $\mu < d \leq 1/8$, then

$$\frac{d}{e(d) - d} \geq \frac{1}{1 + 5d} > 1 - 5d$$

by $e(d) \leq 2d + 5d^2$. If $1/8 < d < 1 - \sqrt{3}/2$, then $e(d) < (1 - d)/2$ and hence

$$\frac{d}{e(d) - d} > \frac{2d}{1 - 3d} > 1 - 5d.$$

The last comparison is equivalent to $15d^2 - 10d + 1 < 0$; its smaller root $(5 - \sqrt{10})/15$ is less than $1/8$. In the low- c row-3 range $1 - \sqrt{3}/2 \leq d < 1/5$, the other endpoint is $(h(1 - d), 1 - h(1 - d))$, and the chord coefficient is

$$\frac{1 - 2h(1 - d)}{h(1 - d) - d}.$$

On $3/4 \leq c \leq \sqrt{3}/2$, direct differentiation of the formula for h shows that $h(c)$ is nonincreasing. Moreover,

$$h\left(\frac{3}{4}\right) = \frac{3}{2(\sqrt{6} + 1)} < \frac{7}{16}.$$

Here low c gives $d \geq 1 - \sqrt{3}/2 > 2/15 > 1/16$, and hence

$$h(1 - d) < \frac{7}{16} < \frac{3 + d}{7}.$$

Thus the displayed chord coefficient is greater than $1/3$, while $1 - 5d < 1/3$. For $d \geq 1/5$, extensivity alone is stronger than the row-3 estimate.

Applying these chord bounds first with $d = A$ and then with $d = m$ proves, without an omitted selected branch,

$$\begin{aligned} p_1 &> L_1 := (2 - 4A)H - (1 - 4A)A, \\ p_2 &:= G_{1-m}(p_1) > L_2 := (2 - 5m)L_1 - (1 - 5m)m. \end{aligned}$$

Finally let

$$D_0 = P - RA, \quad D_h = A(4R - 1) + 10RA^2 - \eta, \quad x = Rw.$$

Feasibility puts $D_0 \leq D \leq D_h$. We first prove that the actual feasible value satisfies $D < 1/10$. Since

$$P = \frac{xE}{1 + E} < \frac{x}{2}, \quad \eta = \frac{x}{1 + E} > \frac{x}{2},$$

the contrary assumption $D \geq 1/10$ and $A + wD < P$ give

$$A < \bar{A} := \frac{w(5R-1)}{10}.$$

The function $D_h(A)$ is increasing, so

$$D \leq D_h(A) < D_h(\bar{A}) < U(R) := \bar{A}(4R-1) + 10R\bar{A}^2 - \frac{x}{2}.$$

Direct expansion yields

$$\frac{1}{10} - U(R) = -\frac{R}{10}P_4(R), \quad P_4(R) = 25R^4 - 60R^3 + 26R^2 + 22R - 14.$$

For $y = 2R - 1 \in [0, 1]$,

$$16P_4(R) = 25y^4 - 20y^3 - 106y^2 + 124y - 39 \leq -101y^2 + 124y - 39 < 0.$$

The last quadratic has discriminant -380 . Hence $U(R) < 1/10$, a contradiction, and therefore $D < 1/10$.

It remains to compare L_2 with $2D + 3D^2$. Put

$$J(D) = L_2(D) - \frac{23}{10}D.$$

Since

$$L_2 - 2D - 3D^2 - J(D) = 3D \left(\frac{1}{10} - D \right) > 0,$$

it is enough to prove $J(D) > 0$. Exact differentiation gives

$$J'(D) = \frac{S(R, A)}{10R^2}, \quad S(R, A) = 100A^2 - (40R + 50)A + R(20 - 23R).$$

The inequality $D_0 \leq D_h$ is equivalent to

$$x \leq A(5R - 1 + 10RA).$$

Because $A < P < x/2 \leq 1/8$,

$$A > L := \frac{4x}{25R - 4}.$$

Also $\partial_A S = 200A - 40R - 50 < -45$, and direct substitution gives

$$S(R, L) = -\frac{Rq_3(R)}{(25R - 4)^2},$$

where

$$q_3(R) = 8775R^3 - 14260R^2 + 7928R - 1120.$$

For $y = 2R - 1$,

$$8q_3(R) = 8775y^3 - 2195y^2 + 997y + 3007 \geq 812 > 0.$$

Thus $J'(D) < 0$ and $J(D) \geq J(D_h)$.

At $D = D_h$ one has $H = 2A + 5A^2$. Write $L_{2,h} = L_2(D_h)$. Expansion gives

$$L_{2,h} - A(1 + 4R) = \frac{A}{R^2}Q(R, A),$$

where

$$\begin{aligned} Q(R, A) = & 100A^3R - 40A^2R^2 - 30A^2R + 12AR^2 - 15AR + 5A \\ & - 4R^3 + 5R^2 - R. \end{aligned}$$

On $0 \leq A \leq x/2$,

$$\partial_A^2 \mathcal{Q} = 20R(30A - 4R - 3) < 0.$$

Its two endpoint values are

$$\mathcal{Q}(R, 0) = x(4R - 1) > 0, \quad \mathcal{Q}\left(R, \frac{x}{2}\right) = \frac{x}{2}p(R),$$

where

$$p(R) = 25R^5 - 30R^4 + 20R^3 - 3R^2 - 7R + 3.$$

For $y = 2R - 1$,

$$32p(R) = 25y^5 + 65y^4 + 90y^3 + 106y^2 - 35y + 5 > 0,$$

because the terms of degree at least three are nonnegative and $106y^2 - 35y + 5$ has discriminant -895 . Concavity therefore gives $L_{2,h} > A(1 + 4R)$.

On the other hand, $A < P = \eta E$ and $A < P < x/2$ imply

$$\frac{D_h}{A} = 4R - 1 + 10RA - \frac{\eta}{A} < C_0 := 4R - 2 + 5Rx - \frac{x}{2}.$$

Here we used

$$\frac{1}{E} > 1 + \frac{x}{2}, \quad (1 - x) \left(1 + \frac{x}{2}\right)^2 = 1 - \frac{x^2(3 + x)}{4} < 1.$$

Finally,

$$1 + 4R - \frac{23}{10}C_0 = \frac{h_3(R)}{20}, \quad h_3(R) = 230R^3 - 253R^2 - 81R + 112.$$

Since $h_3''(R) = 46(30R - 11) \geq 184$, strong convexity at $R_0 = 20/23$ gives

$$h_3(R) \geq \frac{788}{529} - \frac{(17/23)^2}{368} = \frac{289695}{194672} > 0.$$

Consequently

$$J(D_h) > A \left(1 + 4R - \frac{23}{10}C_0\right) > 0,$$

and hence

$$p_2 > L_2 > 2D + 3D^2.$$

The low root $e(D)$ is the smaller root of

$$H_D(z) = z^2 - (1 - D)z + (1 - D)^2 - (1 - D)^4,$$

and direct substitution gives $H_D(2D + 3D^2) = D^2(8D^2 + 19D - 2) < 0$. Hence $e(D) < 2D + 3D^2 < p_2$. The one-hit part of Lemma 5.2, applied to the row-2 map, gives

$$G_{1-D}(p_2) \geq 1 - e(D) > 1 - X,$$

because

$$e(D) < 2D + 3D^2 < \frac{23}{100} < \frac{1}{2} < X.$$

This is (82). The actual-row and terminal inequalities in Proposition 3.5 now give the contradiction. $\square$

Proposition C.2 (Positive-gap all-Vd0, $N_+ = 1$). *If the center role is CE1 or CE2, every vertex role is Vd0, and $N_+ = 1$, then no cover of H can have a boundary gap in the actual open vertex-role traces.*

Proof. The midpoint argument used above makes T_0 uniquely supercritical. In an assumed cover, every boundary gap in the vertex-role traces must lie in a center trace. CE1 therefore has the one-gap state excluded by Proposition C.1. For CE2, the exactly-one-gap state is Proposition 3.7; if both center intervals contain gaps, the endpoint bounds and exact exits are those of Lemma D.8, so the two far outgoing reaches sum to less than one. Rows 2, 3, 4 would then have to cover more than three boundary units, contradicting their three nonsupercritical caps. This exhausts the one- and two-gap states, including singleton gaps. The zero-gap state is intentionally owned by the center-independent obstruction of Strategy 4. $\square$

APPENDIX D. NONSUPERCRITICAL, T3-LIKE, AND VD1/VD2 DEMAND BRANCHES

D.1. The exactly-one-T3 branch.

Proposition D.1 (Exactly one T3-like row). *Assume the center role is CE1 or CE2, $N_+ = 1$, exactly one vertex role is T3-like, and no role is Vd1 or Vd2. Then the roles do not cover H .*

Proof. Normalize the center midpoint to M_0 . A T3-like row misses its own midpoint, a supercritical row misses all its local midpoints, and a Vd0 row cannot rescue a neighboring midpoint. Consequently, after reflection,

$$T_0 \text{ is T3-like, } M_1 \in T_0, \quad T_1 \text{ is uniquely supercritical,}$$

and $T_2, \dots, T_5$ are nonsupercritical Vd0 rows.

In V_0 wedge coordinates put

$$1 \leq D \leq \frac{2}{\sqrt{3}}, \quad E = \sqrt{4 - 3D^2}, \quad R = \frac{D + E}{2},$$

and let a be the $e_{5,0}$ reach of T_0 . Its interval on r_1 is

$$[c, u], \quad c = \frac{D(1 + a) - 1}{R}, \quad u = 1 - \frac{R}{D} + a,$$

where $E/(2D) \leq a \leq (4 - 3D + E)/(4D)$. Its own-radial exit is $x_0 = aD/R$. For $B(c) = \mathcal{B}(c)$ and $A(c) = 1 - B(c)$, direct substitution gives

$$c - \frac{x_0(2 - x_0)}{1 + x_0} = \frac{2N(D, a)}{(D + E)(2Da + D + E)},$$

where

$$N = 4D^2a^2 + D^2a + D^2 - DEa + DE - 2Da - D - E.$$

This convex quadratic is nonnegative: on $1 \leq D \leq 8/7$ its minimum on the allowed interval is at $a = E/(2D)$, and squaring reduces the sign to $4(D - 1)(7D^3 - 10D^2 - 8D + 12) \geq 0$; on $8/7 \leq D \leq 2/\sqrt{3}$ its critical value is positive because $(9D - 10)E + 9D^2 - 6D - 4 > 0$. Before using monotonicity, note explicitly that

$$x_0 = \frac{aD}{R} \leq \frac{4 - 3D + E}{4R} \leq \frac{1}{2};$$

the second inequality is equivalent, using $2R = D + E$, to $D \geq 1$. Also $c \leq 1/2$ because $M_1 \in [c, u]$, and hence $0 \leq A(c) = 1 - B(c) \leq 1/2$. Since $X(2 - X)/(1 + X)$ is increasing on $[0, 1/2]$, this proves

$$(85) \quad x_0 \leq A(c).$$

Radial coverage forces the supercritical row's own demand $c_1 \geq c$. Lemma D.6 gives $b_1 < B(c_1) \leq B(c)$. On $e_{5,0}$ the far row T_5 is forced to reach at least $h \geq B(c)$. Indeed this is $h \geq 1 - a$ unless a CE2 interval hides the boundary gap. In the hiding case the side model gives $T \leq x_0$, and (85) again gives $h \geq 1 - T \geq B(c)$. Therefore

$$\sum_{i=2}^5 (a_i + b_i) \geq (1 - b_1) + 1 + 1 + 1 + h > 4,$$

contrary to the four row caps. The case $c = 1/2$ is already impossible because the strict supercritical feasible set is empty there. $\square$

D.2. The nonsupercritical CE1/CE2 boundary packages. Figure 7 gives a common schematic for the package proved below.

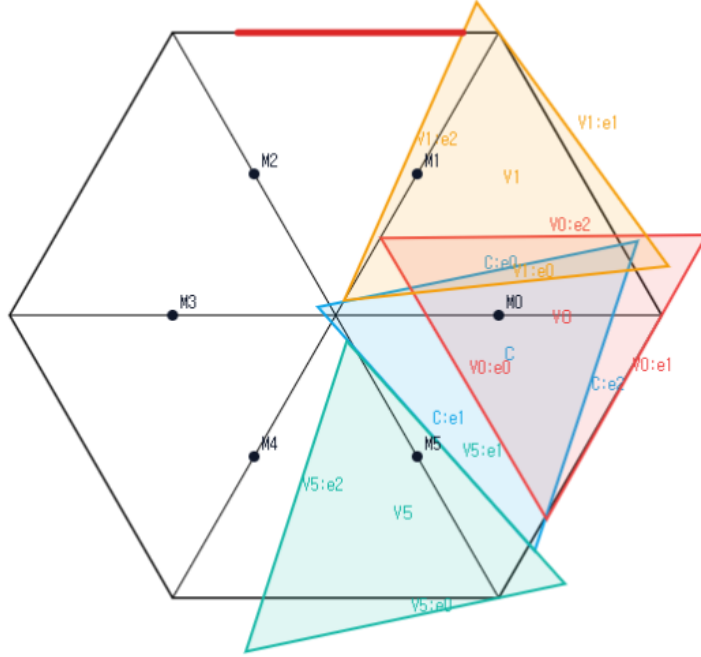


FIGURE 7. Schematic view of the common CE1/CE2, $N_+ = 0$, all-Vd0 boundary package, not to scale. The center role and three representative Vd0 vertex roles are shown; the remaining three roles are omitted for readability. The simultaneous placement is not a candidate cover, and no proof depends on visual inspection.

Proposition D.2 (All-Vd0 boundary loss). *Assume that T_C is CE1 or CE2, every vertex role is Vd0, and*

$$A_i + B_i \leq 1 \quad (0 \leq i \leq 5).$$

Then the seven roles do not cover H .

Proof. If the center boundary traces are already covered by the open vertex roles, then those six roles cover all of ∂H . Put $W_i = U_i \cap \partial H$. These are relatively

open subsets of the connected circle ∂H , each of length at most one. At least two are nonempty, and a finite relatively open cover of a connected space cannot have its nonempty members pairwise disjoint. Thus two overlap on a positive-length relatively open set, whence

$$6 = \mathcal{H}^1(\partial H) < \sum_{i=0}^5 \mathcal{H}^1(W_i) \leq \sum_{i=0}^5 (A_i + B_i) \leq 6,$$

a contradiction.

Suppose next that T_C is CE1 and its trace $[s, t] \subset e_{0,1}$ contains a boundary gap. Put $u = 1 - t$ and $w = t - s$. Boundary coverage gives $B_0 \geq s$ and $A_1 \geq u$. Since the center has no trace on $e_{5,0}$,

$$A_0 + B_5 \geq 1, \quad A_0 + B_0 \leq 1,$$

and hence $B_5 \geq B_0 \geq s$. The exact CE1 formulas, after putting $R = \lambda$ in the orientation selected there, have the form

$$c_1 = u/R, \quad c_5 = 1 - \left(u - \frac{R}{1 + \sqrt{1 - R + R^2}} - \frac{R}{1 - R} w \right).$$

All hypotheses of Lemma D.7 hold. Therefore the outgoing reaches of rows 1 and 5 satisfy $\widehat{b}_1 + \widehat{b}_5 < 1$, where

$$\widehat{b}_1 \leq F_{c_1}(u), \quad \widehat{b}_5 \leq F_{c_5}(s).$$

Rows 2, 3, 4 must then supply boundary length at least

$$(1 - \widehat{b}_1) + 1 + 1 + (1 - \widehat{b}_5) > 3,$$

contrary to their three row caps.

Now let T_C be CE2. If exactly one of $[x, u]$, $[y, v]$ contains a gap, reflect so that it is $[y, v]$. The interval $[x, u]$ is covered by its endpoint roles, so the same endpoint propagation gives inputs y and $1 - v$. With

$$R = \frac{x}{x + y}, \quad s = y, \quad u' = 1 - v, \quad w' = v - y,$$

the CE2 coupling and exit formulas are exactly those of Lemma D.7. The preceding three-row contradiction applies unchanged. If both center intervals contain gaps, put

$$p = 1 - u, \quad q = 1 - v, \quad r = \frac{x}{x + y}, \quad w = \frac{y}{x + y}.$$

The far endpoint roles have inputs at least p and q , with radial demands p/w and q/r . By Lemma D.8, their two outgoing reaches sum to less than one, and the same middle three-row budget is impossible. These are the no-, one-, and two-active-interval states, and are exhaustive. $\square$

We next treat the additional T3-like rows in the $N_+ = 0$ branch. The following local facts also reappear later.

Lemma D.3 (T3-like midpoint and side tradeoff). *Let T_i be T3-like. Then $M_i \notin T_i$. After applying the closed-trace domination of Proposition 2.6, suppose the selected adjacent support is on r_{i+1} and M_{i+1} belongs to the dominated trace. If b is its boundary reach toward V_{i+1} , c its exit on its own arm, and ℓ its entry on r_{i+1} , all measured from the relevant vertex, then*

$$(86) \quad b + c < 1, \quad b + \ell > 1.$$

Consequently two adjacent T3-like traces that cover each other's midpoint, whose selected supports point toward one another, cannot cover both outer half-arms if no other role has positive support there.

Proof. Apply Proposition 2.6 and reflect if necessary. Its translated Type-II majorant has parameters

$$0 < t < 1, \quad z = \sqrt{1-t+t^2}, \quad \alpha = 0, \quad 0 < \beta < z.$$

Its wedge vertex has

$$Q_x = \frac{z-t\beta}{1-t+t^2} > 1.$$

Hence

$$t\beta < z - z^2.$$

The majorant's own-radial exit is $c = \beta/(1-t)$. Since

$$(1-z)(1+z) = t(1-t),$$

we obtain

$$c < \frac{z-z^2}{t(1-t)} = \frac{z}{1+z} < \frac{1}{2}.$$

Here $z < 1$ because $0 < t < 1$. Thus the translated majorant misses M_0 , and its closed-trace domination implies that the original T3-like role misses M_0 as well.

For the tradeoff, the translated T3-like trace has the exhaustive Type-II form

$$\begin{aligned} (1-t)x + ty &\geq 0, \\ \beta + tx - y &\geq 0, \quad 0 < t < 1, \quad z = \sqrt{1-t+t^2}, \quad 0 < \beta < z. \\ \beta + x - (1-t)y &\leq z, \end{aligned}$$

For the selected support on r_1 , substitution in the Type-II half-planes gives

$$b = z - \beta, \quad c = \frac{\beta}{1-t}, \quad \ell = \frac{\beta + 1 - z}{1-t}.$$

The condition $M_1 \in T$ is $\beta \leq z - (1+t)/2$. Therefore

$$b + c \leq z + \frac{t}{1-t} \left(z - \frac{1+t}{2} \right) = 1 - \frac{(1-z)^2}{2(1-t)} < 1,$$

whereas

$$b + \ell - 1 = \frac{t(1-z+\beta)}{1-t} > 0.$$

For a crossed adjacent pair, coverage forces $c_i \geq \ell_{i+1}$ and $c_{i+1} \geq \ell_i$. Combining these two inequalities with (86) gives both $b_i < b_{i+1}$ and $b_{i+1} < b_i$, a contradiction. $\square$

Lemma D.4 (The support-isolated four-label inequality). *Assume T_C is CE1 or CE2, $N_+ = 0$, no role is Vd1 or Vd2, and, after normalization and reflection, T_0 is T3-like with selected adjacent support on r_1 . Then the two endpoint rows in the isolated configuration satisfy*

$$(87) \quad F_{c_5^*}(a_5^*) + F_{c_1^*}(a_1^*) < 1,$$

where a_1^*, a_5^* are the exact boundary inputs and c_1^*, c_5^* the exact radial demands defined below.

Proof. Use the CE1 side variables $0 < \lambda < 1$, $\rho = \sqrt{1 - \lambda + \lambda^2}$ and center slacks

$$X = F_0(O), \quad Y = F_2(O).$$

Then

$$a_C = \lambda - Y, \quad \lambda s = X + Y + 1 - \rho,$$

and Proposition 3.1 gives

$$\rho - X - Y > \frac{1}{2}, \quad \frac{Y}{\lambda} < \frac{1}{2}, \quad \frac{X}{1 - \lambda} < \frac{1}{2}.$$

Consequently the apparent minima in the two radial exits collapse to

$$\gamma_1 = \frac{Y}{\lambda}, \quad \gamma_5 = \frac{X}{1 - \lambda}.$$

For CE2 the second boundary interval is $[S, T]$, where

$$S = \frac{X + Y + 1 - \rho}{1 - \lambda}, \quad T = X + \lambda.$$

The T3-like normal form supplies $D > 1$, $0 < R < 1$, and $\alpha, p \geq 0$ with

$$R^2 - DR + D^2 = 1, \quad \alpha + p = 1/D, \quad q = 1 - p = \alpha + (D - 1)/D < 1/2.$$

Its interval on r_1 , measured from V_1 , is

$$[c, u], \quad c = Dq/R, \quad u = q + (1 - R)/D.$$

For an initial trace $[0, p]$ and a center interval $J = [L, U]$, define

$$\Theta(p, J) = \begin{cases} 1 - p, & J = \emptyset \text{ or } p < L, \\ 1 - \max\{p, U\}, & p \geq L. \end{cases}$$

Then

$$a_1^* = \Theta(p, [s, t]), \quad a_5^* = \Theta(\alpha, [S, T]),$$

with the second interval empty in CE1, and

$$c_5^* = 1 - \gamma_5, \quad c_1^* = \begin{cases} c, & 1 - u \leq \gamma_1 \leq 1 - c, \\ 1 - \gamma_1, & \text{otherwise.} \end{cases}$$

If $a_1^* + a_5^* > 1$, the two caps already imply $F_{c_5^*}(a_5^*) + F_{c_1^*}(a_1^*) < 1$. Assume the hard region $a_1^* + a_5^* \leq 1$. The following table is an exhaustive audit of the four possible first labels; the middle column records the exact decisive inequality.

first label	right labels	conclusion
L	Full	$e(\gamma_5) < a_1^*$
L	L, T_-	$F_{c_5^*}(a_5^*) + F_{c_1^*}(a_1^*) < 1$
L	T_+^{hi}	$e(\gamma_5) < a_1^*$
T_-	L, T_-	$F_{c_5^*}(a_5^*) + F_{c_1^*}(a_1^*) < 1$
T_-	Full, T_+^{hi}	$F_{c_5^*}(a_5^*) < a_1^*$
Full	all	impossible
T_+^{hi}	Full, T_+^{hi}	impossible
T_+^{hi}	L, T_-	$F_{c_5^*}(a_5^*) + F_{c_1^*}(a_1^*) < 1$

We verify the nonnormal entries. Put $\kappa = (1 - \rho)/(1 - \lambda)$. From $X + (1 - \lambda)Y < \rho(1 - \rho)$ one obtains

$$(88) \quad \gamma_1 \geq a_C \implies e(\gamma_5) < a_C,$$

$$(89) \quad S \leq Y/\lambda \implies e(\gamma_5) < S.$$

For (88), one has $\gamma_5 < a_C - \kappa$ and $a_C - \kappa < 1/8$; then (79) applies, and the remaining comparison is

$$(a_C - \kappa) + 5(a_C - \kappa)^2 \leq \kappa.$$

After squaring its positive sides, the difference is $\lambda(23\lambda^3 + 58\lambda^2 + 7\lambda + 72) > 0$. For (89), combine the premise with the center inequality and put $x = \lambda$. This gives $0 < x < 3/8$ and

$$X < X_* = \frac{(1 - \rho)(\rho(1 - 2x) - x(1 - x))}{1 - x - x^2}.$$

Set

$$\delta = \frac{1 - \rho}{1 - x} = \frac{x}{1 + \rho}, \quad \eta = \frac{X}{1 - x}, \quad f = \rho(1 - 2x) - x(1 - x), \quad d = 1 - x - x^2.$$

Then $\gamma_5 \leq \eta < \delta f/d$. Moreover

$$\frac{f}{d} < 1 - 2x,$$

because, after multiplication by positive denominators, this is just

$$\rho < d + \frac{x(1 - x)}{1 - 2x} = 1 + \frac{2x^3}{1 - 2x}.$$

Hence $\eta < \delta(1 - 2x) < 1/8$. If $\eta = 0$, the assertion is immediate. Otherwise,

$$S \geq \frac{X + 1 - \rho}{1 - 2x} = \frac{1 - x}{1 - 2x}(\eta + \delta) > 2\eta \left(\frac{1 - x}{1 - 2x} \right)^2.$$

On the other hand,

$$5\eta < 5\delta(1 - 2x) < 2 \left[\left(\frac{1 - x}{1 - 2x} \right)^2 - 1 \right].$$

For the middle sign use $\rho > 1 - x$ and

$$2(2 - 3x)(2 - x) - 5(1 - 2x)^3 = (4x - 3)(10x^2 - 6x - 1) > 0 \quad (0 < x < 3/8).$$

Indeed both factors are negative; for the quadratic use $10x^2 \leq 15x/4 < 6x + 1$. The low-root estimate (79) now gives

$$e(\gamma_5) \leq 2\gamma_5 + 5\gamma_5^2 \leq 2\eta + 5\eta^2 < S,$$

proving (89) without any polynomial coefficient list. The L separation follows from its exact polynomial: if its output is $z = e(\eta)$ at input $1 - \beta$, then

$$\beta \geq z + \eta.$$

Indeed the selector polynomial $P(x, z) = x^4 - 4x^3 + 5x^2 - (2 + z)x + z - z^2$ is strictly decreasing on the low range and vanishes at $x = \eta$. Equations (88)–(D.2), with the three possibilities for Θ , prove every first- L entry. For a first T_- label the exact root test

$$T_-(a, c) < z \iff c^2(a + z)^2 - (a^2 - ac + c^2) > 0$$

is increasing in z . Substitution of a_1^* , followed by the center inequality, proves $F_{c_5^*}(a_5^*) < a_1^*$ except in the low-low entries, which are immediate from $T_-(a, c) \leq \min\{a, 1 - a\}$ and $L < 1/2$. First-coordinate Full would force either $X \geq (1 - \lambda)\alpha$ and $Y \geq \lambda - \alpha$, contradicting $s \leq p$, or, in the CE2 overlap case, $S \geq 1/2$, contradicting $S \leq \alpha < 1/2$.

For a first T_+^{hi} label put $r = 1 - \lambda$, $y = Y/\lambda$ and choose β so that

$$m = \sqrt{\beta^2 - \beta + 1}, \quad 1 - \gamma_5 = \frac{m}{r + \beta}, \quad F_{c_5^*}(a_5^*) = \frac{\beta m}{r + \beta}.$$

The exact selector is

$$(90) \quad \beta \geq \max \left\{ r, \frac{1}{2}, \frac{1 - r^2}{1 + 2r} \right\}, \quad r \geq (1 - \beta)(r + \beta)^2.$$

The center inequalities give $y < 1/10$ and, with

$$y_* = \frac{r}{1 - r} \left(\frac{m}{r + \beta} - \frac{1}{2} - \frac{1 - r}{1 + \sqrt{r^2 - r + 1}} \right),$$

give $y < y_*$ and

$$3y_* \leq 1 - \frac{\beta m}{r + \beta}.$$

This proves the right- L entry by $e(y) < 3y$. It excludes right Full and right high by the exact high-input filter. For right T_- with input a_C , direct substitution gives a sum below one. For input q , put $c = 1 - y$, $q = tc$, and let $\hat{b}_1 = F_{c_1^*}(q)$. The selected T_- equations give

$$\hat{b}_1 \leq \kappa := \frac{\sqrt{13} - 1}{6}, \quad q \leq \tau < \frac{93}{200},$$

where $\tau \in (0, 1/2)$ is the unique root of $\tau^4 - 3\tau^3 + 3\tau^2 - 3\tau + 1 = 0$. In this CE2 overlap subcase the support ordering gives $q > S$. The common left-high identities are

$$S = S_0 + \frac{1 - r}{r}y, \quad S_0 = 1 - \frac{m}{r + \beta} + \frac{1 - r}{1 + \sqrt{r^2 - r + 1}}.$$

Since $y > 0$, we have the complete chain

$$S_0 < S < q \leq \tau < \frac{93}{200}.$$

We now replace the former interval subdivision by a one-variable comparison. Write $s = r + \beta$ and $\rho_r = \sqrt{r^2 - r + 1}$. The selected Cell 2 inequality factors as

$$r - (1 - \beta)(r + \beta)^2 = (s - 1)(\beta^2 + r\beta - r).$$

The second factor is nonnegative because $r \leq \beta$ and $\beta \geq 1/2$; if it vanishes, then $r = \beta = 1/2$. Hence $s \geq 1$. It follows that $\rho_r \leq m$, since $\rho_r^2 - m^2 = (r - \beta)(s - 1) \leq 0$.

Define the one-variable lower bound

$$\bar{S}(\beta) = 1 - m + \frac{\beta}{1 + m}.$$

Then

$$S_0 \geq 1 - \frac{m}{s} + \frac{1 - r}{1 + m} \geq \bar{S}(\beta).$$

Indeed, the second difference is

$$(s - 1) \left(\frac{m}{s} - \frac{1}{1 + m} \right) \geq 0.$$

For the last sign use $s \leq 2\beta \leq m(1 + m)$: if $g = 3\beta - \beta^2 - 1$, then $g > 0$ and

$$m^2 - g^2 = -\beta(\beta - 1)(\beta^2 - 5\beta + 5) \geq 0, \quad m^2 + g = 2\beta.$$

Put $B_0 = 5657/10000$, $T = 93/200$, and $\beta_* = 161/250$. If $\beta m/s \geq B_0$, then $\beta m \geq B_0$. The function $\phi(\beta) = \beta m$ is strictly increasing, and the exact comparison

$$B_0^2 - \beta_*^2 m(\beta_*)^2 = \frac{22782769}{62500000000} > 0$$

therefore gives $\beta > \beta_*$. On $[\beta_*, 1]$ put

$$H(\beta) = \frac{107}{200} - Tm - (1 - \beta)^2.$$

Since $H'' = -3T/(4m^3) - 2 < 0$, the function H is concave. Its endpoint values are positive: $H(1) = 7/100$, while, for $a_* = 107/200 - (1 - \beta_*)^2 = 51033/125000$,

$$a_*^2 - T^2 m(\beta_*)^2 = \frac{1693881}{62500000000} > 0.$$

Thus $H > 0$ throughout the interval. The identity

$$(\bar{S}(\beta) - T)(1 + m) = H(\beta)$$

shows that $\beta m/s \geq B_0$ would force $S_0 > T$, a contradiction. Consequently

$$F_{c_5^*}(a_5^*) < \frac{5657}{10000} < \frac{7 - \sqrt{13}}{6}.$$

The final strict comparison follows by squaring $\sqrt{13} < 18029/5000$, whose squared difference is $44841/25000000 > 0$. Thus the final sum is also below one. The table is exhaustive, proving (87). $\square$

Proposition D.5 (The $N_+ = 0$ T3-like branch). *Assume that T_C is CE1 or CE2, $N_+ = 0$, no vertex role is Vd1 or Vd2, and at least one vertex role is T3-like. Then the seven roles do not cover H .*

Proof. Normalize the unique center midpoint to M_0 and simultaneously apply the closed-trace domination of Proposition 2.6 to every T3-like role. If T_0 were not T3-like, a maximal consecutive block of T3-like indices among $1, \dots, 5$ would have to match its roles bijectively to the same block of uncovered midpoints. The matching uses only adjacent indices, because a T3-like role misses its own midpoint and Vd0 roles cannot cover adjacent midpoints. Its first two roles therefore form a crossed pair forbidden by Lemma D.3. Hence T_0 is T3-like; reflect so that it supports r_1 .

There are at most two T3-like roles. Indeed, with $N_+ = 0$ and no Vd1/Vd2 role, the short-role identity gives $q = k$, where k is the number of T3-like roles. If $k \geq 3$, Theorem 4.15 gives an immediate full-skeleton contradiction. Midpoint locality then forces T_2, T_4, T_5 to be Vd0. Consequently r_1 has positive support only from T_C, T_0, T_1 , and r_5 only from T_C, T_5 .

The actual outgoing reaches of T_1, T_5 are bounded by the two safe maps in Lemma D.4; an extra adjacent-ray demand on a possible T3-like T_1 only shrinks its feasible set. Thus their sum is less than one. Rows T_2, T_3, T_4 must cover more than three units of the four intervening boundary edges, while their three nonsupercritical caps sum to at most three. This contradiction proves the proposition. $\square$

Lemma D.6 (Free strict-supercritical envelope). *For $0 \leq c < 1/2$,*

$$\sup\{b : \exists a, (a, b, c) \in \mathcal{A}, a + b > 1\} = \mathcal{B}(c) := \frac{c + \sqrt{c^2 - 8c + 4}}{2}.$$

The supremum is not attained. For $c \geq 1/2$ the strict feasible set is empty. The function $\mathcal{B}$ is strictly decreasing on $[0, 1/2]$.

Proof. Lemma 2.10 excludes $c \geq 1/2$. On the boundary $a = 1 - b$, a support rotation has strict slack precisely when

$$c < \frac{1 - b^2}{2 - b}.$$

The right side decreases from $1/2$ to zero. Solving equality gives the displayed larger root. Equality has no strict support slack, which explains nonattainment. Differentiation gives

$$\mathcal{B}'(c) = \frac{1}{2} \left(1 + \frac{c - 4}{\sqrt{c^2 - 8c + 4}} \right) < 0,$$

since $(4 - c)^2 - (c^2 - 8c + 4) = 12$. $\square$

D.3. Two reusable capped-loss inequalities.

Lemma D.7 (One-side capped loss). *Let $0 < R < 1$, $S = \sqrt{1 - R + R^2}$, and let $s, u, w > 0$ satisfy $s + u + w = 1$ and $0 < u < R$. Put*

$$\delta = \frac{R}{1 + S}, \quad \gamma = u - \delta - \frac{R}{1 - R}w,$$

and assume $\gamma > 0$. Set

$$c_1 = u/R, \quad c_5 = 1 - \gamma.$$

Then

$$(91) \quad F_{c_5}(s) + F_{c_1}(u) < 1.$$

Proof. Put $\eta = (1 - R)\delta = 1 - S$ and

$$\widehat{b}_5 = F_{c_5}(s), \quad \widehat{b}_1 = F_{c_1}(u).$$

The center identity may be written

$$(92) \quad c_5 = \frac{1 - u + \eta - Rs}{1 - R}.$$

The hypotheses imply $1/2 < c_1, c_5 < 1$: for example $u > \delta$ gives $c_1 > 1/(1 + S) > 1/2$, and $\gamma < u - \delta < R - \delta = RS/(1 + S) < 1/2$ gives $c_5 > 1/2$. Thus the four high-radial labels of Proposition B.3 are exhaustive.

Right low labels. If the right label is T_- , its exact frontier equation gives $(u + \widehat{b}_1)^2 = 1 - R + R^2 = S^2$, so $\widehat{b}_1 = S - u$. For a right L label write $\widehat{b}_1 = kc_1$, $0 \leq k \leq 1/2$, and $C = 1 - k + k^2 = c_1^2$. With $X_R = R + k$, the Cell- L selector factors as

$$q = c_1^2(X_R - 1)G_k(X_R), \quad G_k(X) = CX^3 + CX^2 - k(1 - k)X + k^2.$$

For $X \geq k$, $G_k(X) > 0$: when $k > 0$, $G'_k(X) \geq k(1 + 2k - k^2 + 3k^3) > 0$ and $G_k(k) = k^2(1 + k + k^3) > 0$. Hence $q \leq 0$ gives $X_R \leq 1$. If $h_R = 1 - X_R$, then

$$S^2 - (u + \widehat{b}_1)^2 = h_R(1 + 2k^2 + k(1 - k)h_R) \geq 0.$$

Consequently either right low label satisfies

$$(93) \quad \widehat{b}_1 \leq X := S - u.$$

A left low label against right high or Full. For left T_- put $z = s/c_5 = 1 - p$ and $h = \sqrt{1 - p + p^2}$. The component selector gives $0 \leq p \leq 1/2$, the frontier

gives $s + \widehat{b}_5 = h$, and its selector gives $s \leq (1 - p)h$. Equation (92) becomes $c_5(1 - Rp) = 1 - u + \eta$, whence

$$u \geq \eta + (1 - h) + Rph.$$

Put $q_0 = ph$. If $q_0 > R$, then $p(1 - p) > R(1 - R)$ and the preceding lower bound would give $u > R$. Thus $q_0 \leq R$, and

$$\eta + 1 - h > \frac{R(1 - R) + p(1 - p)}{2} \geq (1 - R)q_0.$$

Substitution in the exact formula

$$w = \frac{(1 - R)(1 - u)p - \eta(1 - p)}{1 - Rp}$$

now gives $w < 1 - h$. A right high output has the form $H - u$, where $H = \sqrt{1 - \beta + \beta^2} \leq 1$; for Full take $H = 1$. Hence

$$\widehat{b}_5 + \widehat{b}_1 = h + H - (s + u) = 1 + w - (1 - h) - (1 - H) < 1.$$

For a left L label write

$$c_5 = h(x) := \sqrt{1 - x + x^2}, \quad \widehat{b}_5 = xh(x), \quad 0 \leq x \leq \frac{1}{2},$$

and, along the center line,

$$s(x) = \frac{R - u + \eta + (1 - R)(1 - h(x))}{R}.$$

If $y = s(x)/h(x)$, the exact selector factorization is

$$\frac{q}{h(x)^2} = (y - (1 - x))P_x(y),$$

where

$$P_x(y) = (1 - x + x^2)y^3 + (1 + 2x - 2x^2 + 3x^3)y^2 + (x + 2x^2 - x^3 + 3x^4)y + (x^2 + x^3 + x^5).$$

Every coefficient is nonnegative on $0 \leq x \leq 1/2$, and the polynomial is positive at every nondegenerate point. Thus a selected L point has $s(x) \leq (1 - x)h(x)$. The two curves meet once in $(0, 1/2)$: at $x = 0$ the left side is smaller, while at $x = 1/2$ it is enough to use

$$u \leq \frac{R}{1 + R} \quad (\text{Full}), \quad u \leq \min\{RS, S/2\} \quad (T_+^{\text{hi}}).$$

Writing $e_0 = 1 - \sqrt{3}/2$, the corresponding endpoint differences are

$$\frac{(1 - R)(R^2 + 2Re_0 + 2e_0)}{2(1 + R)} > 0$$

for Full, and respectively $[2e_0(1 - R) - R^3]/2 \geq 0$ for $R \leq 1/2$ and $(1 - R)(3R + 4e_0 - 2)/4 \geq 0$ for $R \geq 1/2$ in the high case. At the unique transition x_0 , the L frontier is the $q = 0$ endpoint of the preceding T_- branch. Since $xh(x)$ increases, every selected left L output is no larger than that transition output. The preceding strict loss therefore also covers (L, T_+^{hi}) and (L, Full) .

Full and high exclusions. Left Full forces $s < 1/2$ and $\gamma \geq s$. Right Full or right high gives $u \leq 1/2$, but

$$\gamma - s = 2u - 1 - \delta + \frac{1 - 2R}{1 - R}w < 0 :$$

for $R \geq 1/2$ this is immediate, while for $R < 1/2$ use $w < 1 - u$ to bound it by $(u - R)/(1 - R) - \delta < 0$. Thus left Full cannot pair with either label. If the left label is high and the right one Full, then $s < 1/2$ and $u \leq R/(1 + R)$, whereas $\gamma > 0$ and $w > 1/2 - u$ force $u > R/2 + \eta > R/(1 + R)$; the final inequality is equivalent to $2(1 + R) > 1 + S$. This pair is also impossible.

A right low label. Use (93). If $s > X$, the cap gives $\hat{b}_5 + \hat{b}_1 < 1$. Suppose $s \leq X$ and put

$$c_X = X + 2\delta, \quad f(c, z) = c^4 - c^2 + zc - z^2.$$

At the limiting value $u = R$, with $\kappa = \delta$, one has

$$X_0 = \frac{1 - 2\kappa}{1 + \kappa}, \quad c_0 = \frac{1 + 2\kappa^2}{1 + \kappa}.$$

Now $c_0 > \sqrt{3}/2$ because $16\kappa^4 + 13\kappa^2 - 6\kappa + 1 > 0$, and

$$f(c_0, X_0) = \frac{\kappa^2(1 - 2\kappa)^2(4\kappa^4 + 4\kappa^3 + 10\kappa^2 + 6\kappa + 5)}{(1 + \kappa)^4} > 0.$$

Along the u -line, $\frac{d}{du}f(c_X, X) = -4c_X^3 + c_X + X < 0$, so the same two signs hold before the limiting endpoint. Therefore, for $\ell(c) = c(1 - \sqrt{4c^2 - 3})/2$,

$$\ell(c_X) < X < c_X - \ell(c_X), \quad \ell(c_X) < 1 - X.$$

Write $c = h(z) = \sqrt{1 - z + z^2}$ and let z_X satisfy $h(z_X) = c_X$. The actual $c_5(s)$ corresponds to $0 < z \leq z_X < 1/2$, and

$$s(z) = s_0 + \frac{1 - R}{R}(1 - h(z)), \quad s_0 = \frac{R - u + \eta}{R} > 0.$$

The transition input $(1 - z)h(z)$ decreases, and the preceding root ordering gives $s(z_X) < (1 - z_X)h(z_X)$, hence the same strict inequality for all $z \leq z_X$. To check the order of the two boundary coordinates put

$$\phi(z) = s(z) - zh(z), \quad k_R = \frac{1 - R}{R}.$$

Its endpoint values are $s_0 > 0$ and $X - \ell(c_X) > 0$. Moreover

$$\phi'(z) = \frac{(k_R + z)(1 - 2z)}{2h(z)} - h(z).$$

For $k_R \leq 2$ this is nonpositive; for $k_R > 2$ its zero is described by the strictly increasing function $K_0(z) = (2 - 3z + 4z^2)/(1 - 2z)$, so the minimum is again at an endpoint. Thus $zh(z) < s(z)$. Applying the already displayed polynomial P_z gives the strict Cell- L selector, and $s + zh(z) \leq X + \ell(c_X) < 1$. Since $\partial_b F_L = c(1 - 2z) > 0$ and each feasible fiber is an interval, the capped maximum is

$$\hat{b}_5 \leq zh(z) \leq z_X h(z_X) = \ell(c_X) < 1 - X.$$

Together with $\hat{b}_1 \leq X$, this proves the loss for every right low label.

High-high. Let $c_L(z)$ be the unique root in $[\sqrt{3}/2, 1]$ of $f(c, z) = 0$. With $D_c = 4c^3 - 2c + z > 0$, implicit differentiation gives

$$c_L''(z) = \frac{2(c^2 - cz + z^2) + 16c^2 z(c - z)}{D_c^3} > 0.$$

A selected left high label requires $s < 1/2$ and $c_5(s) \leq c_L(s)$. Define

$$s_0 = \frac{R - u + \eta}{R};$$

then $0 < s_0 < s$ and $c_5(s_0) = 1$. The explicitly named function $g(z) = c_5(z) - c_L(z)$ is concave, with $g(s_0) > 0$. The right high bound $u \leq \min\{RS, S/2\}$ gives $c_5(1/2) > 7/8$. For $R \leq 1/2$ this follows after squaring from $17 - 10R + 9R^2 - 64R^3 - 64R^4 > 0$ (a decreasing polynomial with value $9/4$ at $1/2$); for $R \geq 1/2$ it follows from $(3 + R)^2 - 16S^2 = (1 - R)(15R - 7) > 0$. Finally

$$f(7/8, 1/2) = 33/4096 > 0$$

and f increases in c on this range, so $c_L(1/2) < 7/8$ and $g(1/2) > 0$. Concavity contradicts $g(s) \leq 0$. Thus high-high is impossible.

If both labels are low, the cap already gives $\widehat{b}_5 + \widehat{b}_1 \leq s + u = 1 - w < 1$. The preceding five paragraphs cover every other ordered pair of the four labels, proving (91). $\square$

Lemma D.8 (CE2 two-endpoint capped loss). *Let $[x, U]$ and $[y, v]$ be the two exact CE2 traces in the legacy coordinates of Remark 3.2, where U denotes the endpoint called u there. Put*

$$p = 1 - U, \quad q = 1 - v, \quad r = \frac{x}{x + y}, \quad w = \frac{y}{x + y},$$

and reflect if necessary so that $0 < w \leq 1/2$ and $r = 1 - w$. Then

$$(94) \quad F_{p/w}(p) + F_{q/r}(q) < 1.$$

Proof. Put

$$S_0 = x + y, \quad K = \sqrt{1 - wr}, \quad z = \frac{w}{1 + K}, \quad \zeta = \frac{r}{1 + K}, \quad \alpha = \frac{p}{w}, \quad \beta = \frac{q}{r}.$$

Then $D = S_0 K$, $1 - K = wr/(1 + K)$, and the exact midpoint inequalities give $1/2 < \alpha, \beta < 1$. Dividing the coupling equation $(U + v)S_0 - xy = D$ by S_0 , and using successively $x < U$ and $y < v$, gives the strict endpoint inequalities

$$(95) \quad \beta + p > 1 + z, \quad \alpha + q > 1 + \zeta.$$

Define

$$\Phi_w(\alpha) = F_\alpha(w\alpha), \quad \Phi_r(\beta) = F_\beta(r\beta), \quad h(t) = \sqrt{1 - t + t^2}.$$

Thus the desired sum is $\Phi_w(\alpha) + \Phi_r(\beta)$.

Selected branch parametrizations. For a side of weight $\sigma \in \{w, r\}$, every L label has

$$(96) \quad c = h(k), \quad F_c(\sigma c) = kh(k), \quad 0 \leq k \leq w,$$

on either side. On the large side the range follows from

$$Q_{\text{cell}}/h(k)^2 = (k - w)G_k(r + k),$$

where $G_k(X) = h(k)^2 X^3 + h(k)^2 X^2 - k(1-k)X + k^2 > 0$ for $X \geq k$. The remaining selected parametrizations are

$$(97) \quad \alpha = \frac{h(a)}{w+a}, \quad \Phi_w(\alpha) = a\alpha, \quad r \leq a < 1,$$

$$(98) \quad \beta = \frac{h(b)}{r+b}, \quad \Phi_r(\beta) = b\beta, \quad r \leq b < 1,$$

$$(99) \quad \beta = \frac{K}{r+t}, \quad \Phi_r(\beta) = t\beta = K - r\beta, \quad w \leq t \leq r.$$

For example, the small-high selector factors as

$$h(a)^2 a \left(\frac{w}{(w+a)^2} - (1-a) \right), \quad w - (1-a)(w+a)^2 = (a-r)(a^2 + wa - w),$$

which gives $a \geq r$; the other two factorizations are $(b-w)(b^2+rb-r)$ and $(t-w)(r^2-wt)$, respectively. These formulas specify the selected geometric components, not discarded algebraic roots.

The small-side T_- label is absent for $w < 1/2$. Large-side Full is incompatible with the first inequality in (95). Small-side Full can pair only with L : either high or T_- on the large side has $\beta \leq K$, which would force $p > (1+r)z > w/(1+w)$, contrary to Full. Thus, up to the symmetric $w = r = 1/2$ tie, the exact list is

$$(100) \quad (\text{Full}, L), (T_+^{\text{hi}}, L), (L, L), (L, T_-), (L, T_+^{\text{hi}}), (T_+^{\text{hi}}, T_-), (T_+^{\text{hi}}, T_+^{\text{hi}}).$$

Low-low sums are at most $K < 1$: an L output is at most wK , while (99) is at most rK . For Full- L , put

$$b_{\text{thr}} = 1 + z - \frac{w}{1+w}, \quad \kappa = \frac{w}{1 + (1+w)z}.$$

Full and (95) give $\beta > b_{\text{thr}}$ and $(1+\kappa)b_{\text{thr}} = 1+z$. Direct reduction yields

$$b_{\text{thr}}^2 - h(\kappa)^2 = \frac{w^2(A(w) + KC(w))}{D_0(w, K)} > 0,$$

where

$$A(w) = 8 - 5w + 22w^2 + 6w^3 + 5w^4 + 5w^5 - w^6,$$

$$C(w) = 8 - w + 14w^2 + 4w^3 - 2w^4 + w^5,$$

$$D_0(w, K) = ((1+w)(1+K))^2(1+K+w+w^2)^2.$$

Both numerator polynomials are positive on $[0, 1/2]$. Since h decreases and $(1+k)h(k)$ increases there, $\beta = h(k) > b_{\text{thr}} > h(\kappa)$ implies $(1+k)\beta < 1+z$; combined with (95), this gives $k\beta < p$ and hence the strict Full- L loss.

It remains to verify the four pairs containing a high label on the small side.

(L, T_+^{hi}) . Use parameters $0 \leq k \leq w$ and $r \leq t < 1$:

$$\alpha = h(k), \quad \Phi_w = kh(k), \quad \beta = \frac{h(t)}{r+t}, \quad \Phi_r = \frac{th(t)}{r+t}.$$

The first output and the last output increase in their parameters, while $h(t)/(r+t)$ decreases. If $w \leq 2/5$, then

$$\Phi_w + \Phi_r < wK + \frac{1}{2-w} < 1;$$

after using $K \leq 1 - wr/2$, the positive cleared gap is $P(w) = w^4 - 3w^3 + 4w^2 - 6w + 2$, which decreases on $[0, 2/5]$ and has $P(2/5) = 46/625$. If $w \geq 2/5$, the first endpoint inequality gives $\beta > r + z \geq 3/4$. Since $h(3/4)/(r + 3/4) = \sqrt{13}/(7 - 4w) < 3/4$, monotonicity forces $t < 3/4$. Therefore

$$\Phi_w + \Phi_r < \frac{\sqrt{3}}{4} + \frac{3\sqrt{13}}{20} < 1.$$

(T_+^{hi}, T_-) . Use a, t from (97) and (99). Both demands are at most K . The first endpoint inequality gives

$$\alpha > \frac{1 + z - K}{w} = \frac{1 + r}{1 + K} =: a_0.$$

Thus $K > 1 - w^2$, which forces $w > 1/3$, and comparison at $a = 2/3$ forces $a < 2/3$. The function

$$G(a) = \frac{(1 + a)h(a)}{w + a}$$

decreases on $[r, 2/3]$, because its derivative numerator $2a^3 + (4w - 1)a^2 + (1 - w)a + w - 2$ is at most its value $-7/54$ at $(2/3, 1/2)$. Hence $G(a) \leq (1 + r)K$. Using the second endpoint inequality and $\Phi_r = K - r\beta$ gives

$$\Phi_w + \Phi_r < (2 + r)K - 1 - \zeta < 1.$$

After multiplication by $1 + K$, the last positive gap is $r(3w - w^2 - K)$; its squared comparison is

$$(3w - w^2)^2 - K^2 = w^4 - 6w^3 + 8w^2 + w - 1 > 0$$

for $w > 1/3$ (value $1/81$ at $1/3$, positive derivative thereafter).

$(T_+^{\text{hi}}, T_+^{\text{hi}})$. Use parameters a, b from (97)–(98). The endpoint inequalities first force $w > 2/5$: for $w \leq 2/5$, the asserted contrary inequality $K(w + 1/(2r)) \leq 1 + z$ reduces, after multiplication by $2r(1 + K)$, to

$$K(2w^2 - 4w + 1) + 2w^4 - 4w^3 + w^2 - w + 1 > 0,$$

which is immediate according as the coefficient of K is nonnegative or, using $K \leq 1$, from the decreasing polynomial $2w^4 - 4w^3 + 3w^2 - 5w + 2 \geq 172/625$.

For $F_c(t) = (1 + t)h(t)/(c + t)$, the derivative numerator $2t^3 + (4c - 1)t^2 + (1 - c)t + c - 2$ is increasing on the present rectangle, so F_c has at most one interior minimum. Endpoint comparison gives

$$\Phi_w + \alpha \leq \frac{2}{1 + w}, \quad \Phi_r + \beta \leq \frac{2}{1 + r}.$$

The nontrivial endpoint signs are exactly

$$4 - (1 + r)^2(1 + w)^2K^2 = -w^2(w - 1)^2(w^2 - w - 3) > 0$$

and

$$16r^2 - (1 + r)^4h(r)^2 = (1 - r)(r^5 + 4r^4 + 7r^3 + 9r^2 - 4r - 1) > 0;$$

the bracket is increasing on $[1/2, 1]$ and equals $13/32$ at $1/2$. Weighting the two strict endpoint inequalities by w/K^2 and r/K^2 gives

$$\alpha + \beta > \frac{2K + 1}{K(1 + K)}.$$

Consequently

$$\Phi_w + \Phi_r < \frac{2}{1+w} + \frac{2}{1+r} - \frac{2K+1}{K(1+K)} < 1.$$

The numerator of the last gap is $3Kwr - Q(w)$, where $Q(w) = w^4 - 2w^3 + 7w^2 - 6w + 2 > 0$. Its squared difference is

$$D(w) = -w^8 + 4w^7 - 9w^6 + 13w^5 - 41w^4 + 65w^3 - 55w^2 + 24w - 4.$$

Here $D'(w) = (1-2w)H(w)$ with

$$H(w) = 4w^6 - 12w^5 + 21w^4 - 22w^3 + 71w^2 - 62w + 24 > 0$$

on $[2/5, 1/2]$; use $71w^2 - 62w + 24 \geq 743/71$ and the remaining cubic block at least $-11/4$. Thus D increases and $D(2/5) = 4904/390625 > 0$, proving the claim.

(T_+^{hi}, L) . Use the fully specified parameters

$$\alpha = \frac{h(t)}{w+t}, \quad p = w\alpha, \quad b = t\alpha, \quad \frac{1}{2} \leq t < 1,$$

and

$$\beta = H = h(k), \quad \widehat{b}_L = kH, \quad 0 \leq k \leq w.$$

Since $p + b = h(t)$, the first endpoint inequality gives

$$(101) \quad 1 - (b + kH) > G := 2 + z - h(t) - (1 + k)H.$$

Put

$$p_0 = 1 + z - K = \frac{w(2-w)}{1+K},$$

let $t_0 \in (1/2, 1)$ be the unique point with $p(t_0) = wh(t_0)/(w+t_0) = p_0$, and put $b_0 = t_0 p_0/w$. The exact endpoint lemma is

$$(102) \quad b_0 < 1 - wK.$$

To verify it, set $b_* = 1 - wK$ and $T_* = wb_*/p_0$. One has $0 < T_* < 1$, and

$$p_0^2(w + T_*)^2 - w^2h(T_*)^2 = -\frac{w^3(KE(w) + J(w))}{(1+K)^2(2-w)^2} > 0,$$

where

$$E(w) = 2w^5 + 3w^3 - 20w^2 + 23w - 6, \\ J(w) = w^6 + 8w^5 - 22w^4 + 12w^3 + 6w^2 + 2w - 6.$$

Here $J \leq -111/64$; if $E > 0$, then $KE + J < E + J < 0$, because $E + J$ is increasing and equals $-139/64$ at $w = 1/2$. Thus $p(T_*) < p(t_0)$. Since $p(t)$ decreases, $t_0 < T_*$ and (102) follows.

Finally put $P = 1 + z - p$. If $P < K$, then $t < t_0$ and $k \leq w$, so

$$G > 2 + z - h(t_0) - (1 + w)K = 1 - b_0 - wK > 0.$$

If $P \geq K$, then $K \leq P \leq P_1 := 1 + z - w/(1+w) \leq 1$ and there is $a \in [0, w]$ with $h(a) = P$. Since $P < H$, one has $k < a$ and $(1+k)H < (1+a)P = P + \ell(P)$, where $\ell(P) = P(1 - \sqrt{4P^2 - 3})/2$. Hence

$$G > \mathcal{D}(P) := 1 - b(P) - \ell(P).$$

This function is concave. Indeed $\ell''(P) > 0$, and the high output b is convex in $p = 1 + z - P$: with $A_t = 2 + w - (1 + 2w)t > 0$ and $Q_t = 8wt^2 + 4t^2 - 8wt - 13t - w + 4 < 0$,

$$\frac{d^2b}{dp^2} = -\frac{2h(t)(t+w)^3Q_t}{w^2A_t^3} > 0.$$

At $P = K$, (102) gives $\mathcal{D}(K) = 1 - b_0 - wK > 0$. At $P = P_1$, set $\kappa = w/(1 + (1 + w)z)$. The Full- L comparison already proved gives $P_1 > h(\kappa)$, hence $\ell(P_1) < \kappa P_1 = w/(1 + w)$, while $b(P_1) = 1/(1 + w)$. Thus $\mathcal{D}(P_1) > 0$, and concavity gives $G > 0$ throughout. Equation (101) proves the final hard pair.

The exact list (100) is now exhausted, proving (94). $\square$

D.4. Half-edge radial envelope. The adjacent Vd1/Vd2 placement uses one upper bound for a single admissible-set fiber. Its half-edge domain is essential; outside that domain the linear cap is false.

Lemma D.9 (Half-edge rational radial envelope). *Suppose*

$$\frac{1}{2} \leq p < 1, \quad 0 \leq h \leq p, \quad p + h < 1.$$

Then

$$(103) \quad c_{\max}(p, h) \leq 1 - \frac{h}{3}.$$

The inequality is strict when the selected admissible-set component is $\mathcal{A}_T$.

Proof. Put

$$s = p + h, \quad d = p^2 + ph + h^2, \quad q = s^4 - s^2 + ph.$$

Because $s < 1$, one has $d = s^2 - ph < 1$, so only the two nonsupercritical cells of Proposition B.2 occur.

First suppose $q \leq 0$. Since $p \geq h$, the selected radial value is the unique root $C_L(h) \in [\sqrt{3}/2, 1]$ of

$$P_h(c) := c^4 - c^2 + hc - h^2 = 0,$$

with $C_L(0) = 1$. This cell forces $h < 3/8$. Indeed, if $h \geq 3/8$, then $p \geq 1/2$ and $s \geq 7/8$, while

$$\frac{\partial q}{\partial p} = 4s^3 - 2s + h \geq \frac{167}{128} > 0$$

and

$$\frac{d}{dh} q \left(\frac{1}{2}, h \right) = h(4h^2 + 6h + 1) > 0.$$

Thus

$$q(p, h) \geq q \left(\frac{1}{2}, \frac{3}{8} \right) = \frac{33}{4096} > 0,$$

a contradiction.

At $c_0 = 1 - h/3$,

$$P_h(c_0) = \frac{h}{81} (h^3 - 12h^2 - 63h + 27).$$

The cubic decreases on $[0, 3/8]$ and has value $891/512$ at $3/8$, so this quantity is positive for $h > 0$. Also $c_0 > 7/8 > \sqrt{3}/2$ and

$$P'_h(c) = c(4c^2 - 2) + h > 0 \quad (c \geq \sqrt{3}/2).$$

Hence $C_L(h) < c_0$. The case $h = 0$ gives equality.

Now suppose $q > 0$. The selector keeps the smaller geometric root

$$C_T(p, h) = \frac{2p}{1 + \sqrt{4s^2 - 3}}.$$

Put

$$r = \sqrt{4s^2 - 3}, \quad \gamma = \frac{p - h}{s}.$$

The exact identity

$$\frac{q}{s^2} = \frac{r^2 - \gamma^2}{4}$$

gives $0 \leq \gamma < r$. Write $\gamma = wr$ with $0 \leq w < 1$. Then

$$p = \frac{s(1 + wr)}{2}, \quad h = \frac{s(1 - wr)}{2},$$

and

$$C_T(p, h) = \frac{s(1 + wr)}{1 + r}.$$

For fixed s, r , define

$$\Psi(w) = C_T(p, h) + \frac{h}{3}.$$

A direct differentiation gives

$$\Psi'(w) = \frac{sr(5 - r)}{6(1 + r)} > 0,$$

so

$$\Psi(w) < \Psi(1) = \frac{s(7 - r)}{6}.$$

The hypothesis $p \geq 1/2$ gives $s(1 + r) \geq 1$. This forces $r > 1/8$; otherwise

$$4s^2(1 + r)^2 = (r^2 + 3)(1 + r)^2 \leq \frac{193}{64} \frac{81}{64} < 4.$$

Finally,

$$144 - (r^2 + 3)(7 - r)^2 = (1 - r)(r^3 - 13r^2 + 39r - 3).$$

On $[1/8, 1]$ the cubic factor is increasing and has value $857/512$ at $1/8$. Hence $s(7 - r)/6 < 1$, and therefore

$$C_T(p, h) + \frac{h}{3} < 1.$$

This proves the strict $\mathcal{A}_T$ estimate. $\square$

Remark D.10 (Exact domain limitation). The hypothesis $p \geq 1/2$ cannot be removed. At

$$p = h = \frac{2}{5}$$

one has $d = 12/25 < 1$, $s = 4/5 < 1$, and $q = -44/625 < 0$, so the selected component is $\mathcal{A}_L$. At $c_0 = 13/15$,

$$P_{2/5}\left(\frac{13}{15}\right) = -\frac{14}{50625} < 0, \quad P_{2/5}(1) = \frac{6}{25} > 0.$$

Since $13/15 > \sqrt{3}/2$ and $P_{2/5}$ is strictly increasing on the selected root interval,

$$c_{\max}\left(\frac{2}{5}, \frac{2}{5}\right) > \frac{13}{15}.$$

Thus Lemma D.9 is used only on its stated half-edge domain.

D.5. The exactly-one-Vd1/Vd2 placements. We now assume CE2, $N_+ = 1$, and exactly one row is Vd1 or Vd2. The corner normal form of Proposition 4.6 will be used in the following form: for some $t > 0$ and $d = \sqrt{t^2 + t + 1}$,

$$(104) \quad x - (t+1)y \leq a, \quad ty - (t+1)x \leq tb, \quad tx + y \leq d - a - tb.$$

In particular $a + b < 1/2$ for a positive-support row.

Lemma D.11 (Vd1 adjacent rescue). *Assume an original open-role skeleton cover in the CE2 exact- M_0 branch, with $T_C \cap \{M_0, \dots, M_5\} = \{M_0\}$. Suppose T_0 is the unique Vd1 row, $M_1 \in T_0$, T_1 is uniquely supercritical, and $T_2, \dots, T_5$ are nonsupercritical Vd0 rows. Suppose also that these are the only positive-adjacent-support roles. Then the roles do not cover the perimeter together with r_1 .*

Proof. From (104), the r_1 interval is

$$[c, u], \quad c = \frac{t(1-b)}{t+1}, \quad u = \frac{d-a-tb-1}{t}.$$

The midpoint inequalities and absence of positive r_5 support force $t \geq 1$. They also give

$$a \leq F(t, c) := d - 1 - \frac{3t}{2} + c(t+1) \leq L(c) := \sqrt{3} - \frac{5}{2} + 2c.$$

With $A(c) = 1 - \mathcal{B}(c)$, squaring positive sides gives

$$2L(c) \leq A(c),$$

because the difference is $4(20c^2 + (18\sqrt{3} - 52)c + 47 - 24\sqrt{3}) > 0$ on $[0, 1/2]$. Writing $\delta = 1 - u$, the same normal form yields

$$\frac{a}{a+\delta} \leq \frac{L(c)}{L(c)+1/2} \leq 2L(c) \leq A(c).$$

Thus $a \leq A(c)$. If the CE2 interval on $e_{5,0}$ hides the gap after T_0 , its side parameters X, Y, λ, ρ satisfy $Y \geq \lambda\delta$ and $S \leq a$, whence

$$T = X + \lambda \leq a + \lambda(u - a) \leq \frac{a}{a+\delta} \leq A(c).$$

In every other ordering its forced tail starts no later than a . Consequently the far-row reach is always $h \geq \mathcal{B}(c)$.

Coverage of r_1 gives the supercritical row a radial demand at least c . Therefore Lemma D.6 gives $b_1 < \mathcal{B}(c) \leq h$. The four remaining row caps would then have to satisfy

$$\sum_{i=2}^5 (a_i + b_i) \geq (1 - b_1) + 1 + 1 + 1 + h > 4,$$

a contradiction. □

Lemma D.12 (Vd1-supercritical replacement).

Assume an original open-role perimeter-and-radial-skeleton cover in the CE2 exact- M_0 branch and the complementary no-additional-positive-support case. Let T_i , $i \neq 0$, be uniquely supercritical, and let the unique Vd1 row, distinct from T_0 and adjacent to T_i , contain M_i . Assume every other vertex row is nonsupercritical Vd0 and the center has no boundary trace on the shared edge of this pair. Then the pair can be replaced by two open nonsupercritical Vd0 roles preserving all boundary and radial demands used by Proposition D.2.

Proof. Normalize the Vd1 row at V_0 and the supercritical row at V_1 . By the shared-edge hypothesis, no center trace intervenes on $e_{0,1}$; by the no-additional-positive-support hypothesis, no third vertex role has positive boundary or radial support in the handoffs used below. In (104), midpoint rescue and one-sided support imply $t \geq 1$ and, with

$$c = \frac{d - a - tb}{t + 1}, \quad \lambda = \frac{t(1 - b)}{t + 1}, \quad \mu = \frac{d - a - tb - 1}{t},$$

give the strict consequences

$$(105) \quad a + c < 1, \quad a < \lambda \leq \frac{1}{2}, \quad b < \frac{1}{2}, \quad \mu < 1 - a.$$

The shared edge forces the supercritical input $a_i \geq 1 - b > 1/2$, and radial coverage forces $c_i \geq \lambda$. A direct consequence of Proposition B.2 is

$$(x, y, z) \in \mathcal{A}, \quad x \geq 1/2, \quad z \leq 1/2 \implies y \leq 1 - z.$$

Indeed, substituting $x = 1/2, y = 1 - z$ in the selected supercritical polynomial gives $z^2(2z - 5)(2z - 1)/4 \geq 0$, and the polynomial is increasing in both relevant coordinates. Hence $b_i \leq 1 - c_i \leq 1 - \lambda$ and $a + b_i < 1$. Open radial handoff further gives

$$c_i^{\text{req}} < \max\{c_i, \mu\} < \max\{1 - b_i, 1 - a\}.$$

Choose $a < p_1 < p_2 < 1 - b_i$ within these strict margins. In local coordinates use the unit triangles

$$\Delta_p^- = \text{conv}\{(0, 1 - p), (1, 1 - p), (0, -p)\}, \quad p \leq 1/2,$$

and

$$\Delta_p^+ = \text{conv}\{(p, 0), (p, 1), (p - 1, 0)\}, \quad p \geq 1/2.$$

Their incident reaches are $p, 1 - p$, their own-radial reach is respectively $1 - p, p$, and they have zero adjacent support. Small translations, chosen below every strict margin above, make the two replacements open Vd0 roles with boundary sum < 1 while preserving the shared boundary and both radial handoffs. Thus the datum reduces to Proposition D.2. $\square$

D.6. Exact-demand terminal routing.

Proposition D.13 (Exact-demand terminal branches). *The exact-demand method proves the following terminal exclusions:*

- (1) CE1 or CE2, $N_+ = 0$, all vertex roles Vd0;
- (2) CE1 or CE2, $N_+ = 0$, with at least one T3-like role and no Vd1/Vd2;
- (3) CE1 or CE2, $N_+ = 1$, all vertex roles Vd0, with at least one boundary gap;
- (4) CE1 or CE2, $N_+ = 1$, exactly one T3-like role and no Vd1/Vd2;
- (5) the CE2, $N_+ = 1$, exactly-one-Vd1/Vd2 branch.

All gap alternatives include singleton gaps.

Proof. The five items are respectively Propositions D.2, D.5, C.2, D.1, and A.5. The strictness at open handoffs appears in the weak endpoint bounds, the nonattained supercritical envelope, and the strict replacement margins; no positive-length assumption is made about a V-gap itself. $\square$

APPENDIX E. THE DIRECT NINE-POINT OBSTRUCTION

This section supplies the exact frontier, witness, sign, and cap-overlap calculations for the direct nine-point obstruction.

This section proves a center-class-independent theorem. Its hypotheses are that all six vertex roles are Vd0, that they cover the full boundary, and that exactly one actual boundary row is supercritical. The theorem will therefore apply both to the CE0 branch and to the zero-gap subcases of CE1 and CE2. Six common radial points and three asymmetric points are excluded directly from all six actual vertex roles and are therefore forced into the center role. Only the unique supercritical row uses an AB -union; no nonsupercritical model core is introduced.

E.1. The finite caliper criterion. We first record the finite caliper certificate used below. It enters the proof through the exact statement and verification argument given here; neither a plot nor an unrecorded numerical convention is used.

Let R and J denote counterclockwise rotations through $2\pi/3$ and $\pi/2$, respectively. For a nonempty finite set $S \subset \mathbb{R}^2$ and a vector N , put

$$(106) \quad W_S(N) = \sum_{k=0}^2 \max_{z \in S} \langle z, R^k N \rangle.$$

Theorem E.1 (Finite caliper certificate). *If S has at least two distinct points, then S is contained in a closed unit equilateral triangle if and only if there are distinct $p, q \in S$ and $\varepsilon \in \{-1, 1\}$ for which*

$$(107) \quad W_S(\varepsilon J(q - p)) \leq \frac{\sqrt{3}}{2} \|p - q\|.$$

If S consists of one point, it is contained in equilateral triangles of arbitrarily small positive side length.

For $S_x = \{O, A, B, x\}$, every clause of (107) is a finite polynomial clause of degree at most four in the coordinates of A, B, x ; at most $12 \cdot 4^3$ clauses are required, and repeated point labels merely delete zero-length pairs.

Proof. For a unit vector n , write

$$G_S(n) = \sum_{k=0}^2 h_S(R^k n), \quad h_S(n) = \max_{z \in S} \langle z, n \rangle.$$

The three supporting half-planes with normals n, Rn, R^2n form an equilateral triangle of side $2G_S(n)/\sqrt{3}$. Hence S fits in a closed unit equilateral triangle precisely when $\min_{\|n\|=1} G_S(n) \leq \sqrt{3}/2$.

On an open normal-direction cell on which the three support points are fixed, $G_S(\theta) = c \cos \theta + d \sin \theta$ and $G_S'' = -G_S < 0$ unless all points coincide. Thus a minimum occurs at a cell boundary, where some support line contains two distinct points p, q . After a cyclic rotation of the three normals, its normal is $\varepsilon J(q - p)/\|p - q\|$. Homogeneity yields (107). A nondegenerate segment is checked directly with its two normals.

For the polynomial expansion fix (p, q, ε) , put $N = \varepsilon J(q - p)$ and $N_k = R^k N$, and choose support labels $z_0, z_1, z_2 \in S_x$. The support cell is

$$(108) \quad \langle z_k, N_k \rangle \geq \langle z, N_k \rangle \quad (z \in S_x, \quad k = 0, 1, 2),$$

with $\|p - q\|^2 > 0$. Since $W_S(N) \geq 0$, the remaining test is

$$(109) \quad 4 \left(\sum_{k=0}^2 \langle z_k, N_k \rangle \right)^2 \leq 3 \|p - q\|^2.$$

The points are affine-linear in the parameters, so (108)–(109) have degree at most four. Six unordered pairs, two signs, and 4^3 support-label triples give the asserted finite exhaustive list. $\square$

E.2. Corner-clipped AB -unions. Let e_A, e_B be unit vectors with angle 120° , put

$$O = 0, \quad A = ae_A, \quad B = be_B, \quad C = \{ue_A + ve_B : u, v \geq 0\},$$

and define

$$(110) \quad \mathcal{R}(a, b) = C \cap \bigcup \{T : T \text{ is a closed unit equilateral triangle and } O, A, B \in T\}.$$

The finite description in Theorem E.1 applies to the four-point set $\{O, A, B, x\}$ and makes membership in (110) an exact finite semialgebraic condition of degree at most four.

The strict supercritical branch admits a simpler closed form. In the next statement write $x = ue_A + ve_B$ and $\|x\|^2 = u^2 + v^2 - uv$.

Theorem E.2 (Strict supercritical AB -union). *Assume*

$$a + b > 1, \quad \rho = a^2 + ab + b^2 < 1,$$

put $h = \sqrt{3}/2$, $D = \sqrt{4\rho - 3}$, and define

$$\begin{aligned} \alpha &= \frac{h(a + 2b - aD)}{2\rho}, & \beta &= \frac{h(a - b + (a + b)D)}{2\rho}, \\ \gamma &= \frac{h(-a + b + (a + b)D)}{2\rho}, & \delta &= \frac{h(2a + b - bD)}{2\rho}. \end{aligned}$$

Then all four coefficients are positive and

$$(111) \quad \mathcal{R}(a, b) = (C \cap \mathcal{D}_A \cap \mathcal{H}_A) \cup (C \cap \mathcal{D}_B \cap \mathcal{H}_B),$$

where

$$\begin{aligned} \mathcal{D}_A &= \{(u, v) : (u - a)^2 + v^2 - (u - a)v \leq 1\}, \\ \mathcal{D}_B &= \{(u, v) : u^2 + (v - b)^2 - u(v - b) \leq 1\}, \\ \mathcal{H}_A &= \{(u, v) : \alpha(u - a) + \beta v \leq 0\}, \\ \mathcal{H}_B &= \{(u, v) : \gamma u + \delta(v - b) \leq 0\}. \end{aligned}$$

Its non-axis frontier consists, in counterclockwise order, of the A -circle, the A -line, the B -line, and the B -circle.

Proof. The identities

$$4\rho - 3(a + b)^2 = (a - b)^2, \quad (a + b)^2 D^2 - (a - b)^2 = 4\rho((a + b)^2 - 1)$$

give $0 < D < 1$ and positivity of the four coefficients.

We apply the caliper criterion of Theorem E.1. For a point $x = (u, v)$ in the relative interior of C but outside $\triangle OAB$, the four hull vertices occur in the order

O, B, x, A . The two cone-axis hull edges cannot certify a unit triangle: their support sums, divided by h , are respectively

$$\max\{a, u\} + \max\{b, v - u\}, \quad \max\{b, v\} + \max\{a, u - v\},$$

and are at least $a + b > 1$. It remains to test the Ax and Bx calipers.

For the Ax caliper put $p = u - a$ and $r^2 = p^2 + v^2 - pv$. Exact evaluation of the three supports gives

(112)

$$\frac{G_A}{h} = \frac{\max\{r^2, -ap + (a+b)v\} + N_A}{r}, \quad N_A = \begin{cases} 0, & p \leq 0, \\ ap, & 0 \leq p \leq v, \\ (a+b)p - bv, & p \geq v. \end{cases}$$

No support label is omitted in these three sectors. If $p \leq 0$, $G_A \leq h$ is equivalent to

$$r \leq 1, \quad -ap + (a+b)v \leq r.$$

The second inequality is the half-plane condition in (111), because

$$r^2 - (-ap + (a+b)v)^2 = \frac{(cp + dv)(c'p + d'v)}{4\rho},$$

where

$$c = a + 2b - aD, \quad d = a - b + (a+b)D, \quad c' = a + 2b + aD, \quad d' = a - b - (a+b)D,$$

and $c, c', d > 0 > d'$. Thus the low sector is exactly $\mathcal{D}_A \cap \mathcal{H}_A$. In the middle sector $0 < p \leq v$ one has $r \leq v$, whereas the numerator in (112) is at least $(a+b)v > r$, so no fit is possible.

In the high sector $p \geq v$, the inequalities $G_A \leq h$ imply

$$(113) \quad r^2 + (a+b)p - bv \leq r, \quad bp + av \leq r.$$

Write $x - A = ry(\theta)$, $0 \leq \theta \leq 60^\circ$, with cone coordinates

$$y(\theta) = te_A + we_B, \quad t = \frac{2}{\sqrt{3}} \cos(\theta - 30^\circ), \quad w = \frac{2}{\sqrt{3}} \sin \theta.$$

Then (113) gives

$$r \leq f(\theta) = 1 - (a+b)t + bw, \quad S(\theta) = bt + aw \leq 1.$$

Let n_B be the unit vector with dual coordinates (γ, δ) . The identities

$$b\gamma + (a+b)\delta = h, \quad (a+b)\gamma + a\delta = \frac{h}{2}(1+D), \quad b\delta - a\gamma = \frac{h}{2}(1-D)$$

show that the unique angle θ_B of this normal satisfies $S(\theta_B) = 1$ and $f(\theta_B) = (1-D)/2$. The function S has at most one critical point, a maximum, so $S(\theta) \leq 1$ implies $\theta \leq \theta_B$. On the feasible part $f, f' \geq 0$; hence $f(\theta) \cos(\theta_B - \theta)$ is increasing. Consequently

$$\langle n_B, x - B \rangle \leq a\gamma - b\delta + \frac{h}{2}(1-D) = 0.$$

The diameter bound also gives $\|x - B\| \leq 1$. Thus every high sector fit belongs to $\mathcal{D}_B \cap \mathcal{H}_B$. Reflection exchanges a, u with b, v and supplies the corresponding result for the Bx caliper. This proves (111) in the cone interior; the axes follow by taking limits, since both sides are closed.

For completeness, the line–circle junctions are obtained by solving the two displayed line and circle equations. The line normals satisfy

$$\alpha^2 + \alpha\beta + \beta^2 = h^2, \quad \gamma^2 + \gamma\delta + \delta^2 = h^2,$$

and the line determinant is

$$\alpha\delta - \gamma\beta = \frac{3(1-D)(D+3)}{16\rho} > 0.$$

The successive cone slopes are

$$+\infty, \quad \frac{2}{1-D}, \quad \frac{a(1-D)+2b}{2a+b(1-D)}, \quad \frac{1-D}{2}, \quad 0;$$

the two middle comparisons reduce to $a(4 - (1-D)^2) > 0$ and $b(4 - (1-D)^2) > 0$. This proves the asserted four-piece order. $\square$

E.3. Exact-one handoffs and common demands.

Lemma E.3 (Exact-one handoff chain). *Assume the six vertex roles cover ∂H and exactly one actual row is supercritical. Rotate its index to 4. The strict handoffs supplied by Proposition 2.9 satisfy*

$$x_3 < x_2 < x_1 < x_0 < x_5 < x_4.$$

With

$$a = 1 - x_3, \quad b = x_4, \quad p = 1 - b, \quad q = 1 - a,$$

one has

$$(114) \quad 0 < a, b < 1, \quad a + b > 1, \quad a^2 + ab + b^2 < 1,$$

and every selected row obeys

$$a_i \geq p, \quad b_i \geq q.$$

Proof. Every selected row other than 4 is strictly nonsupercritical, so $x_i < x_{i-1}$ for $i \neq 4$, which is precisely (E.3). In particular x_3 is the global minimum and x_4 the global maximum. The two anchors of row 4 lie in its open role; their mutual squared distance is $a^2 + ab + b^2$, strictly below the diameter squared of the closed unit triangle. This proves (114). Finally every $x_j \in [x_3, x_4]$, so $1 - x_{i-1} \geq 1 - x_4 = p$ and $x_i \geq x_3 = q$. $\square$

E.4. Six directly forced radial points.

Put $h = \sqrt{3}/2$ and

$$s = p + q, \quad m = \min(p, q), \quad M = \max(p, q), \quad \chi = s^4 - s^2 + pq.$$

The strict domain (114) gives

$$(115) \quad pq > (1-s)(2-s), \quad m > 1-s, \quad s > 1-m.$$

Let $c_* = c_{\max}(p, q)$ be the exact radial envelope of Proposition B.2. On the present subcritical domain, equation (75) becomes

$$(116) \quad c_* = \begin{cases} C_L(m), & \chi \leq 0, \\ \frac{2M}{1 + \sqrt{4s^2 - 3}}, & \chi > 0, \end{cases}$$

where $C_L(m)$ is the unique root in $[\sqrt{3}/2, 1]$ of

$$z^4 - z^2 + mz - m^2 = 0.$$

At $\chi = 0$ the two expressions agree and equal s .

Lemma E.4 (Direct radial forcing). *One has*

$$(117) \quad 0 < c_* < 1, \quad c_* > 1 - m.$$

Assume that every U_i is Vd0 and that $U_C, U_0, \dots, U_5$ cover H . Then, for every i ,

$$D_i = (1 - c_*)V_i \in U_C.$$

Consequently U_C contains the centered closed disk

$$(118) \quad \mathcal{D}_\eta = \{x : \|x\| \leq \eta\}, \quad \eta = h(1 - c_*).$$

Proof. On the C_L branch, $C_L(m) < 1$ because the defining polynomial is positive at 1; if $s < \sqrt{3}/2$, then $c_* \geq \sqrt{3}/2 > s > 1 - m$, while if $s \geq \sqrt{3}/2$ its value at s is $\chi \leq 0$, so $c_* \geq s > 1 - m$. On the other branch put $E = \sqrt{4s^2 - 3}$. The selector gives $sE > M - m$, whence $c_* < s < 1$. Moreover, with

$$A_0 = \frac{2M}{1 - m} - 1,$$

one has

$$A_0^2 - E^2 = \frac{4(1 - s)\{(1 - m)(1 - 2m) + m(2 - m)(1 - s)\}}{(1 - m)^2} > 0.$$

Thus $A_0 > E$ and $c_* > 1 - m$. This proves (117).

Fix i . If $D_i \in U_i$, openness supplies $\varepsilon > 0$ with $c_* + \varepsilon < 1$ such that $(1 - c_* - \varepsilon)V_i \in U_i$. The same role contains its two selected handoff anchors. Since their incoming and outgoing demands dominate p and q , convexity and passage to the closure give a closed unit triangle realizing $(p, q, c_* + \varepsilon)$. This contradicts the definition of $c_* = c_{\max}(p, q)$. Hence $D_i \notin U_i$.

If D_i belonged to U_{i-1} or U_{i+1} , a small plane ball inside that open role would meet the adjacent radial arm r_i in a positive-length interval. This contradicts the Vd0 definition. For the remaining roles put $r = 1 - c_* > 0$. Directly,

$$\|rV_i - V_{i\pm 2}\|^2 = 1 + r + r^2 > 1, \quad \|rV_i - V_{i+3}\| = 1 + r > 1.$$

Since $V_j \in U_j$ and two points of an open unit triangle are at distance strictly below 1, these inequalities exclude D_i from the other three vertex roles. Thus no vertex role contains D_i . The point lies in $\text{int}(H)$ by (117), so the assumed cover forces $D_i \in U_C$.

The six points D_i form a regular hexagon of circumradius $1 - c_*$. Their convex hull contains its centered incircle of radius $h(1 - c_*)$, and convexity of U_C gives (118). $\square$

We next define the three asymmetric witnesses. At V_4 use local coordinates

$$P = V_4 + u(V_5 - V_4) + v(V_3 - V_4), \quad \mathbf{q}(u, v) = u^2 + v^2 - uv.$$

The outgoing demand is b and the incoming demand is a . Accordingly, set

$$\begin{aligned} \alpha &= \frac{h(b + 2a - bD)}{2\rho}, & \beta &= \frac{h(b - a + (a + b)D)}{2\rho}, \\ \gamma &= \frac{h(-b + a + (a + b)D)}{2\rho}, & \delta &= \frac{h(2b + a - aD)}{2\rho}, \end{aligned}$$

where now $\rho = a^2 + ab + b^2$ and $D = \sqrt{4\rho - 3}$. The two line pieces of Theorem E.2 are

$$L_-(\lambda) = (b - \beta\lambda, \alpha\lambda), \quad L_+(\mu) = (\delta\mu, a - \gamma\mu).$$

They meet at $\lambda_* = \mu_* = 8h\rho/(3(D+3))$ in

$$(119) \quad J = \left(\frac{2b+a-aD}{D+3}, \frac{b+2a-bD}{D+3} \right).$$

Put

$$E_- = (1-b, 2-b), \quad E_+ = (2-a, 1-a).$$

The actual adjacent handoffs have local centers

$$C_5(t) = (1+t, t), \quad 1-a \leq t \leq b, \quad \text{and} \quad C_3(y) = (y, 1+y), \quad 1-b \leq y \leq a.$$

The restrictions of the two unit-circle equations are

$$(120) \quad \begin{aligned} g_-(\lambda) &= \frac{3}{4}\lambda^2 - 3(\alpha + b\beta)\lambda + 3b^2 - 3b + 2, \\ g_+(\mu) &= \frac{3}{4}\mu^2 - 3(\delta + a\gamma)\mu + 3a^2 - 3a + 2. \end{aligned}$$

Let λ_- and μ_+ be their first roots; explicitly,

$$\begin{aligned} \lambda_- &= 2(\alpha + b\beta) - \frac{2}{3}\sqrt{9(\alpha + b\beta)^2 - 9b^2 + 9b - 6}, \\ \mu_+ &= 2(\delta + a\gamma) - \frac{2}{3}\sqrt{9(\delta + a\gamma)^2 - 9a^2 + 9a - 6}. \end{aligned}$$

Lemma E.5 (Moving adjacent-role signs). *Throughout (114),*

$$(121) \quad \lambda_* < \lambda_- < h^{-1}, \quad \mu_* < \mu_+ < h^{-1}.$$

Moreover $J_u, J_v < 1/2$. The point $L_+(\mu_+)$ satisfies

$$(L_+(\mu_+))_u + (L_+(\mu_+))_v < 3 - 2a,$$

and the reflected bound holds for $L_-(\lambda_-)$. Writing $F(P, t) = \mathbf{q}(P - C_5(t)) - 1$, one has, for every $t \in [1-a, b]$,

$$(122) \quad F(L_+(\mu_+), t) \geq 0, \quad F(J, t) > 0, \quad F(L_-(\lambda_-), t) > 0.$$

Under the reflection $(u, v, a, b) \leftrightarrow (v, u, b, a)$, the analogous signs hold for every $y \in [1-b, a]$: the first-root point $L_-(\lambda_-)$ has nonnegative sign against $C_3(y)$, and the other two witnesses have strictly positive sign.

Proof. Put

$$U = a + b - 1, \quad z = a - b, \quad D^2 = z^2 + 6U + 3U^2.$$

Then $0 < U < 1/6$ and $z^2 < \Omega := 1 - 6U - 3U^2$. We give the fixed-line calculation for the circle centered at $E_+ = (2-a, 1-a)$; reflection gives the calculation for E_- . Put

$$F(P, t) = \mathbf{q}(P - (1+t, t)) - 1, \quad t_0 = 1-a,$$

so this circle is $F(P, t_0) = 0$. At the junction, exact expansion gives

$$(D+3)^2 F(J, t_0) = 3D(z^2 + Uz + 1 - 3U) + P_J,$$

where

$$P_J = z^4 + 5z^2 + 6Uz^2 + 3U^2z^2 + 9Uz + 3U + 15U^2 > 0;$$

indeed $z^2 + Uz + 1 - 3U \geq 1 - 3U - U^2/4 > 0$, and $5z^2 + 9Uz + 15U^2$ is positive definite. Thus $F(J, t_0) > 0$. Also

$$J_u + J_v = \frac{(1+U)(3-D)}{3+D} < 1.$$

Indeed this inequality is $3U < D(2 + U)$, which follows from $D^2 \geq 3U(2 + U)$ and $(2 + U)^3 > 3U$. Since

$$\frac{\partial F}{\partial t}(P, t) = 1 + 2t - P_u - P_v,$$

$F(J, t)$ is strictly increasing for $t \geq 0$, and therefore is positive throughout $[1 - a, b]$.

On the opposite line L_- , the derivative at the junction, after multiplication by a positive factor, has at $t = 1 - a$ the numerator

$$N_0 = D(9 + 3z^2 + 6Uz - 9U^2) + P_0,$$

where

$$P_0 = 18U^3 + 6U^2z^2 + 63U^2 + 18Uz^2 + 18Uz + 54U \\ + 2z^4 + 9z^2 + 9.$$

The coefficient of D is at least $9 - 6U - 9U^2 > 0$, while $P_0 \geq 9 + 54U - 18U > 0$. Hence $N_0 > 0$. At the other endpoint $t = b$, the corresponding numerator is

$$N_1 = D(9 + 3z^2 + 6U - 3U^2) + P_1,$$

where

$$P_1 = 9U^4 - 3U^3z + 45U^3 + 9U^2z^2 - 6U^2z + 72U^2 \\ - Uz^3 + 21Uz^2 + 9Uz + 45U + 2z^4 + 9z^2 + 9.$$

Its D -coefficient is positive and, using $|z| < 1$,

$$P_1 \geq 9 + 45U - 9U - U - 6U^2 - 3U^3 > 0.$$

The junction derivative is affine in t , so it is positive throughout $[1 - a, b]$. Since the restriction of F to L_- is a convex quadratic, is positive at the junction, and is increasing there, it has no intersection on the opposite line side for any actual adjacent handoff.

On the correct line L_+ , the value at its outer endpoint h^{-1} has, after multiplication by a positive factor, numerator

$$A_0 = P_2 - 4DU(1 + U + z),$$

where

$$P_2 = 3U^4 + 6U^3z + 6U^3 + 4U^2z^2 + 12U^2z + 12U^2 \\ + 2Uz^3 + 6Uz^2 + 2Uz + 6U + z^4 + 2z^2 - 3.$$

For fixed U , replace the odd powers of z by $x = |z|$. The resulting polynomial $P_2^+(U, x)$ satisfies

$$\partial_x P_2^+ = 2(3U^3 + 4U^2x + 6U^2 + 3Ux^2 + 6Ux + U + 2x^3 + 2x) > 0,$$

and therefore its supremum for $z^2 < \Omega$ occurs at $x = \sqrt{\Omega}$. There

$$P_2^+(U, \sqrt{\Omega}) = 4U(U + \sqrt{\Omega} - 3) < 0.$$

As $1 + U + z = 2a > 0$, this gives $A_0 < 0$. The positive junction value and negative endpoint value force the first root strictly between them, proving the L_+ half of (121); reflection proves the other half.

For the coordinate bounds, the exact identities

$$D^2 - (3U - z)^2 = 6U(1 + z - U), \quad D^2 - (3U + z)^2 = 6U(1 - z - U)$$

give $J_u, J_v < 1/2$. It remains to control the coordinate sum along L_+ . For $E = L_+(h^{-1})$, the sign of $3 - 2a - E_u - E_v$ is the sign of

$$D(6U + 2z + 6) + B(U, z),$$

where

$$B(U, z) = -9U^3 - 9U^2z - 9U^2 - 3Uz^2 - 18Uz + 3U - 3z^3 + 3z^2 - 3z + 3.$$

Here

$$B_z = -3(3z^2 + 2Uz + 6U - 2z + 3U^2 + 1) < 0,$$

because the quadratic in parentheses has discriminant $-32U^2 - 80U - 8 < 0$. Hence its minimum on $z^2 < \Omega$ is at $z = \sqrt{\Omega}$, where $B(U, \sqrt{\Omega}) = 6(1 - 3U - \sqrt{\Omega}) > 0$ because $(1 - 3U)^2 - \Omega = 12U^2$. Thus $E_u + E_v < 3 - 2a$. Also $J_u + J_v < 1 < 3 - 2a$, and the coordinate sum is affine on the line segment, so the same strict bound holds at the first root. Now

$$\frac{\partial F}{\partial t}(P, t) = 1 + 2t - P_u - P_v.$$

Since $F(L_+(\mu_+), 1-a) = 0$, the coordinate-sum bound makes its t -derivative strictly positive for $t \geq 1 - a$. Together with the no-opposite-line result, this proves (122). Reflection gives both the bound on L_- and the complete $C_3(y)$ statement. $\square$

Define the local and global witnesses by

$$(123) \quad Q_-^{\text{loc}} = L_-(\lambda_-), \quad Q_0^{\text{loc}} = J, \quad Q_+^{\text{loc}} = L_+(\mu_+),$$

and by the displayed conversion from (u, v) to the plane.

Lemma E.6 (Direct asymmetric forcing). *Assume that $U_C, U_0, \dots, U_5$ cover H . Then*

$$Q_-, Q_0, Q_+ \in \text{int}(H) \setminus \bigcup_{i=0}^5 U_i,$$

and consequently all three witnesses belong to U_C .

Proof. The root order and positivity of the line coordinates give $0 < u, v < 1$ for all three points. Each belongs to $\mathcal{R}(b, a)$ and hence lies in a closed unit triangle containing V_4 , so $\mathbf{q}(u, v) \leq 1$; hence $|u - v| < 1$ and all three points lie in $\text{int}(H)$.

The closure of U_4 is one of the triangles represented by $\mathcal{R}(b, a)$, since it contains V_4, X_3, X_4 . All three witnesses lie on its exact non-axis frontier. If one belonged to the open triangle U_4 , a sufficiently small plane ball about it would lie both in U_4 and in the interior of the corner cone, making the witness an interior point of $\mathcal{R}(b, a)$. This contradiction excludes U_4 .

For the roles U_0, U_1, U_2 , their contained vertices have local coordinates

i	0	1	2
V_i	(2, 1)	(2, 2)	(1, 2).

For $P = (u, v)$,

$$\|P - (2, 1)\|^2 - 1 = \mathbf{q}(u, v) + 2 - 3u, \quad \|P - (1, 2)\|^2 - 1 = \mathbf{q}(u, v) + 2 - 3v.$$

The inequalities $J_u, J_v < 1/2$ and the line order give distance greater than one from V_0 for Q_-, Q_0 and from V_2 for Q_0, Q_+ . For the remaining pair, the first-root circle and the sum bound in Lemma E.5 give

$$\|Q_+ - V_0\|^2 - 1 = a(3 - a - (Q_+)_u - (Q_+)_v) > a^2,$$

and, by reflection, $\|Q_- - V_2\|^2 - 1 > b^2$. All three points are farther than one from $V_1 = (2, 2)$, because writing $u = 1 - \xi$, $v = 1 - \eta$ gives

$$\|P - V_1\|^2 = 1 + \xi + \eta + \xi^2 + \eta^2 - \xi\eta > 1.$$

Since two points in one open unit equilateral triangle have distance strictly less than one, these inequalities exclude U_0, U_1, U_2 .

The actual handoff $X_5 = C_5(x_5)$ belongs to U_5 , and $1 - a < x_5 < b$ by (E.3). The three signs in (122) therefore put every witness at distance at least one from X_5 , excluding U_5 . Similarly, if $y = 1 - x_2$, then $1 - b < y < a$, $X_2 = C_3(y) \in U_3$, and the reflected signs exclude U_3 . Thus no vertex role contains any witness. The assumed cover now forces all three interior witnesses into U_C . $\square$

The nine directly forced points are $D_0, \dots, D_5, Q_-, Q_0, Q_+$. Under a putative cover, Lemmas E.4 and E.6 therefore force the compact set

$$(124) \quad K_{\text{wit}}(a, b) = \mathcal{D}_\eta \cup \{Q_-, Q_0, Q_+\} \subset U_C.$$

E.5. Tangent reduction and exact mixed overlaps. For the nontrivial regime $c_* > 2/3$, we first remove the square radicals in the first-root witnesses. Normalize $\bar{\alpha} = \alpha/h$, and similarly for the other three coefficients, and put

$$\xi = \lambda/h, \quad \zeta = \mu/h, \quad \xi_* = \zeta_* = \frac{8\rho}{3(D+3)}.$$

Let g_-, g_+ be the rescaled forms of (120), and take one Newton step at the junction:

$$\widehat{\xi}_- = \xi_* - \frac{g_-(\xi_*)}{g'_-(\xi_*)}, \quad \widehat{\zeta}_+ = \zeta_* - \frac{g_+(\zeta_*)}{g'_+(\zeta_*)}.$$

Define, in local coordinates,

$$\begin{aligned} A &= \left(b - \frac{3}{4}\bar{\beta}\widehat{\xi}_-, \frac{3}{4}\bar{\alpha}\widehat{\xi}_- \right), \\ B &= J, \\ C &= \left(\frac{3}{4}\bar{\delta}\widehat{\zeta}_+, a - \frac{3}{4}\bar{\gamma}\widehat{\zeta}_+ \right), \end{aligned}$$

and convert them to global vectors.

Lemma E.7 (Newton inner reduction). *The points above have coordinates rational in a, b, D and satisfy*

$$A \in (Q_0, Q_-), \quad C \in (Q_0, Q_+).$$

Consequently

$$\widehat{K} = \text{conv}(\mathcal{D}_\eta \cup \{A, B, C\}),$$

obeys $\widehat{K} \subseteq \text{conv}(K_{\text{wit}})$.

Proof. Each quadratic g_-, g_+ is strictly convex, positive, and decreasing at the junction. Its tangent zero therefore lies strictly between the junction and its first root. This gives the two open-segment inclusions. All displayed coefficients and Newton quotients are rational in a, b, D , and convexity gives the asserted containment. $\square$

Put $W = C + RA$. For a disk $\mathbb{D}(X, r)$ let

$$I(X, r) = \{n : \|n\| = 1, \langle X, n \rangle + r \geq h\}.$$

E.6. The exact global certificate for the mixed overlaps. This subsection proves the two mixed-overlap signs used in Proposition 7.2. It first replaces the exact radial value by rational upper envelopes, then reduces the mixed residuals to eight integer-polynomial signs. Three fixed analytic charts and twenty global Bernstein expansions prove those signs. No interval arithmetic, box subdivision, pruning, or branch-and-bound is used.

Retain the notation of Section E. Thus

$$p = 1 - b, \quad q = 1 - a, \quad s = p + q, \quad m = \min\{p, q\}, \quad M = \max\{p, q\},$$

$$\chi = s^4 - s^2 + mM, \quad h = \frac{\sqrt{3}}{2}, \quad c_* = c_{\max}(p, q).$$

The strict domain gives

$$(125) \quad mM > (1 - s)(2 - s), \quad m > 1 - s, \quad s > \frac{2}{3}.$$

Put

$$F_m(x) = x^4 - x^2 + mx - m^2, \quad \kappa = F'_m(s) = 4s^3 - 2s + m.$$

Since $m > 1 - s$,

$$(126) \quad \kappa > 4s^3 - 3s + 1 = (s + 1)(2s - 1)^2 > 0.$$

E.6.1. Rational radial upper envelopes. Define

$$(127) \quad \bar{c}_L = s - \frac{\chi}{\kappa} \quad (\chi \leq 0), \quad \bar{c}_T = s - \frac{\chi}{1 - m} \quad (\chi \geq 0).$$

Lemma E.8 (Branchwise radial upper bounds). *On the corresponding radial cells,*

$$c_* \leq \bar{c}_L < 1, \quad c_* \leq \bar{c}_T < 1.$$

At $\chi = 0$ both upper bounds equal $s = c_*$.

Proof. On the $C_L(m)$ cell, $F_m(s) = \chi \leq 0$ and $F_m(c_*) = 0$. For $x \geq s > 2/3$ one has $F''_m(x) = 12x^2 - 2 > 0$, so convexity and (126) give

$$0 = F_m(c_*) \geq F_m(s) + F'_m(s)(c_* - s),$$

whence $c_* \leq s - \chi/\kappa = \bar{c}_L$. Let $U = 1 - s$. From (125), $mM > U(1 + U)$, and therefore

$$\begin{aligned} \chi + U\kappa &> U\{m + 3s^3 - s^2 - 3s + 2\} \\ &= U\{m + 1 - 4U + 8U^2 - 3U^3\} > 0. \end{aligned}$$

Indeed $0 < U < 1/3$ and $1 - 4U + 8U^2 - 3U^3 > 1 - 4U + 7U^2 > 0$. Thus $\bar{c}_L < 1$.

On the other cell put

$$E = \sqrt{4s^2 - 3}, \quad m_0 = \frac{s(1 - E)}{2}, \quad M_0 = \frac{s(1 + E)}{2}.$$

Then

$$\chi = (m - m_0)(M_0 - m), \quad s - c_* = \frac{2(m - m_0)}{1 + E}.$$

Hence

$$\frac{\chi}{s - c_*} = \frac{1 + E}{2}(M_0 - m) \leq 1 - m,$$

because $0 \leq E \leq 1$ and $M_0 \leq 1$. This proves $c_* \leq \bar{c}_T$, while $\chi > 0$ gives $\bar{c}_T < s < 1$. $\square$

Reflect so that $z = a - b \geq 0$ and put $U = a + b - 1$. Then

$$\begin{aligned} s &= 1 - U, & m &= \frac{1 - U - z}{2}, & D^2 &= z^2 + 6U + 3U^2 =: K(U, z), \\ \Xi(U, z) &:= 4\chi = 1 - 10U + 21U^2 - 16U^3 + 4U^4 - z^2, \\ G(U, z) &:= 2\kappa = 5 - 21U + 24U^2 - 8U^3 - z. \end{aligned}$$

Consequently

$$(128) \quad \bar{c}_L = 1 - U - \frac{\Xi(U, z)}{2G(U, z)}, \quad \bar{c}_T = 1 - U - \frac{\Xi(U, z)}{2(1 + U + z)}.$$

Neither expression contains the quartic root or the radial radical $\sqrt{4s^2 - 3}$.

E.6.2. The Gram residuals and exact core reduction. On the active cell write $\bar{c}$ for the corresponding expression in (128), and put

$$\bar{\eta} = h(1 - \bar{c}), \quad e = 1 - \bar{c}, \quad t = 2\bar{c} - 1.$$

Lemma E.8 gives $0 < \bar{\eta} \leq \eta = h(1 - c_*)$ when $c_* > 2/3$. Thus the three disks obtained from $\mathbb{D}(C, 2\eta)$, $\mathbb{D}(W, \eta)$, and $\mathbb{D}(RA, 2\eta)$ by replacing η with $\bar{\eta}$ are contained in the disks already present in the Minkowski sum.

Represent a global vector as (x, hy) and put $C' = R^{-1}C$,

$$N_A = \|A\|^2, \quad N_C = \|C\|^2, \quad \nu = \langle A, C' \rangle, \quad \Delta_0 = x_A y_{C'} - y_A x_{C'}.$$

The Gram identity is

$$h^2 \Delta_0^2 = N_A N_C - \nu^2.$$

The two mixed square residuals for the smaller disks factor as

$$(129) \quad \bar{P}_A = h^2 N_A Q_A, \quad Q_A = \Delta_0^2 - \|tA - eC'\|^2,$$

$$(130) \quad \bar{P}_C = h^2 N_C Q_C, \quad Q_C = \Delta_0^2 - \|tC' - eA\|^2.$$

All witness coordinates are rational in U, z, D . Substituting (128) into (129)–(130) and reducing modulo $D^2 - K(U, z)$ gives, for $B \in \{L, T\}$ and $X \in \{A, C\}$,

$$(131) \quad \text{num}(Q_{B,X}) = 32(K + 3)^2 \{A_{B,X}(U, z) + DB_{B,X}(U, z)\},$$

where $A_{B,X}, B_{B,X} \in \mathbb{Z}[U, z]$. Every omitted denominator is strictly positive: it is a positive constant times $(D + 3)^4$, the square of the relevant upper-envelope denominator, and squares of the two nonzero Newton-derivative numerators. The term counts are

(B, X)	$\#A_{B,X}$	$\#B_{B,X}$
(L, A)	347	298
(L, C)	347	298
(T, A)	342	294
(T, C)	341	294.

Since $D \geq 0$, it remains to prove all eight integer polynomials nonnegative.

E.6.3. *Three fixed charts and global positive bases.* Set

$$\mathcal{G}(U) = 1 - 6U - 3U^2, \quad H(U) = 1 - 10U + 21U^2 - 16U^3 + 4U^4,$$

$$U_H = 1 - \frac{\sqrt{3}}{2}, \quad U_{\max} = \frac{2}{\sqrt{3}} - 1.$$

The identities

$$4\chi = H(U) - z^2, \quad H(U) = (U - 1)^2(4U^2 - 8U + 1)$$

and

$$\mathcal{G}(U) - H(U) = 4U(1 - 6U + 4U^2 - U^3) \geq 0 \quad (0 \leq U \leq U_H)$$

show that the radial cells are exactly covered by

$$(132) \quad \begin{aligned} T : & \quad 0 \leq U \leq U_H, \quad 0 \leq z^2 \leq H(U), \\ L_0 : & \quad 0 \leq U \leq U_H, \quad H(U) \leq z^2 \leq \mathcal{G}(U), \\ L_1 : & \quad U_H \leq U \leq U_{\max}, \quad 0 \leq z^2 \leq \mathcal{G}(U). \end{aligned}$$

We use the elementary positive-basis fact that a polynomial whose coefficients in the tensor Bernstein basis on $[0, 1]^2$ are all nonnegative is nonnegative on that square. This is one global polynomial identity, not a subdivision procedure. All coefficients below lie in $\mathbb{Q}(\sqrt{3})$, and their signs are decided exactly by rational comparisons.

For the T cell use

$$(133) \quad U = \frac{(x-1)(3-x)}{4x}, \quad z = y \frac{9-x^4}{8x^2}, \quad 1 \leq x \leq \sqrt{3}, \quad 0 \leq y \leq 1.$$

Here

$$H\left(\frac{(x-1)(3-x)}{4x}\right) = \left(\frac{9-x^4}{8x^2}\right)^2,$$

so the chart covers the whole T cell. Clearing the positive powers of 2 and x , then putting $x = 1 + (\sqrt{3} - 1)r$, gives four global Bernstein expansions on $[0, 1]^2$:

	bidegree	coefficients	zero coefficients
$A_{T,A}$	(88, 22)	2047	2
$B_{T,A}$	(80, 20)	1701	0
$A_{T,C}$	(88, 22)	2047	2
$B_{T,C}$	(80, 20)	1701	0.

Every nonzero coefficient is strictly positive.

For an L -cell polynomial $F(U, z)$, write

$$F(U, z) = E_F(U, w) + zO_F(U, w), \quad w = z^2,$$

and put

$$R_F(U, w) = E_F(U, w)^2 - wO_F(U, w)^2.$$

The implications $E_F \geq 0$ and $R_F \geq 0$ give $E_F \geq \sqrt{w}|O_F|$, hence $F \geq 0$. Use the two fixed charts

$$(134) \quad U = U_H r, \quad w = H(U) + s\{\mathcal{G}(U) - H(U)\}$$

for L_0 , and

$$(135) \quad U = U_H + (U_{\max} - U_H)r, \quad w = s\mathcal{G}(U)$$

for L_1 , with $0 \leq r, s \leq 1$. The global Bernstein expansions of E_F and R_F have the following bidegrees and coefficient counts:

F	$L_0 : E_F$	$L_0 : R_F$	$L_1 : E_F$	$L_1 : R_F$
$A_{L,A}$	(46, 11); 564	(92, 22); 2139	(26, 11); 324	(52, 22); 1219
$B_{L,A}$	(42, 10); 473	(84, 20); 1785	(24, 10); 275	(48, 20); 1029
$A_{L,C}$	(46, 11); 564	(92, 22); 2139	(26, 11); 324	(52, 22); 1219
$B_{L,C}$	(42, 10); 473	(84, 20); 1785	(24, 10); 275	(48, 20); 1029

Every coefficient in these sixteen expansions is strictly positive. Therefore all eight polynomials in (131) are nonnegative on the corresponding radial cells.

Lemma E.9 (Exact mixed-cap overlaps). *In the regime $c_* > 2/3$, the tangent witnesses defined above satisfy*

$$I(C, 2\eta) \cap I(W, \eta) \neq \emptyset, \quad I(W, \eta) \cap I(RA, 2\eta) \neq \emptyset.$$

Proof. Since $D \geq 0$, the eight signs prove $Q_A, Q_C \geq 0$ for $z \geq 0$ by (131); reflection supplies $z \leq 0$. Equations (129)–(130) give the two mixed cap overlaps for the smaller disks, hence for the original disks. $\square$

Exact replay record. The canonical data are in

proof/3XXX_CE0/31XX_Nplus1/310X_all_Vd0/

3105X_self_contained_direct_Vd0_nine_point/3105X_computation/

The module `mixed_overlap_core_polynomials.py` assembles the six numbered data shards. The exact checks

`verify_mixed_overlap_core_derivation.py`

`verify_global_core_positivity.py`

derive the eight core polynomials and verify the twenty global Bernstein identities over $\mathbb{Q}(\sqrt{3})$. They authenticate SHA-256 digest

dc46aaf263655d5159ecd3a81db72ee82477951d06172f4743b248df37209485

and return

`adaptive_subdivision: false`

`branch_and_bound: false`

`result: PASS`

using exact integer, rational, and $\mathbb{Q}(\sqrt{3})$ arithmetic.

E.7. Ray order and four overlaps.

Proposition E.10 (Technical four-overlap verification). *Assume (114) and $c_* > 2/3$. Then*

$$\|A\|, \|C\| > \eta,$$

the rays A, B, C, W, RA occur in that cyclic order, and the four consecutive cap pairs

$$I(A, 2\eta) \leftrightarrow I(B, 2\eta) \leftrightarrow I(C, 2\eta) \leftrightarrow I(W, \eta) \leftrightarrow I(RA, 2\eta)$$

overlap across their intervening short sectors.

Proof. Put $\tau = h - 2\eta = h(2c_* - 1)$. The rays of

$$(136) \quad A, B, C, W, RA$$

occur in this cyclic order from A to RA . Indeed the two line distances give

$$\frac{\text{cross}(A, B)}{\|B - A\|} = \alpha(1 - b) + \beta > 0, \quad \frac{\text{cross}(B, C)}{\|C - B\|} = \gamma + \delta(1 - a) > 0.$$

If $A = (-X, -Y)$ and $C = (-P, -Q)$ in the centered cone basis, then $0 < X, Y, P, Q < 1$, $X, Q > 1/2$, and

$$\frac{\text{cross}(C, RA)}{h} = PX + (Q - P)Y > 0.$$

The last sign is immediate unless $Q < P$ and $X < Y$, in which case it is larger than $Q - P/2 > 0$. The ray of $W = C + RA$ lies between those of C and RA . The same coordinates give

$$(137) \quad \|A\|, \|C\| > h/2 > \eta.$$

For the two equal-radius overlaps it is enough to prove

$$(138) \quad \frac{\text{cross}(A, B)}{\|B - A\|} \geq \tau, \quad \frac{\text{cross}(B, C)}{\|C - B\|} \geq \tau.$$

We record the exact analytic verification. By reflection take $a \leq b$ and set $m = 1 - b$, $M = 1 - a$, $U = a + b - 1$, $s = m + M$, and $w = M - m$. The smaller of the two normalized left sides in (138) is

$$(139) \quad d = \frac{\mathcal{A} + U(2m - 1) + D(\mathcal{A} + U)}{2\rho}, \quad \mathcal{A} = 1 - m + m^2.$$

On the C_L sheet, write $F_L(z) = z^4 - z^2 + mz - m^2$. This defining polynomial evaluated at $r = 1 - m/2 + m^2/2$ is

$$F_L(r) = \frac{m^2(m - 1)}{16}(m^5 - 3m^4 + 11m^3 - 17m^2 + 28m - 12) > 0.$$

Indeed the polynomial in parentheses is increasing on $(0, 1/2)$ and has value $-33/32$ at $1/2$; its derivative is $28 - 34m + m^2(5m^2 - 12m + 33) > 0$. Since $F_L(h) = -(m - h/2)^2 \leq 0$ and the selected root is unique, $2c_* - 1 \leq \mathcal{A}$. Put

$$X_0 = \mathcal{A} + U, \quad Y_0 = 2\rho\mathcal{A} - \mathcal{A} - U(2m - 1).$$

Then

$$d - \mathcal{A} = \frac{D(\mathcal{A} + U) - Y_0}{2\rho},$$

and exact expansion gives

$$\begin{aligned} Y_0 &= 2(m^2 - m + 1)U^2 + (2m^3 - 2m + 3)U \\ &\quad + (m^2 - m + 1)(2m^2 - 2m + 1) > 0. \end{aligned}$$

Furthermore

$$D^2X_0^2 - Y_0^2 = 4m\rho\Phi(U, m),$$

where

$$\begin{aligned} \Phi(U, m) &= (1 - m)(m^2 - m + 2)U^2 \\ &\quad + (2 - m^2)(m^2 - m + 1)U \\ &\quad - m(1 - m)^2(m^2 - m + 1). \end{aligned}$$

In these variables

$$\chi = U^4 - 4U^3 + 5U^2 - (m + 2)U + m(1 - m).$$

Since $U^2(U^2 - 4U + 5) > 0$, the selector $\chi \leq 0$ implies $U > m(1 - m)/(m + 2)$; Φ is increasing in U and at that lower endpoint is

$$\frac{m^2(1 - m)(m^4 - 2m^3 + 6m^2 - 6m + 4)}{(m + 2)^2} > 0.$$

The last quartic equals $(m^2 - m)^2 + 5m^2 - 6m + 4 > 0$. Thus $d > \mathcal{A} \geq 2c_* - 1$.

On the other radial sheet let $E = \sqrt{4s^2 - 3}$. Then $0 \leq w < sE$ and $c_* = (s + w)/(1 + E)$. At fixed U , differentiation of (139) can be checked without an implicit estimate. Namely,

$$d = \frac{1 + D}{2} - UB(D, w), \quad B(D, w) = \frac{(1 + U)(3 + D) + w(1 - D)}{D^2 + 3}.$$

Here $D^2 = w^2 + 6U + 3U^2$, so $0 \leq D' = w/D \leq 1$, and

$$B_w = \frac{1 - D}{D^2 + 3} \geq 0.$$

Writing $N = (1 + U)(3 + D) + w(1 - D)$, direct differentiation gives

$$B_D = \frac{(1 + U - w)(D^2 + 3) - 2DN}{(D^2 + 3)^2} > -\frac{34}{27}.$$

Indeed the first term in the numerator is positive, while $1 + U - w = 2a > 0$, $N < (7/6)4 + 1 = 17/3$, $D < 1$, and $D^2 + 3 \geq 3$. If $B_D < 0$, then $B_D D' \geq B_D$; otherwise $B_D D' \geq 0$. Hence $B' = B_w + B_D D' > -34/27$, and $U < 1/6$ yields

$$d' < \frac{1}{2} + \frac{1}{6} \frac{34}{27} = \frac{115}{162} < 1,$$

whereas $(2c_* - 1)' = 2/(1 + E) \geq 1$. Hence their difference decreases with w . At the boundary $w = sE$, where $\chi = 0$ and $c_* = s$, the preceding sheet applies: this is an admissible point because $h < s < 1$ and $D^2 = 4s^4 - 12s + 9 < 1$. Indeed $s^4 - 3s + 2 = (s - 1)(s^3 + s^2 + s - 2) < 0$ on this interval; the second factor is increasing and already equals $(7h - 5)/4 > 0$ at $s = h$. The preceding sheet gives $d \geq 2s - 1$. Therefore (138) holds on both sheets, and reflection supplies the other line.

It remains to check the two mixed overlaps. On the two radial cells use the exact rational envelopes and global Bernstein identities established in Subsection E.6. Lemma E.9 gives precisely the overlaps for $\mathbb{D}(C, 2\eta)$ with $\mathbb{D}(W, \eta)$ and for $\mathbb{D}(W, \eta)$ with $\mathbb{D}(RA, 2\eta)$.

Thus the four consecutive pairs of caps centered on the rays (136) overlap. $\square$

APPENDIX F. NONSTANDARD SYMBOLS

Standard geometric notation is defined at first use. The following symbols recur across proof mechanisms.

symbol	meaning
CE0, CE1, CE2	The three exhaustive center-role boundary-trace classes.
Vd0, Vd1, Vd2, T3-like	The four exhaustive vertex-role classes.
(A_i, B_i, C_i)	Actual maximal incoming boundary, outgoing boundary, and radial reaches.
(a_i, b_i, c_i)	Selected lower-bound demands; these do not define N_+ .
N_+	The number of actual rows with $A_i + B_i > 1$.
B_c, F_c, G_c	The outgoing feasible envelope, its nonsupercritical cap, and the induced demand map.
$e(d), \psi(x)$	The low-root threshold and the universal selected- T_+ increment $1 - \sqrt{1 - x + x^2}$.
$R, W, E, \alpha, \delta, \Delta_R, \Delta_L$	The common signed CE1/CE2 center variables and trace surpluses.
$f(a, b), G(a, b)$	Normalized maximal retained area and corresponding local area loss.
$c_*, \mathcal{D}_\eta, Q_-, Q_0, Q_+$	The radial envelope, forced disk, and exact asymmetric witnesses in Strategy 4.
$A, B, C, \hat{K}, K_{\text{wit}}$	The Newton inner points, inner witness hull, and exact forced witness set.
$\Lambda(K), I(X, r)$	The least enclosing equilateral side and the associated support cap.